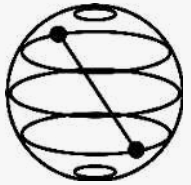
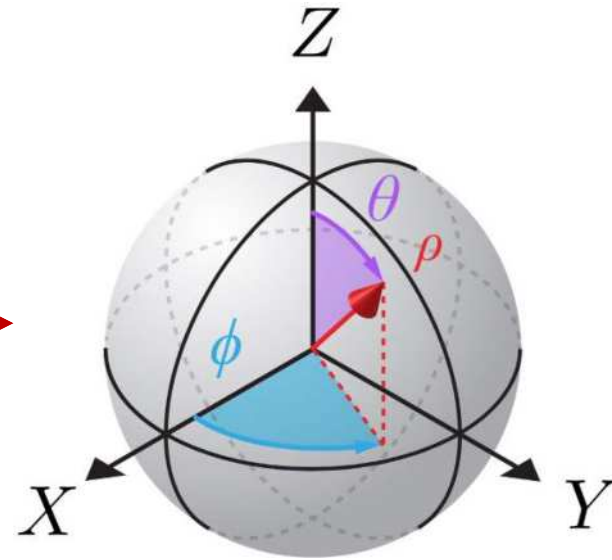
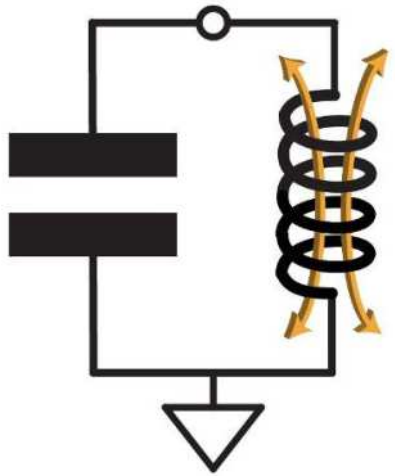


Superconducting Qubits 101

Making Your First Qubit From an Oscillator

Introduction to Circuit
Quantum Electrodynamics (cQED)

Zlatko K. Minev



IBM Quantum
IBM T.J. Watson Research Center, Yorktown Heights, NY



qiskit.org/metal



@zlatko_minev



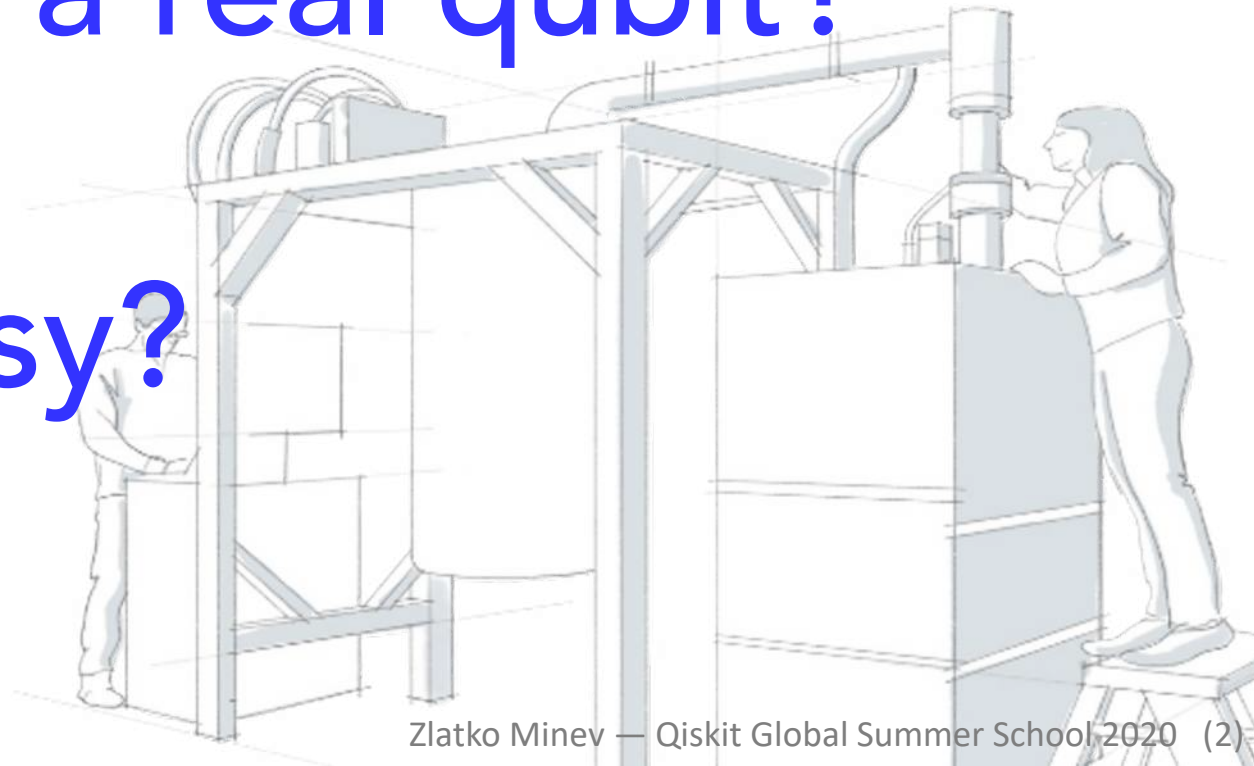
zlatko-minev.com

*Image copyright:
ZKM unless otherwise noted*

What is a real qubit?

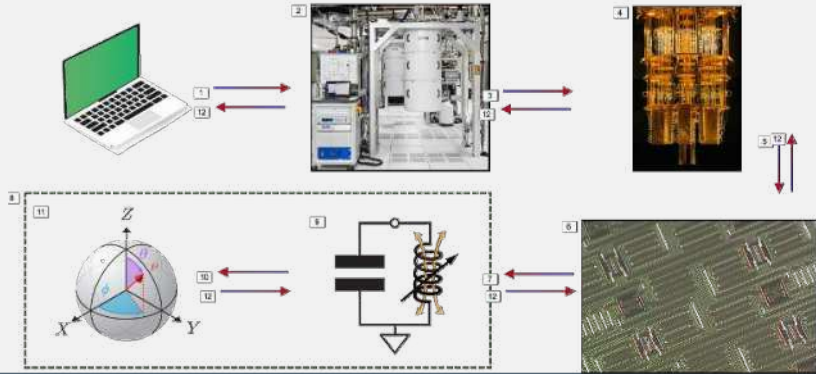
How can you design, control, and measure a real qubit?

Easy?

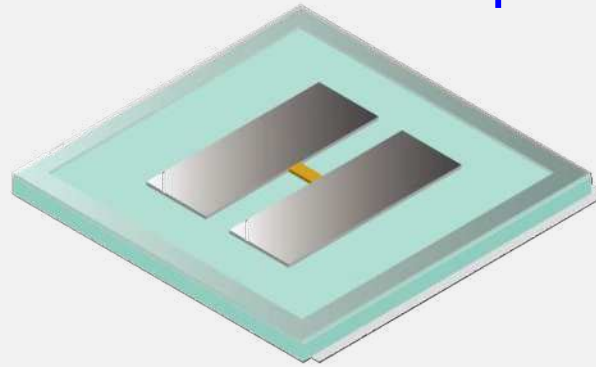


On the road ahead

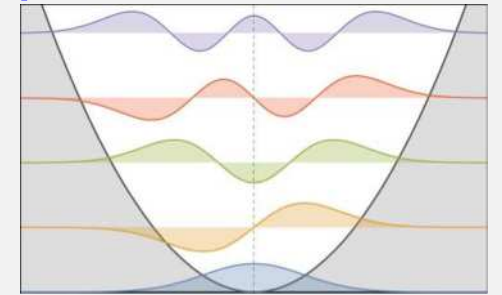
Qubit in the cloud



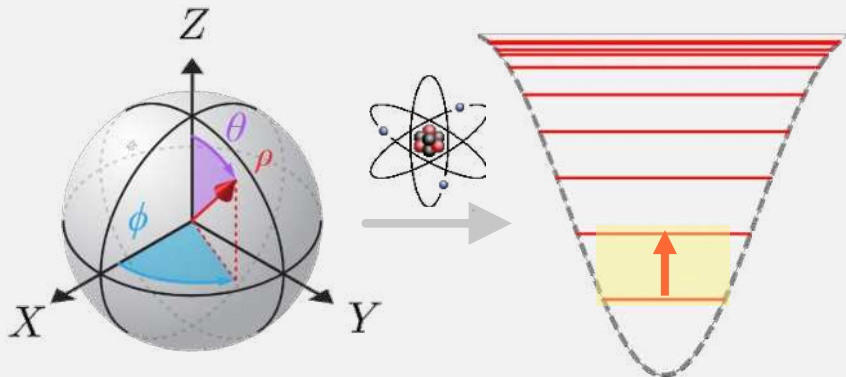
cQED: Transmon qubit



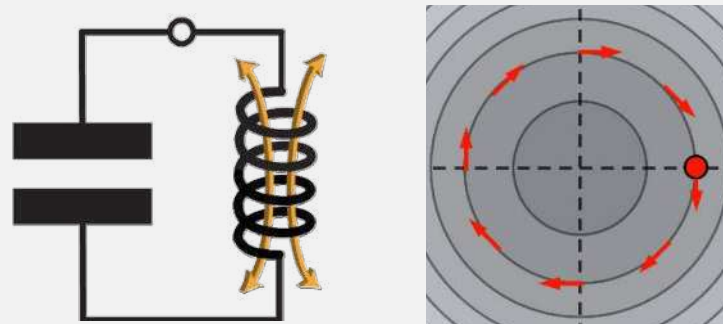
Unveiling the quantum oscillator



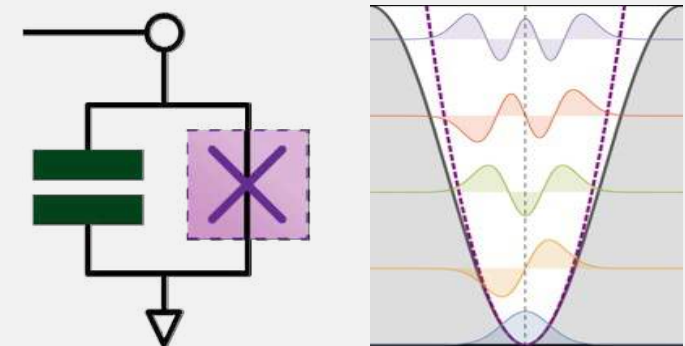
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit



0.0.3

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QDesigns

QComponents

Docs > Qiskit Metal | Quantum Device Design & Analysis (Q-EDA) 0.0.3

Qiskit Metal | Quantum Device Design & Analysis (Q-EDA) 0.0.3

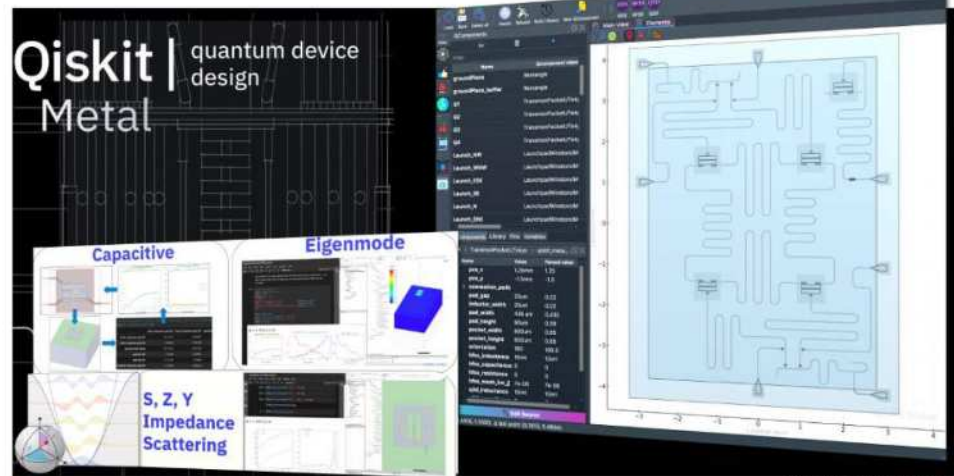
ATTENTION

This is an alpha version of Qiskit Metal, the code is still under development. Please let us know about anything you might want us to add or elaborate upon in the Slack channel [#metal](#) in the [qiskit workspace](#).

HINT

You can open this documentation using

```
import qiskit_metal
qiskit_metal.open_docs()
```

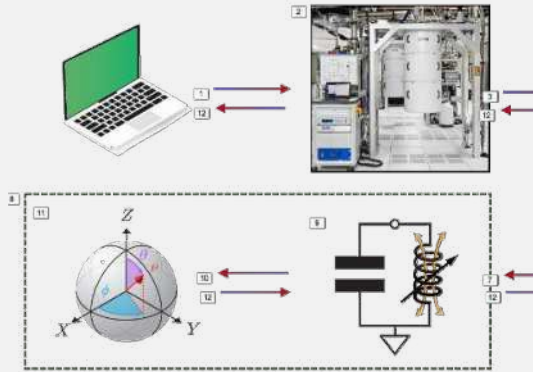


About

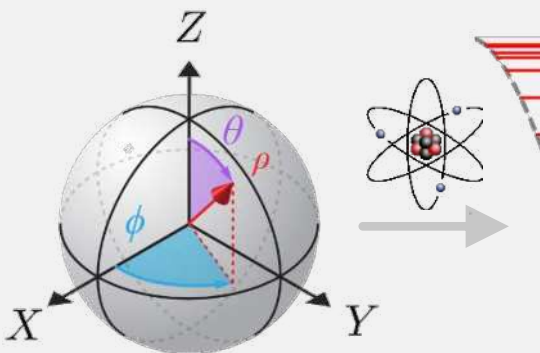
Qiskit Metal | for quantum device design & analysis ('Qiskit Metal') is an open-source framework (and library) for the design of superconducting quantum chips and devices. Call it quantum EDA (Q-EDA) and analysis. Qiskit Metal is:

- Open source
- Community-driven
- A python API and a front-end visual GUI interface

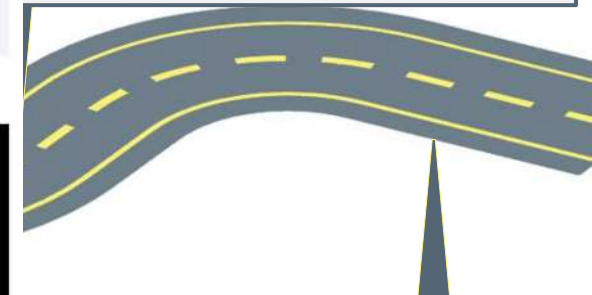
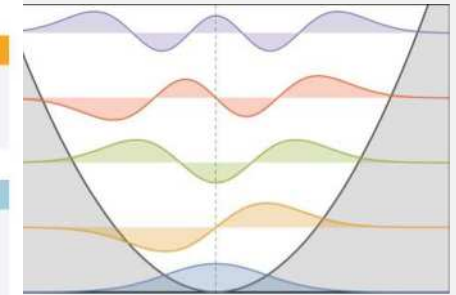
Qubit in the c



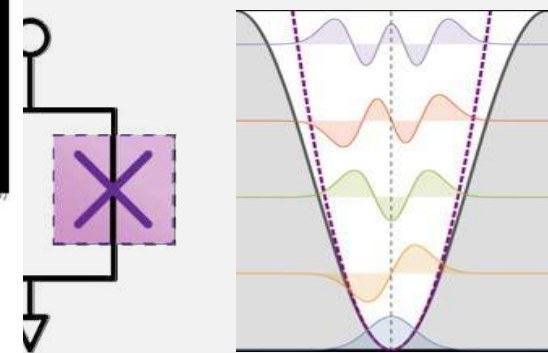
Qubit from atom /



Unveiling the quantum oscillator



ransmon qubit



This Lecture

Introductory and skill reaffirming

Don't need to know much going in, but we will go far

Examples: simplest, most practical examples

Step by step

Ask questions!

Tightly integrated work by Dr. Thomas McConkey!

qiskit.org/metal -> docs -> videos -> E02

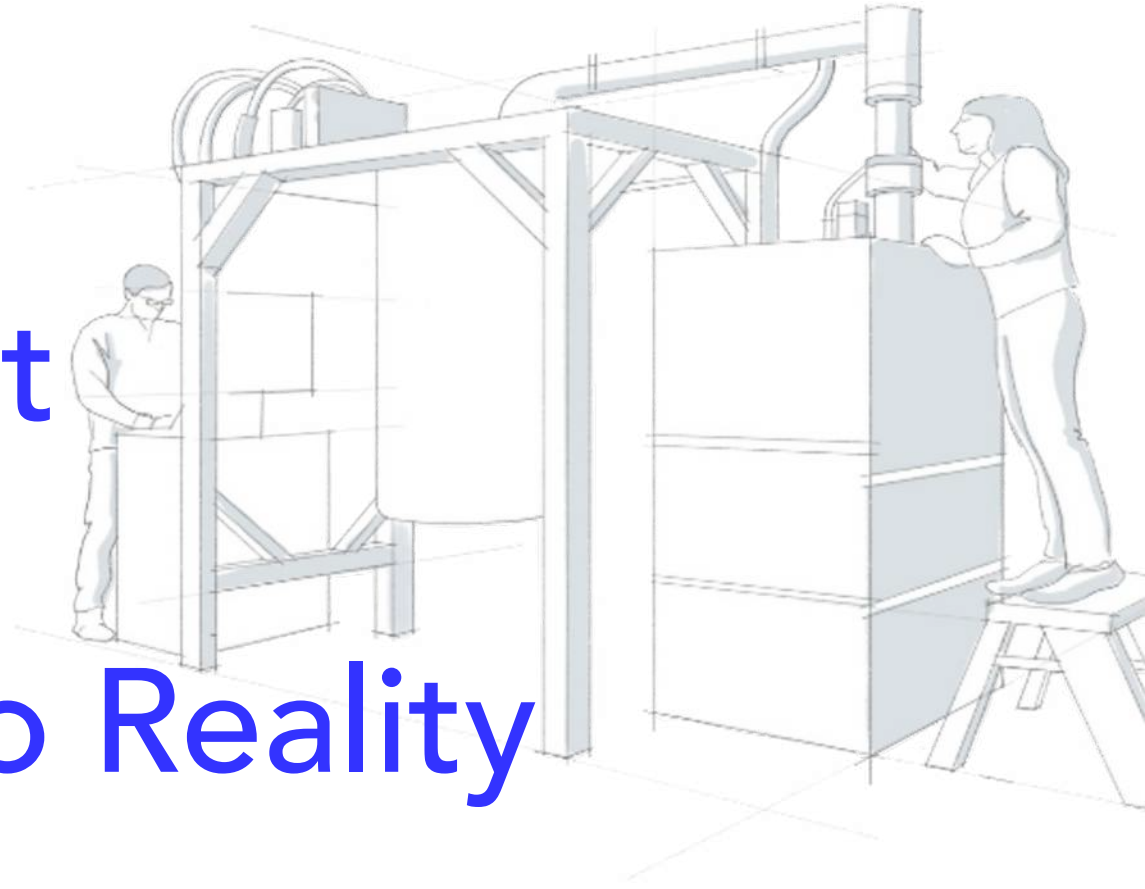


Advanced
material



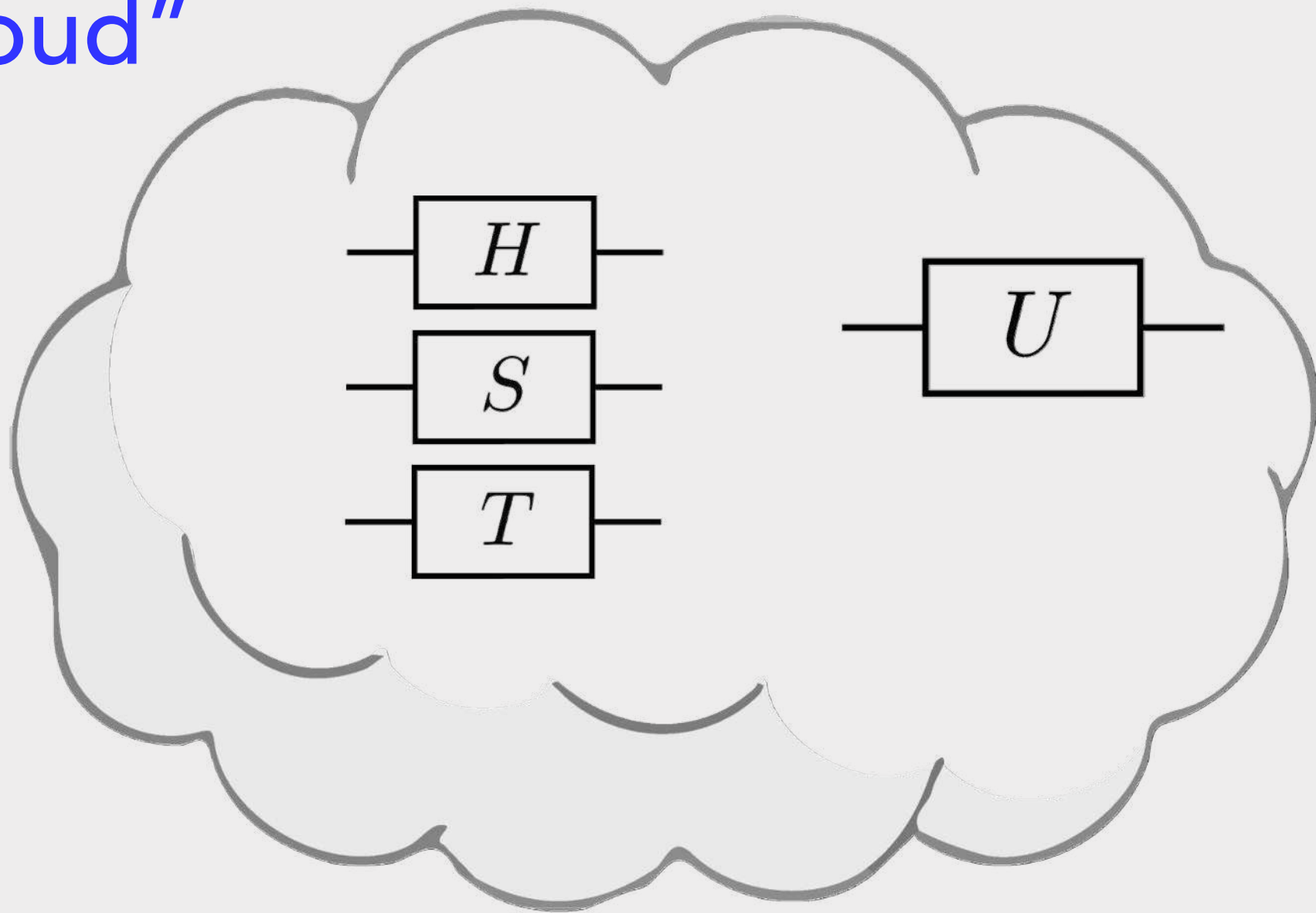
Qubit

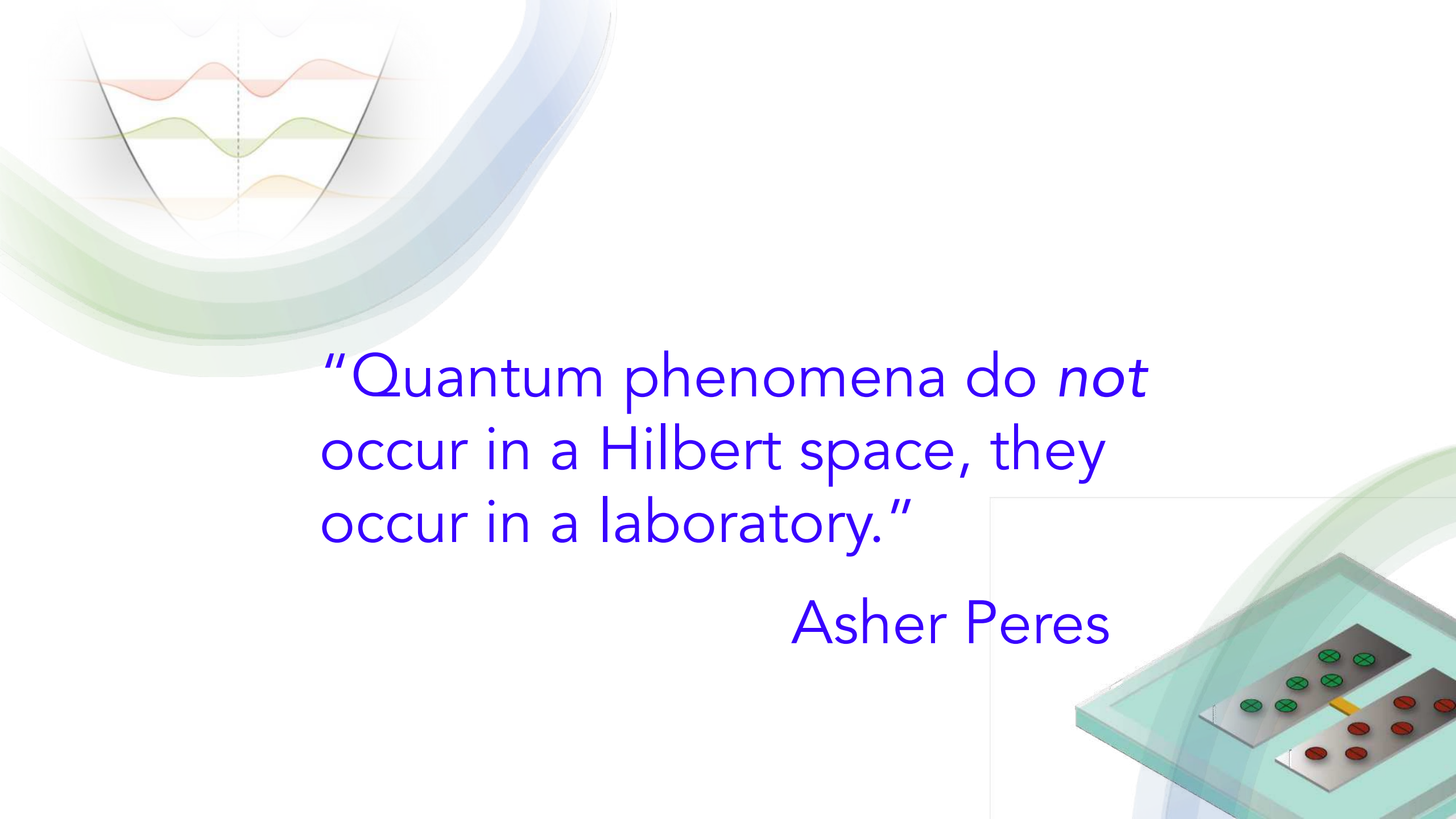
From Idea to Reality



THE BIG PICTURE
before calculations

"Quantum cloud"





"Quantum phenomena do *not* occur in a Hilbert space, they occur in a laboratory."

Asher Peres

Quantum in the cloud or lab

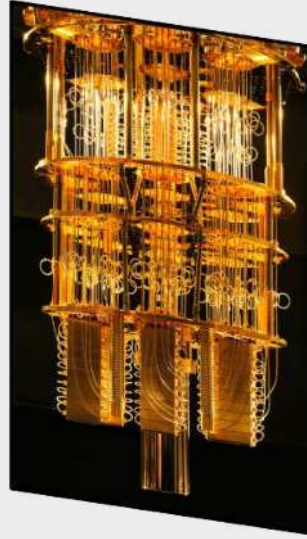
Superconducting qubits



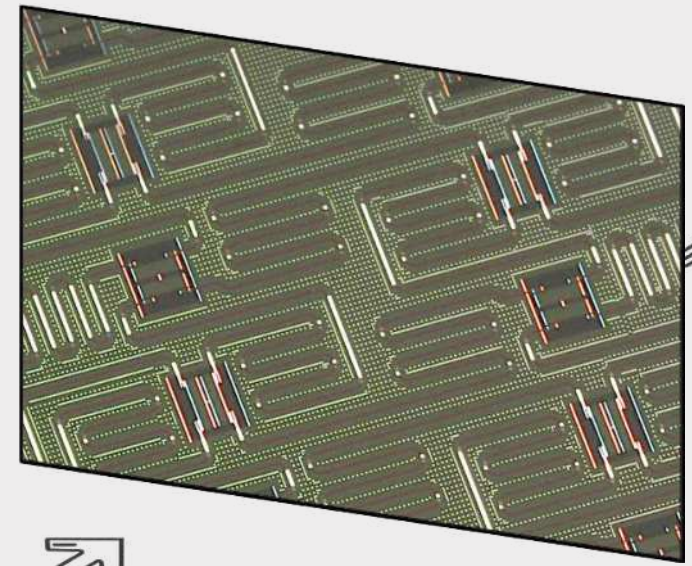
Quantum community



Lab / cloud facility

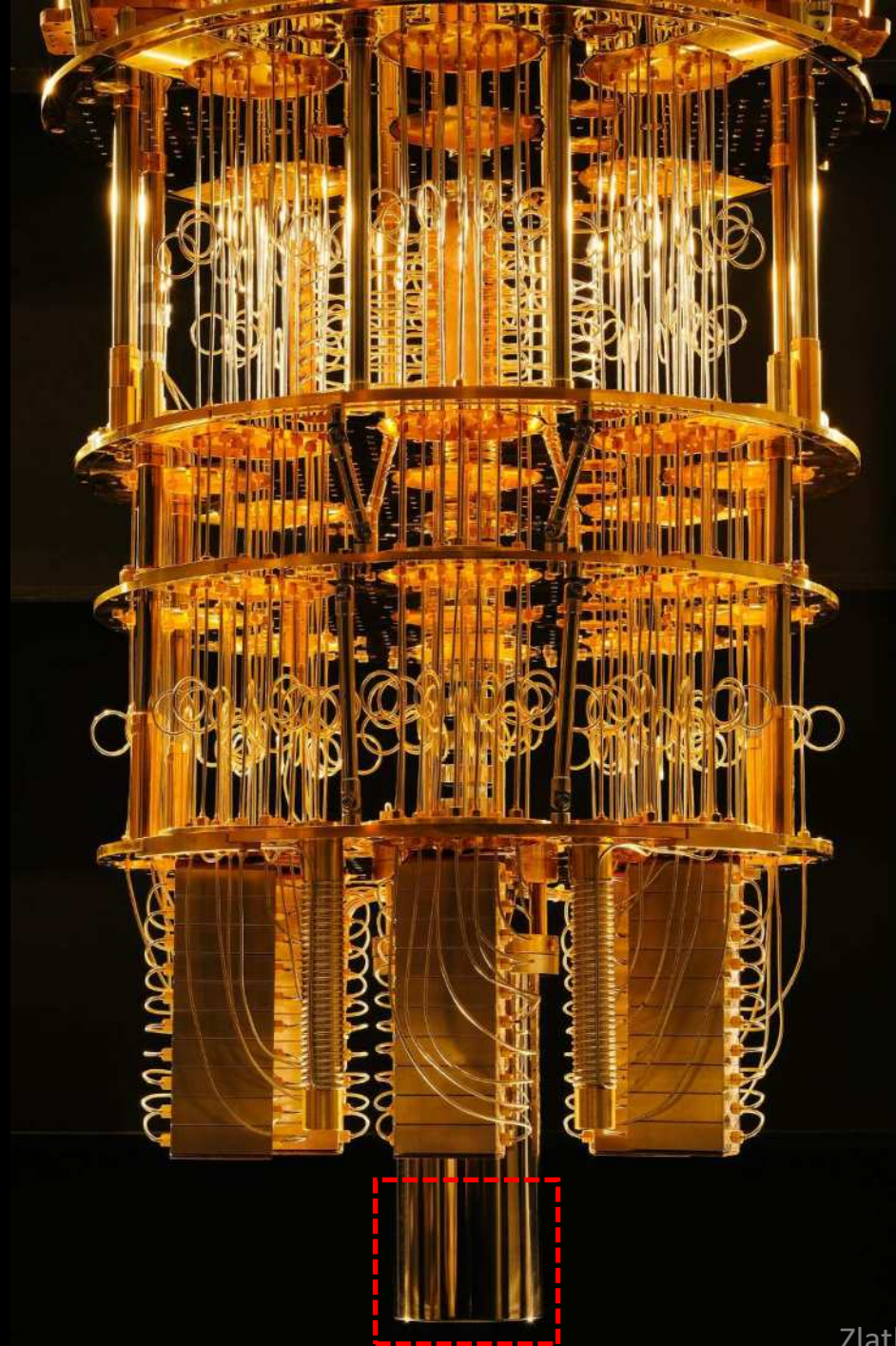


Cryogenic environment

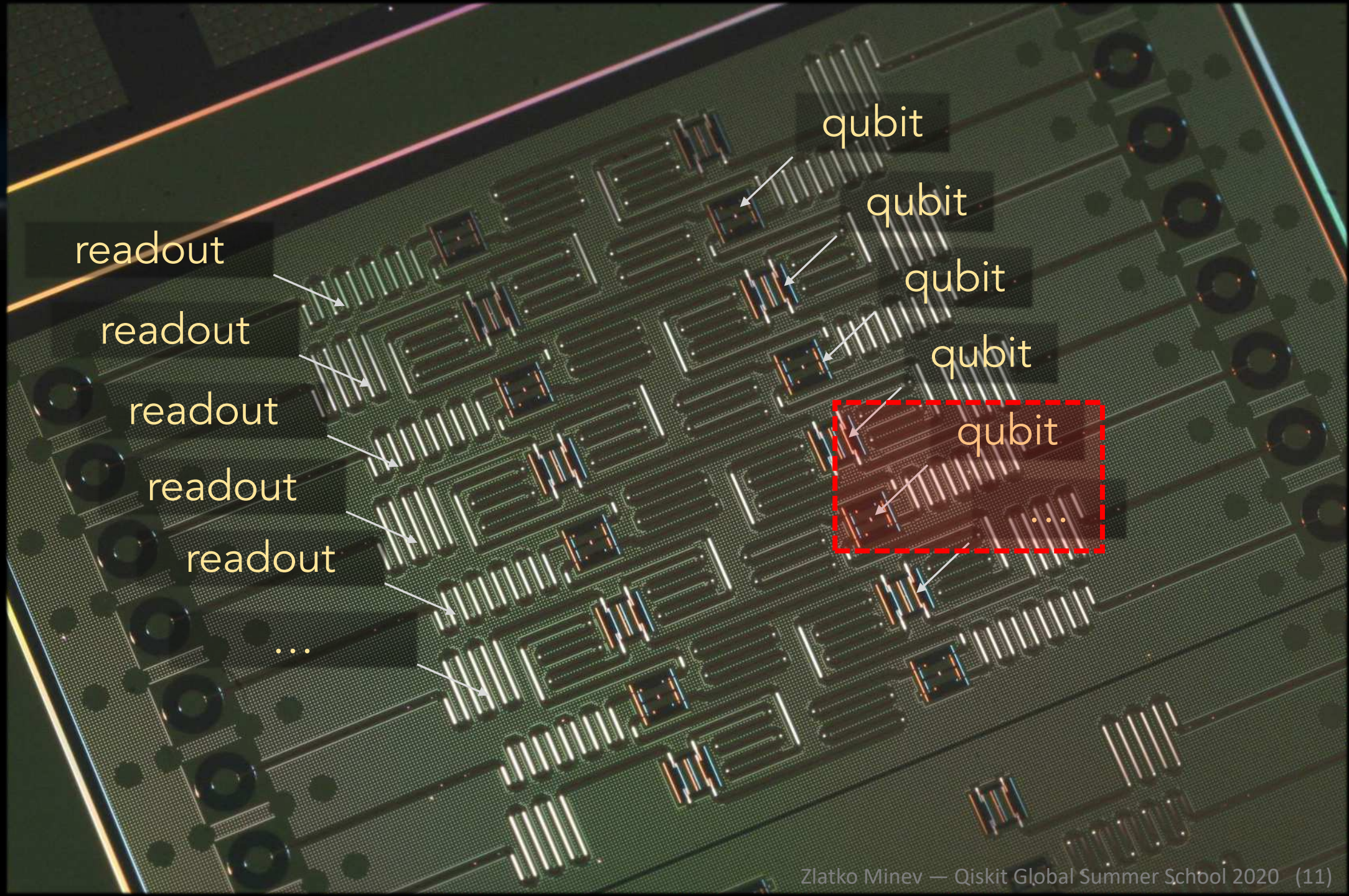
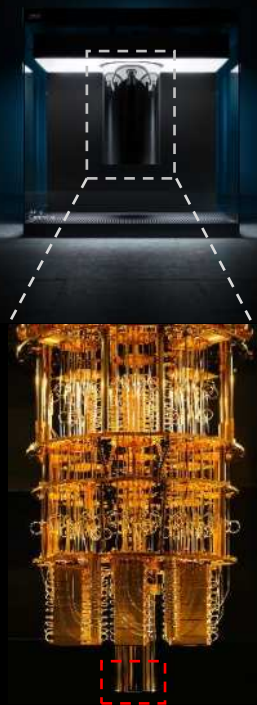


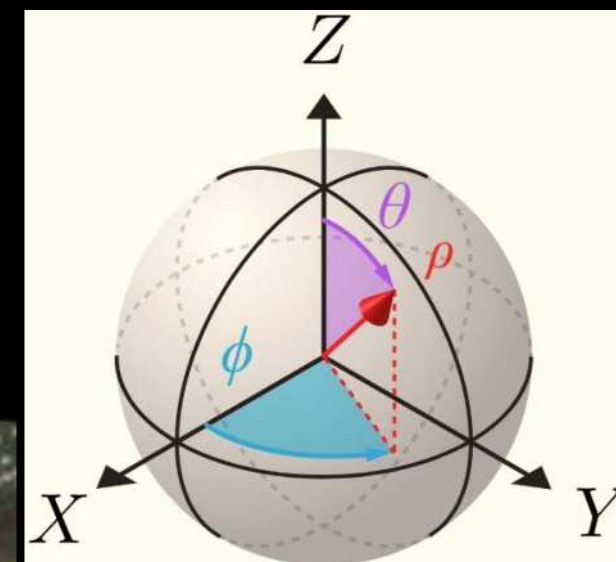
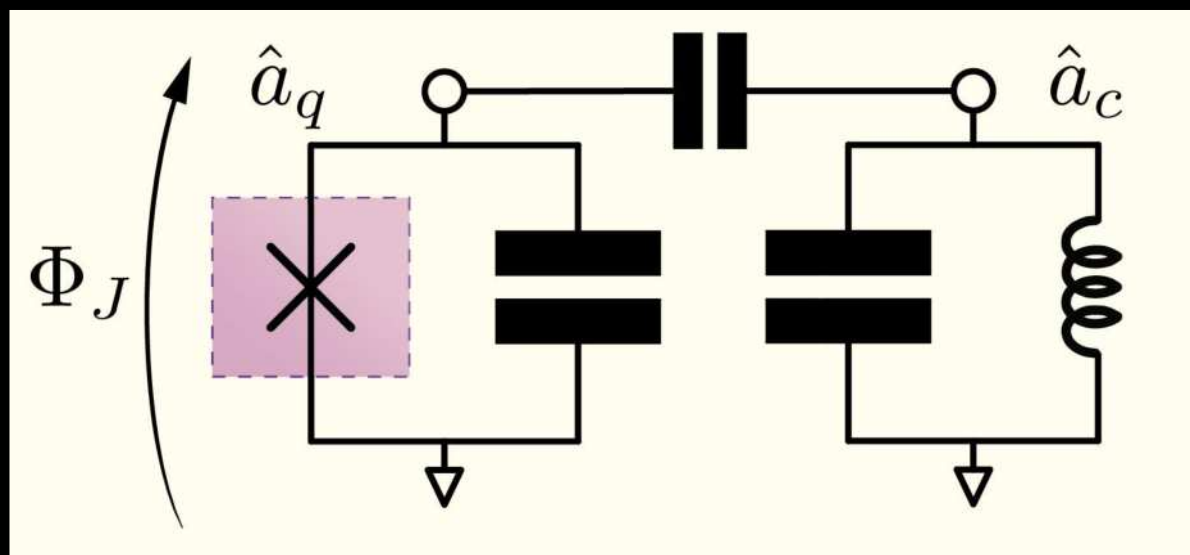
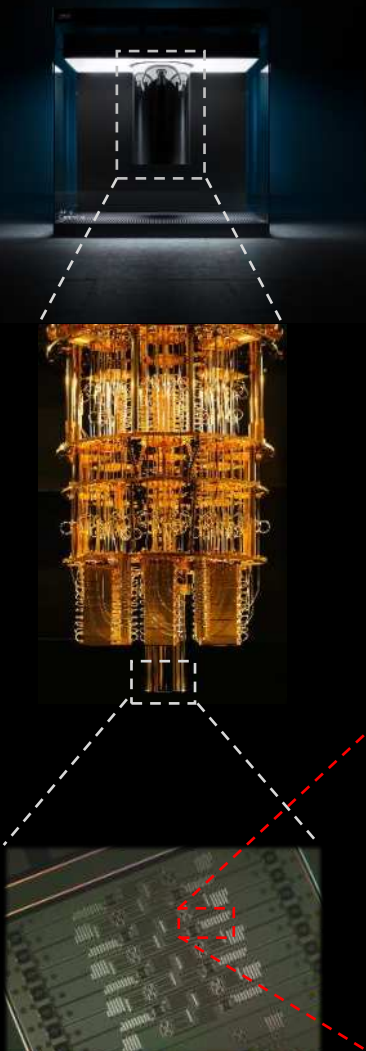
Quantum-device chip



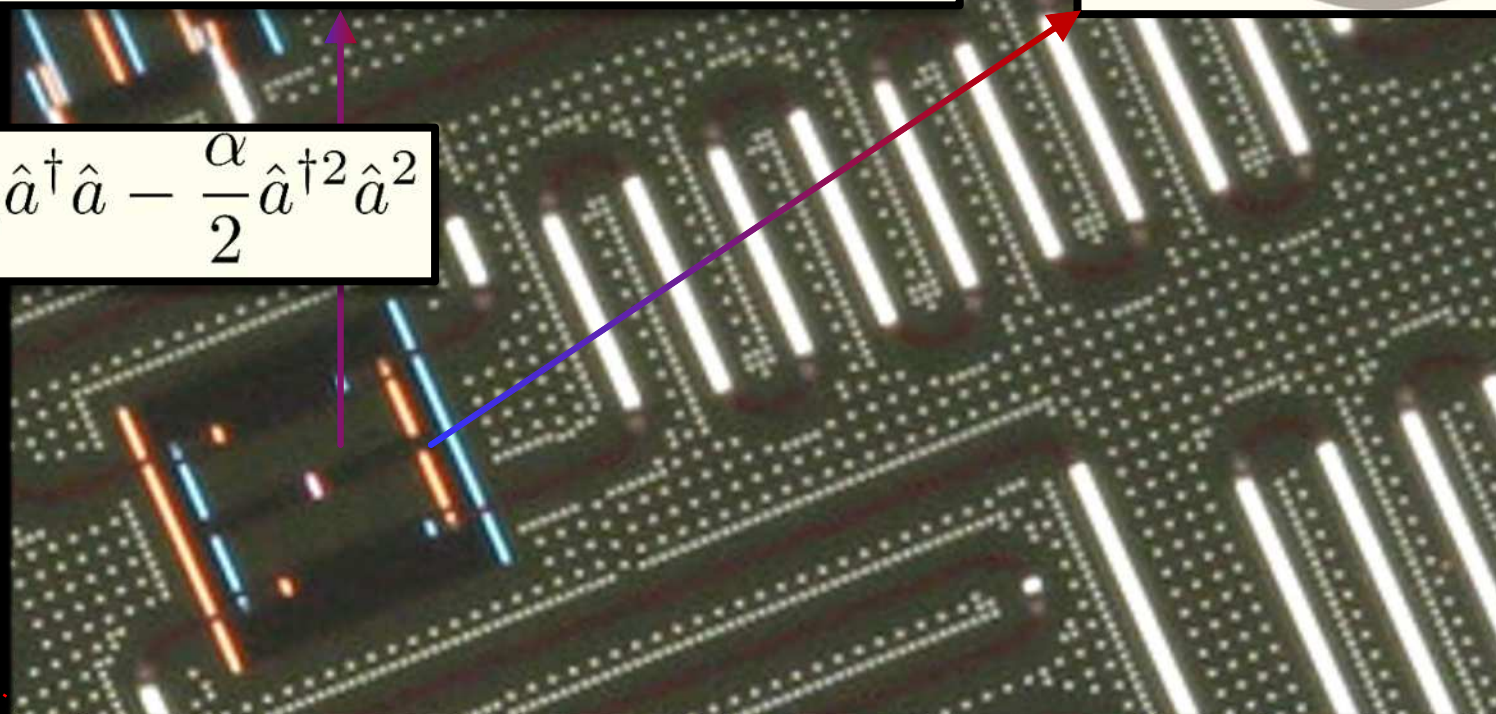


Operation at
15 mK ($-273.13\text{ }^{\circ}\text{C}$)



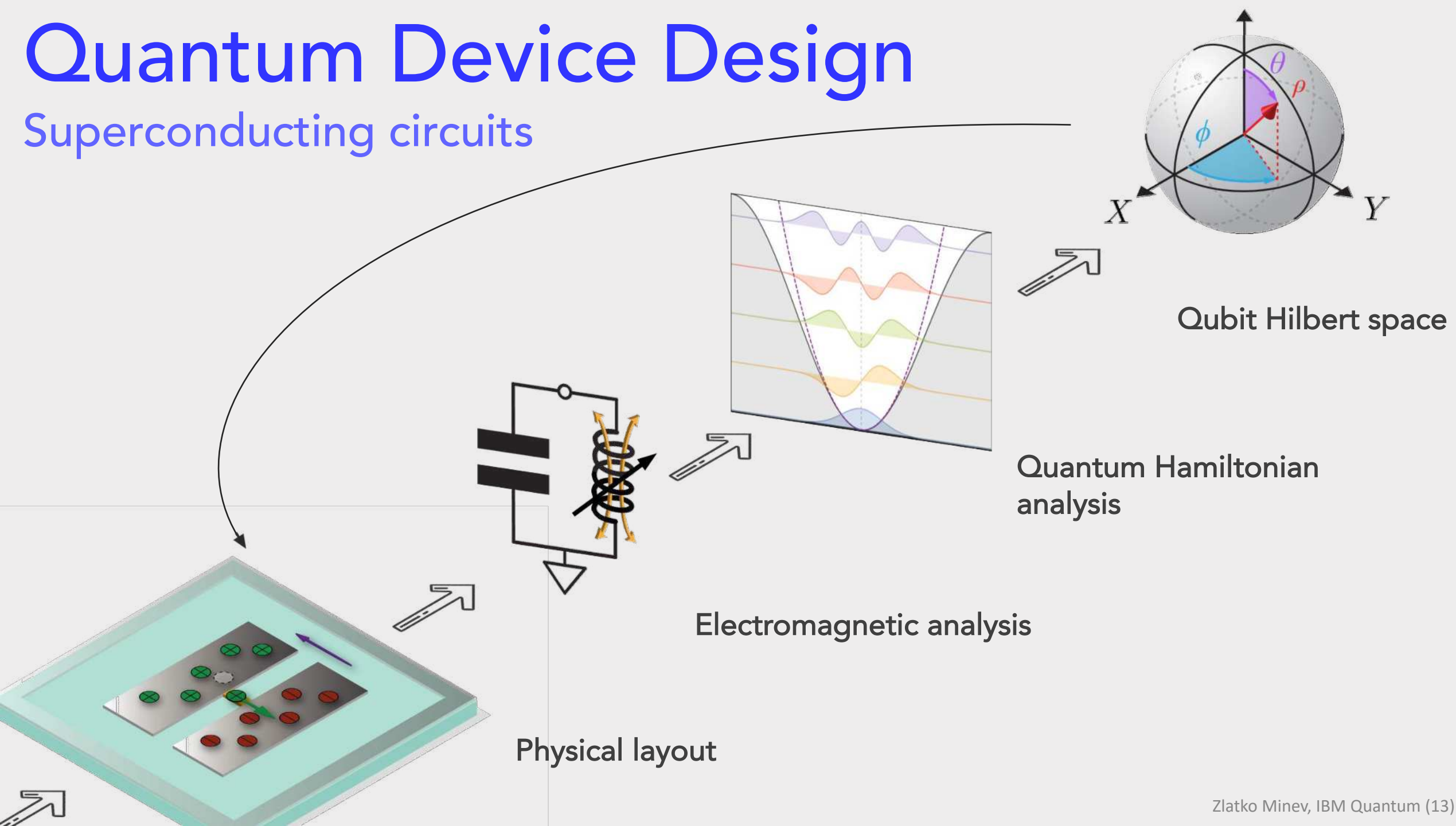


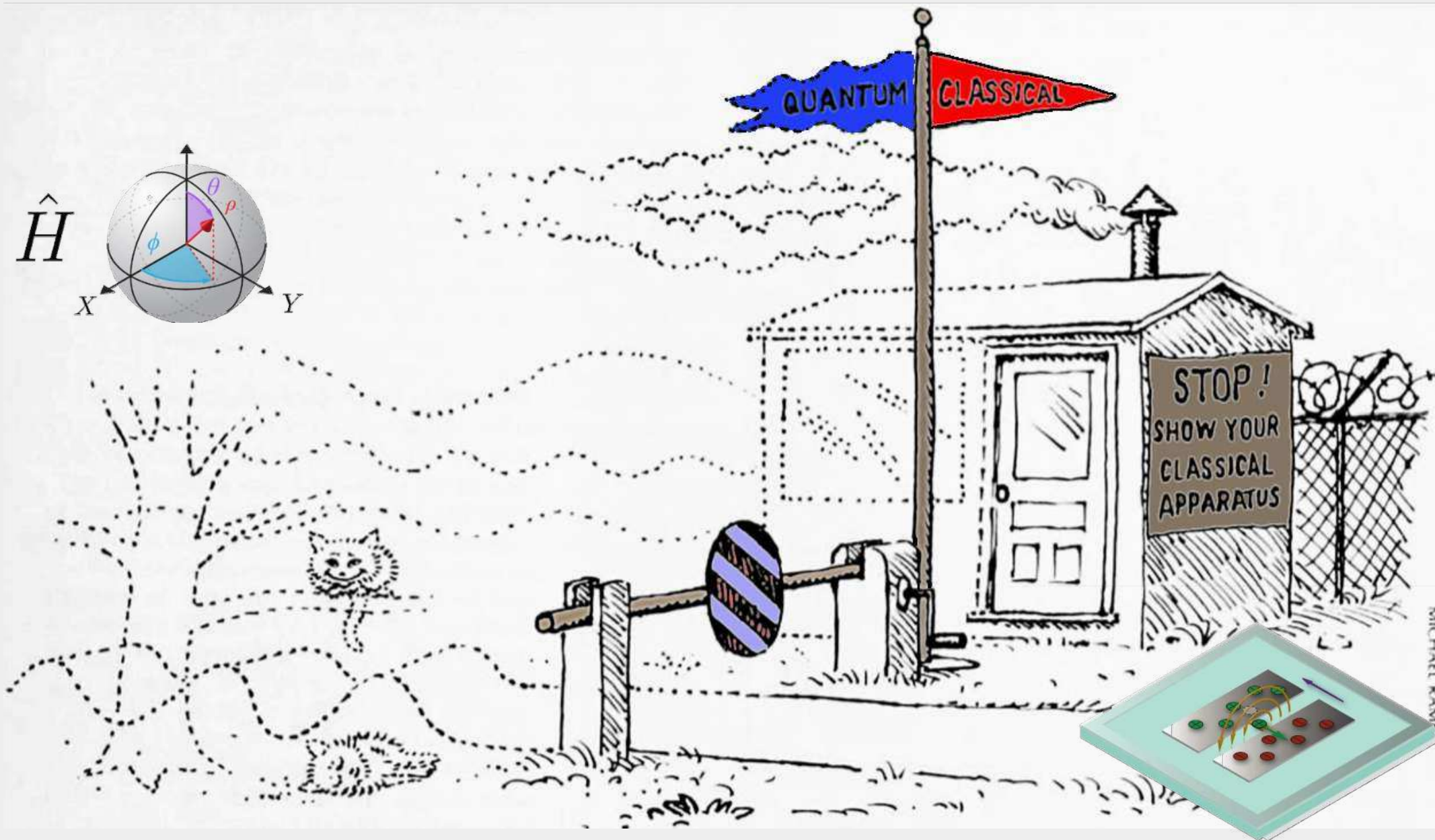
$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$



Quantum Device Design

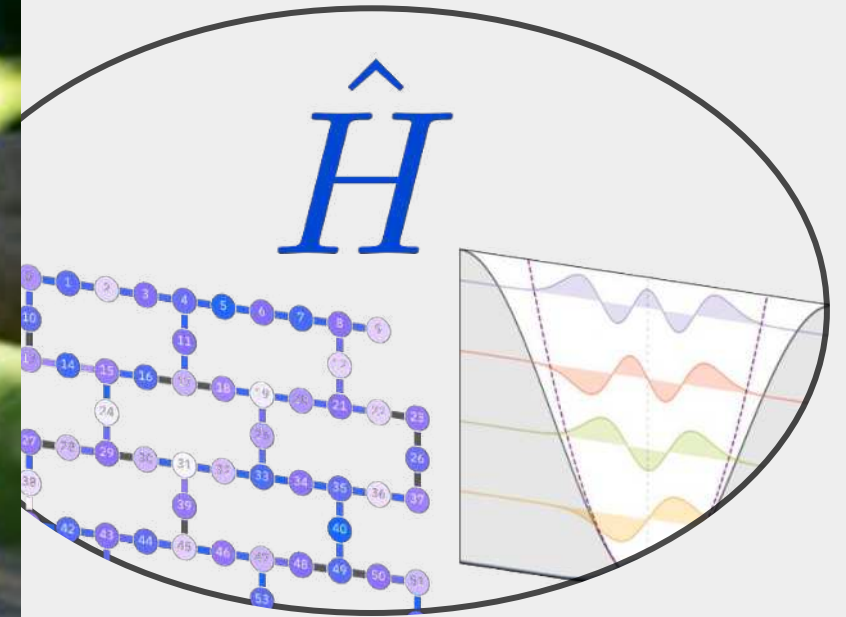
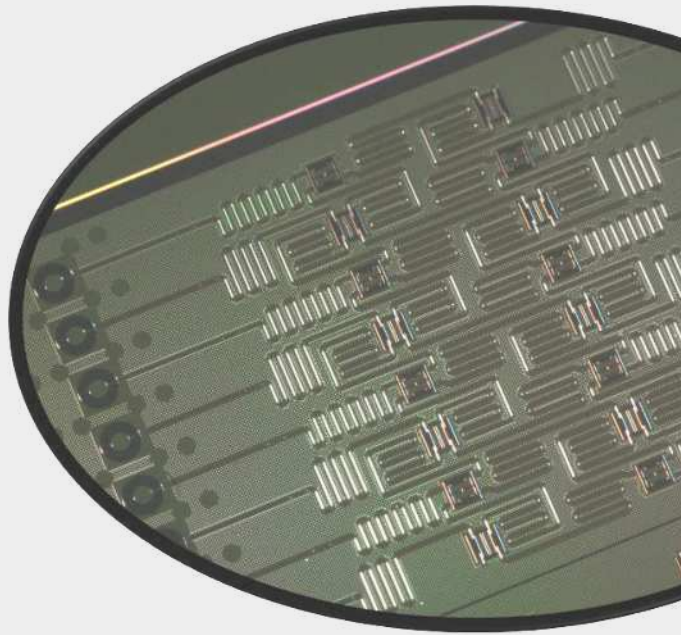
Superconducting circuits





Drawing: Zurek, Physics Today (1991)

Physical Devices $\leftrightarrow$ Quantum Hamiltonian



Quantum chip design flow

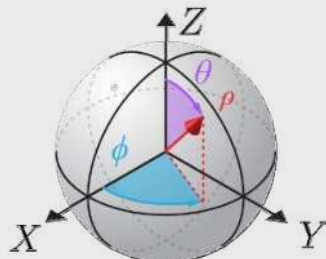
Complexity, Information, Accuracy

Risk, Cost, Time, Resources

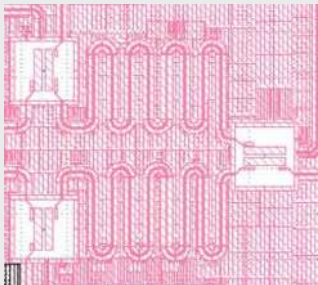
Concept



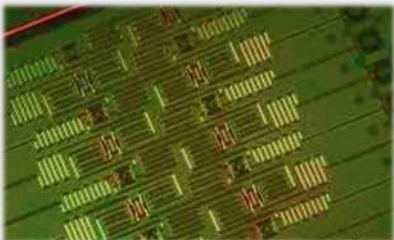
Hamiltonian



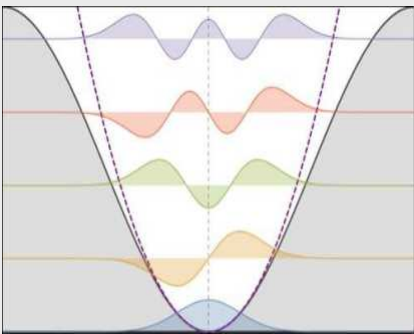
Layout



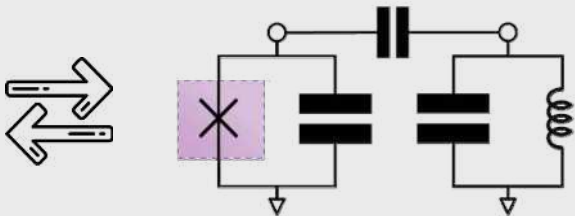
Fabrication



Quantum Analysis



Electromagnetic Analysis



Quantum chip design flow

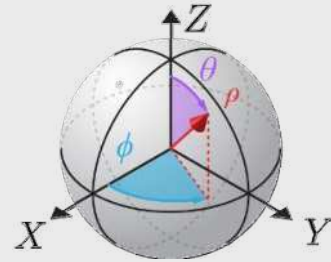
Complexity, Information, Accuracy

Risk, Cost, Time, Resources

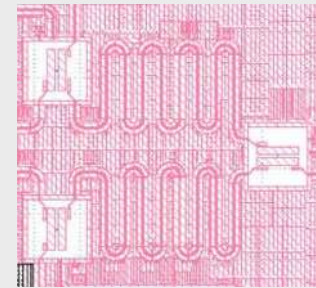
Concept



Hamiltonian



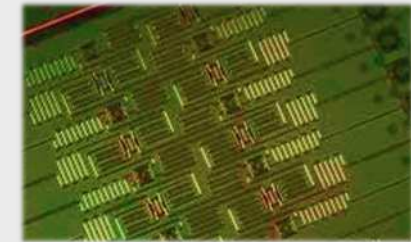
Layout



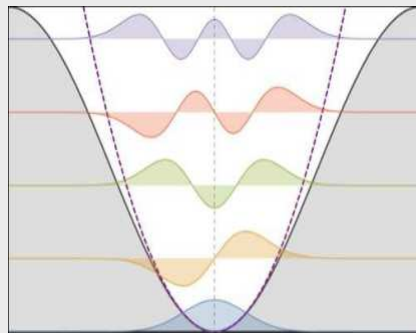
DRC
LVS



Fabrication

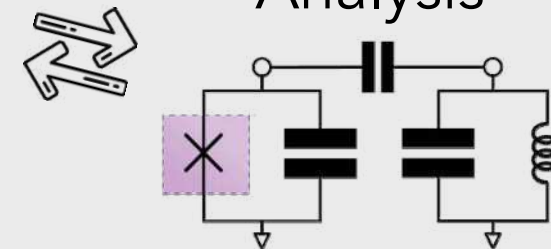


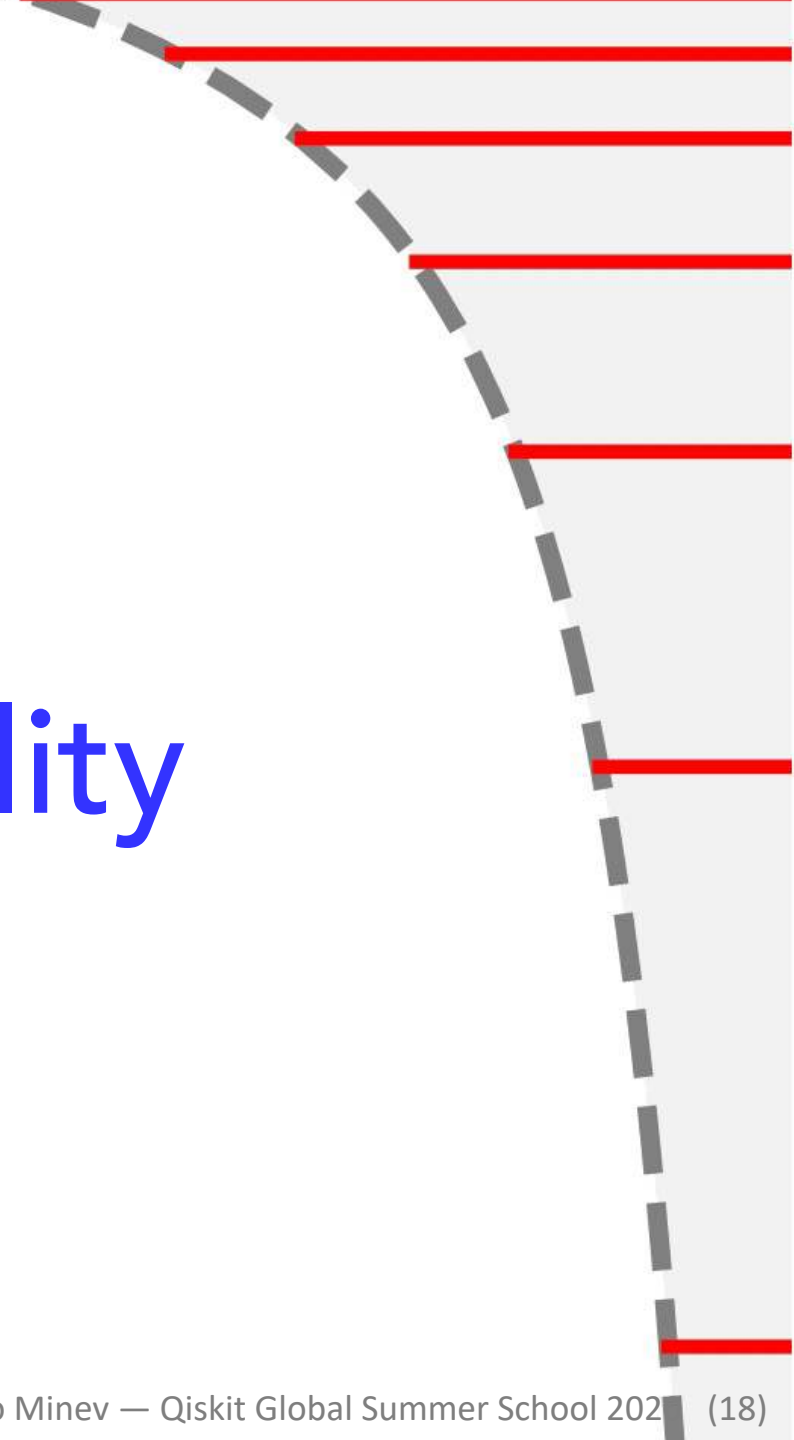
Quantum Analysis



Project
Metal

Electromagnetic
Analysis



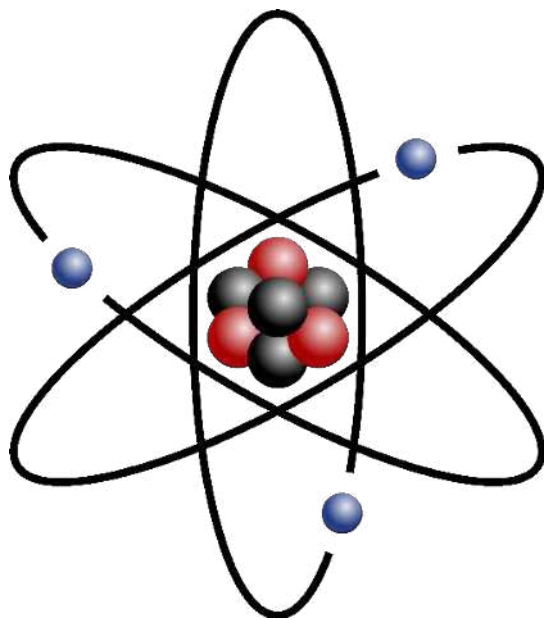


Qubit

From Idea to Reality

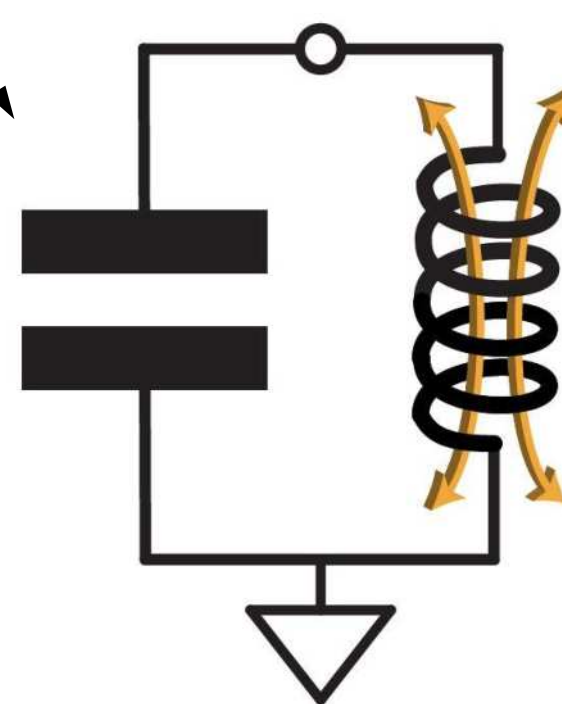
Concepts

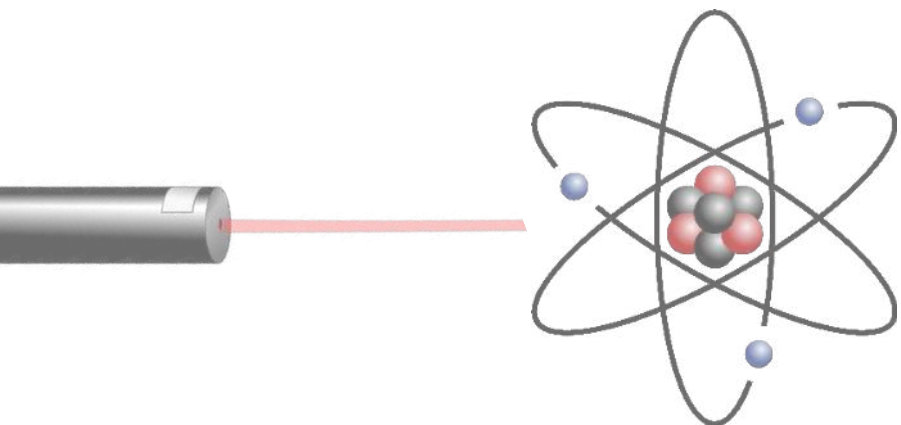
Artificial atoms



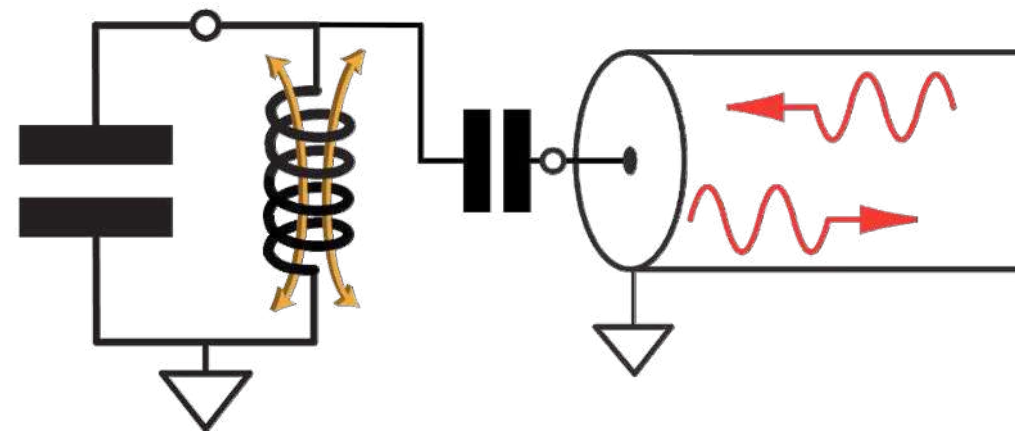
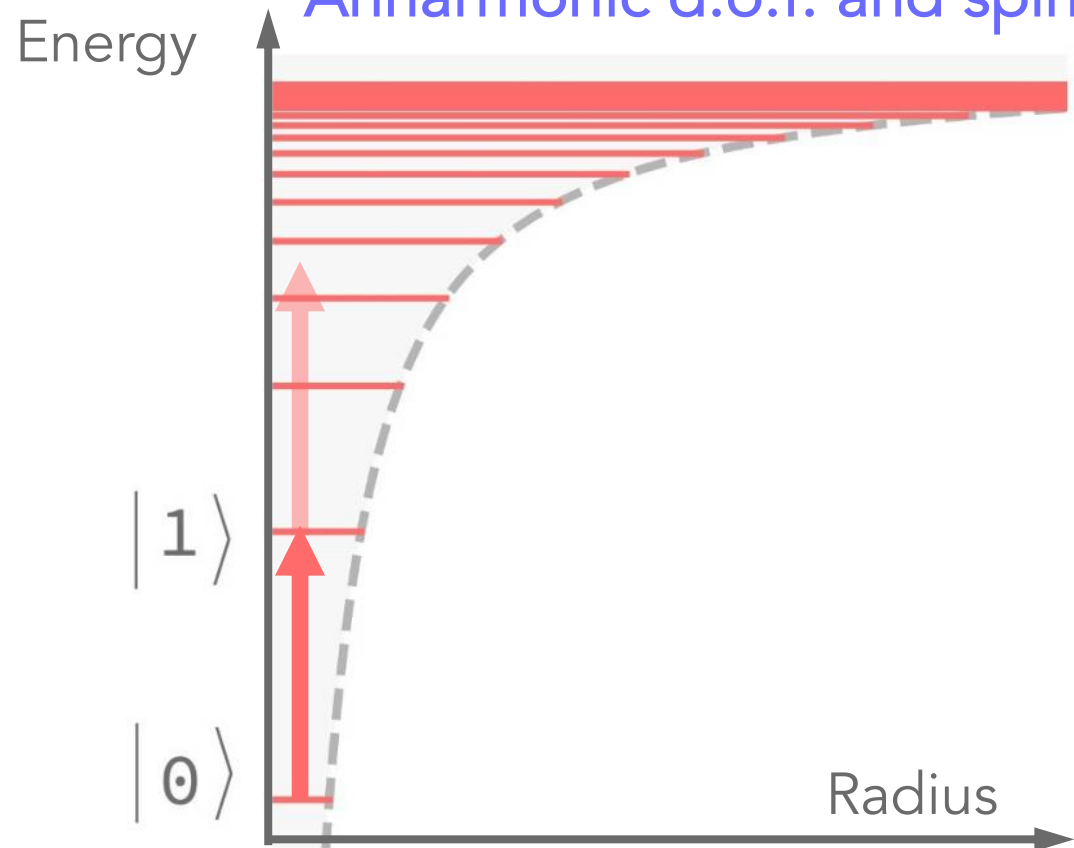
capacitor

inductor

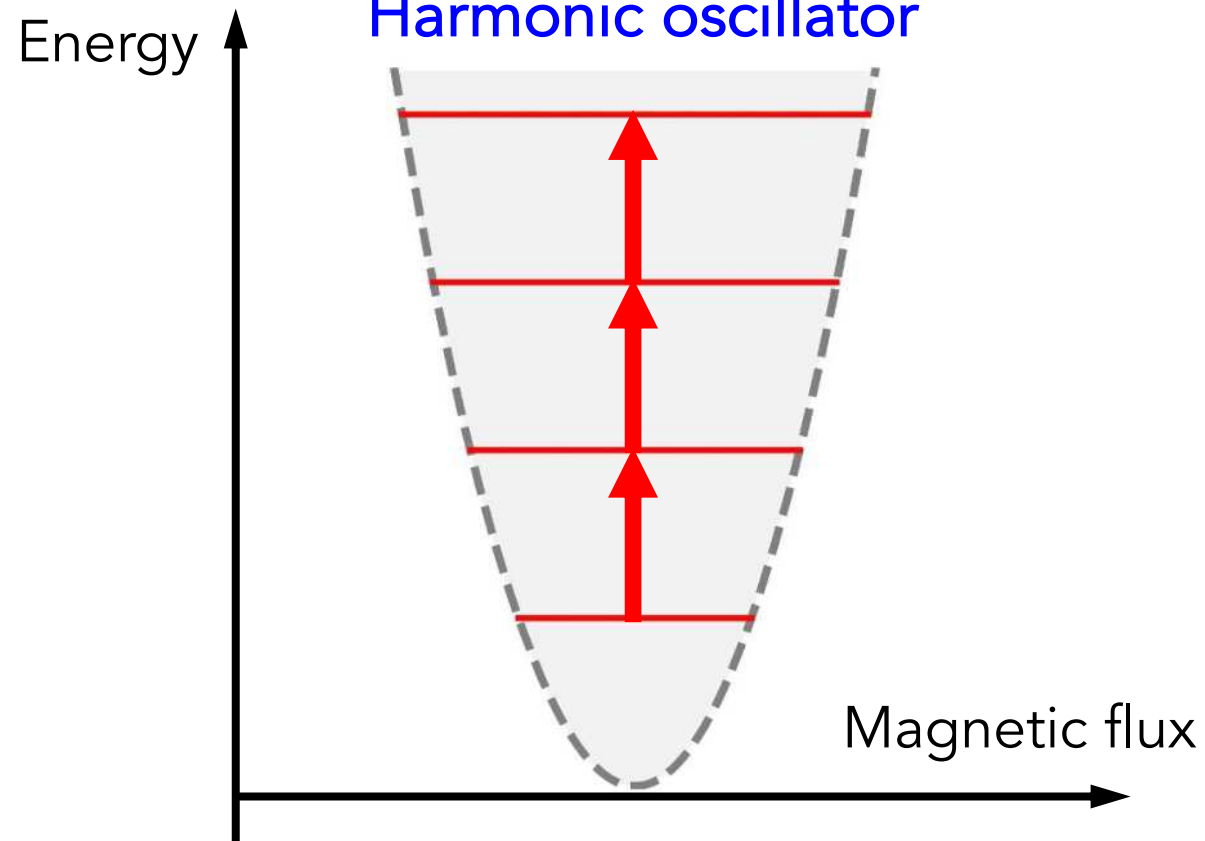


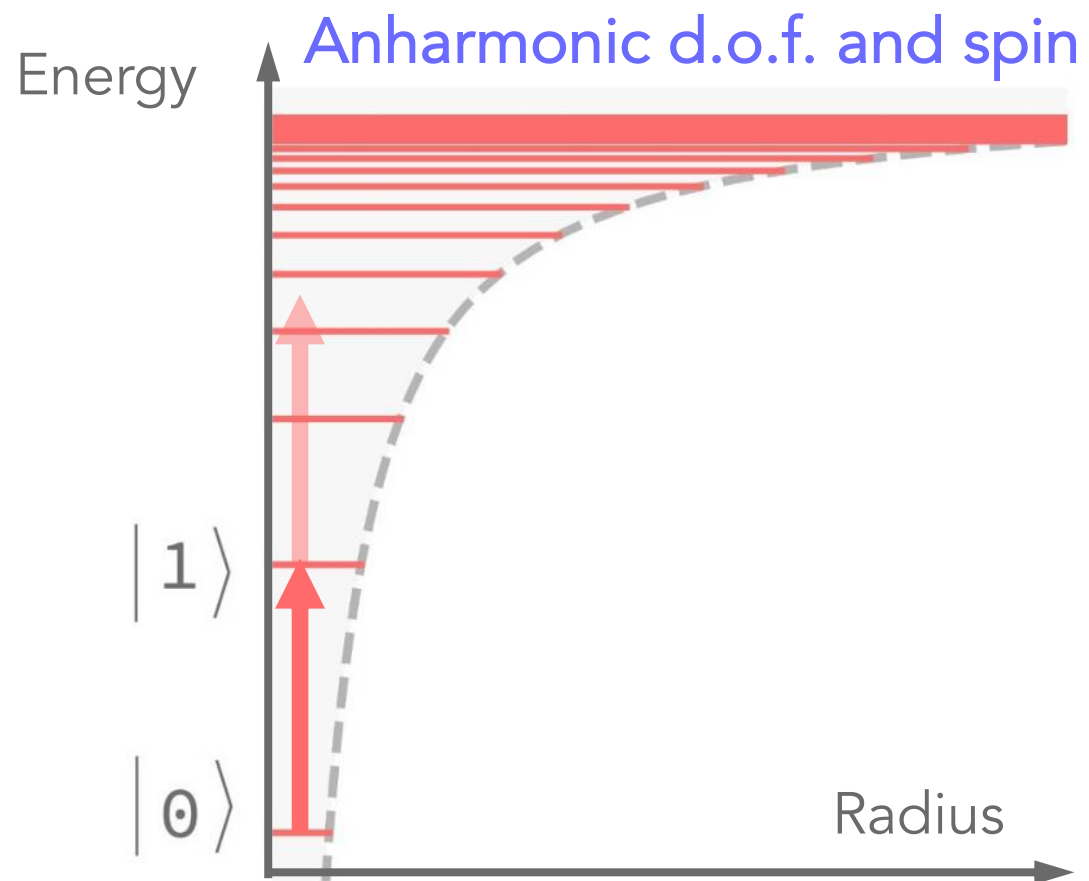
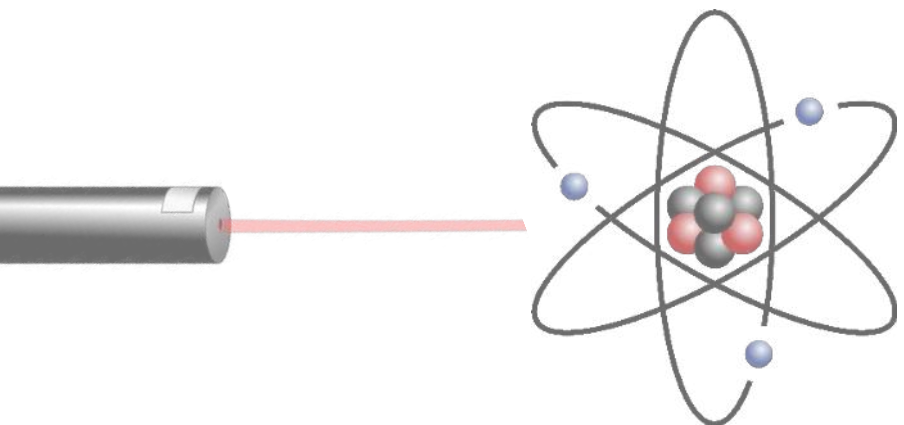


Anharmonic d.o.f. and spin

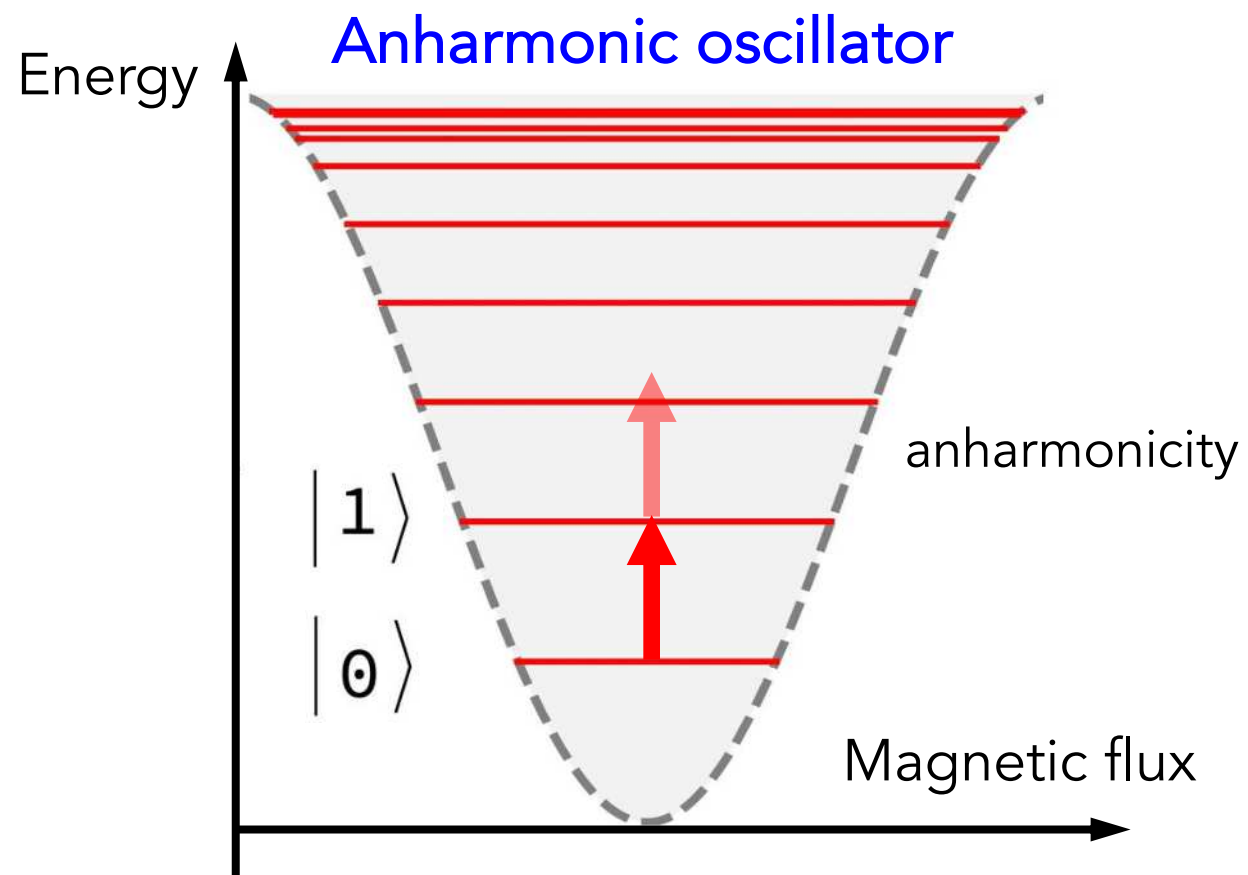
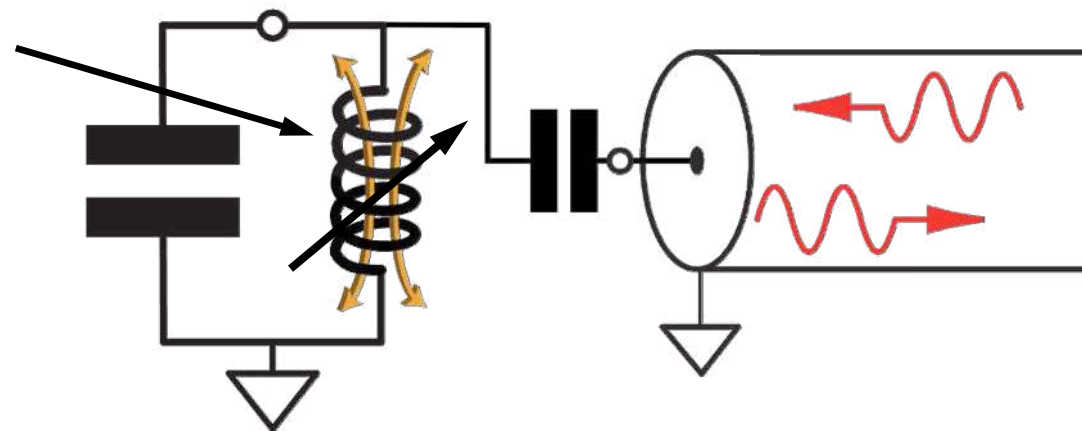


Harmonic oscillator



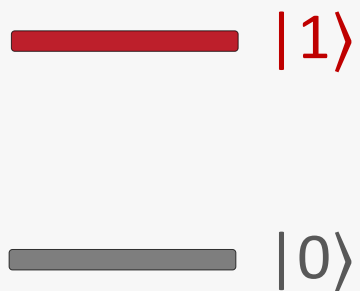


non-linear
inductor

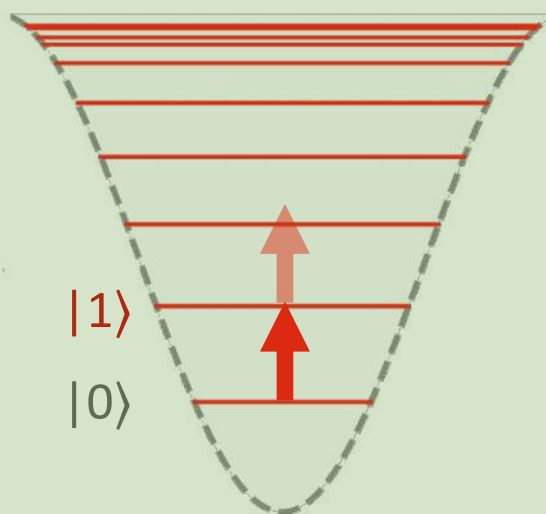


Big-picture connections

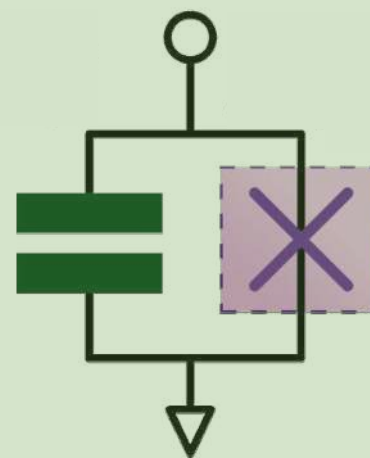
Idealization of
qubit



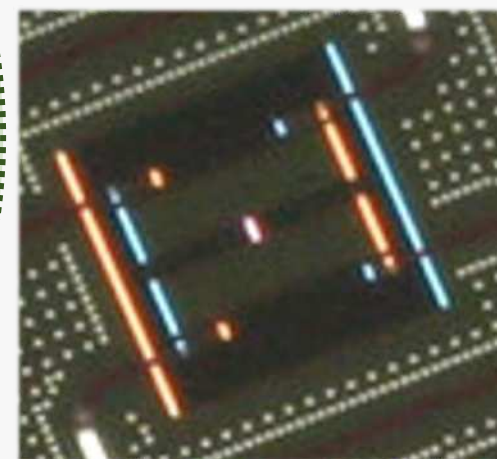
Anharmonic
oscillator



Physical circuit
model



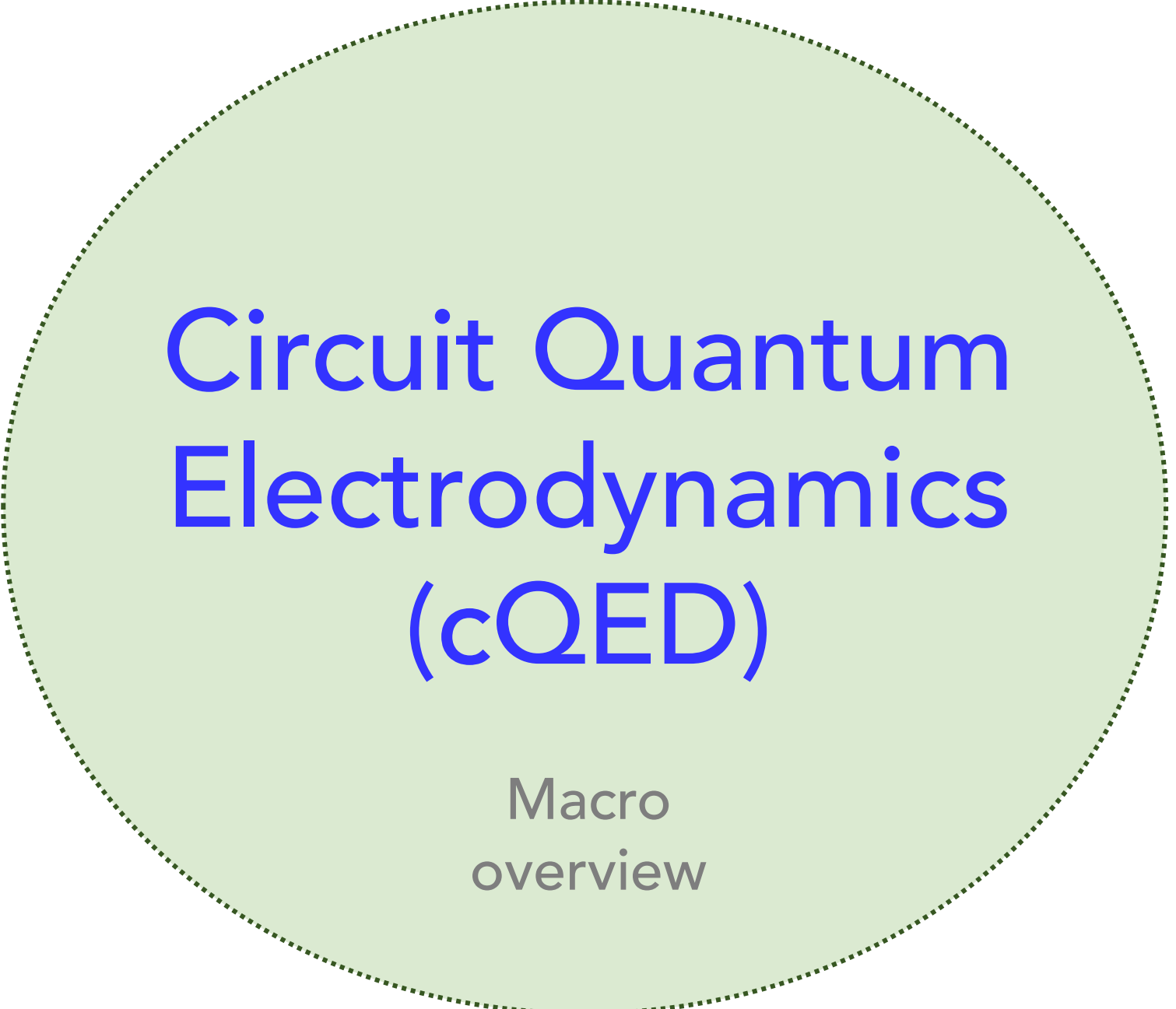
Physical layout



Idealization



Physical reality



Circuit Quantum Electrodynamics (cQED)

Macro
overview

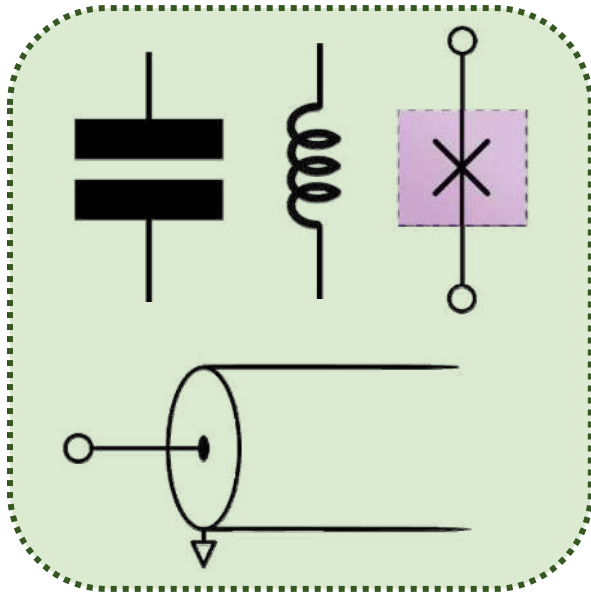
There are two kinds of physicists:

Those who believe all of physics is *spins*.

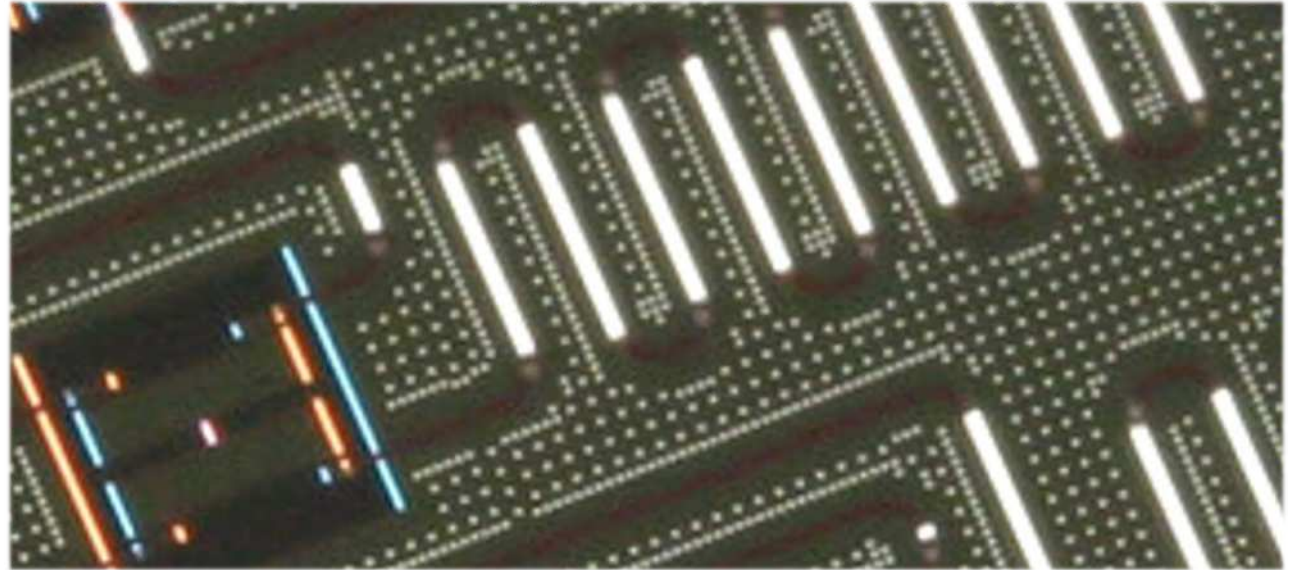
Those who believe all of physics is *oscillators*.

cQED Ingredients

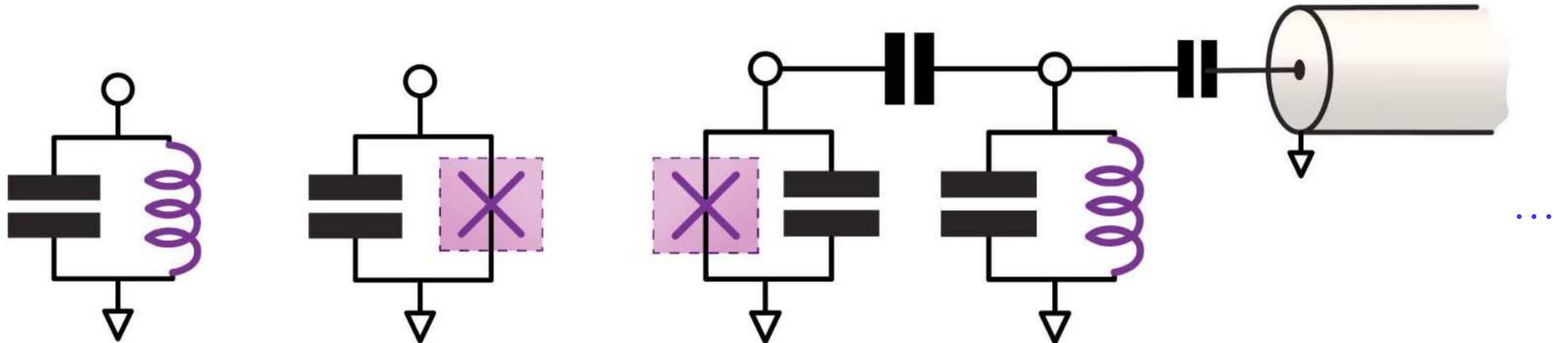
Circuit elements



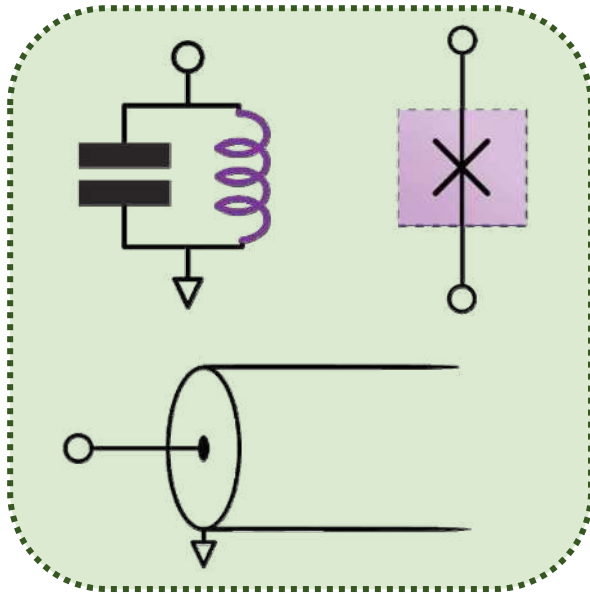
Microwave oscillators



Combine



cQED Ingredients



Small dissipation

Isolation from environment

Low temperature

Nonlinearity

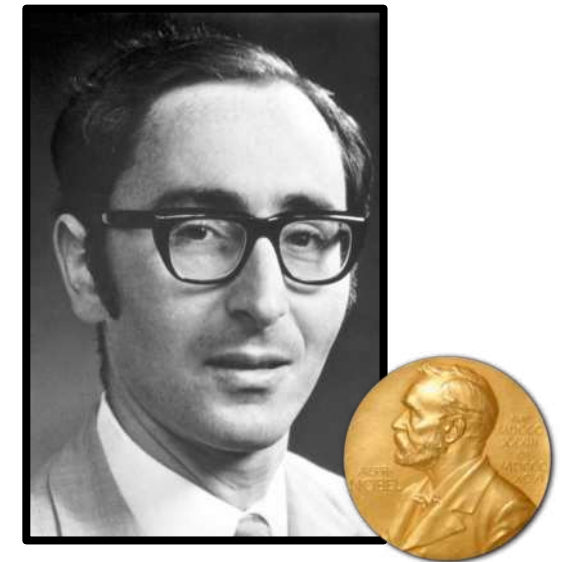
Large vacuum fluctuations

Superconductivity

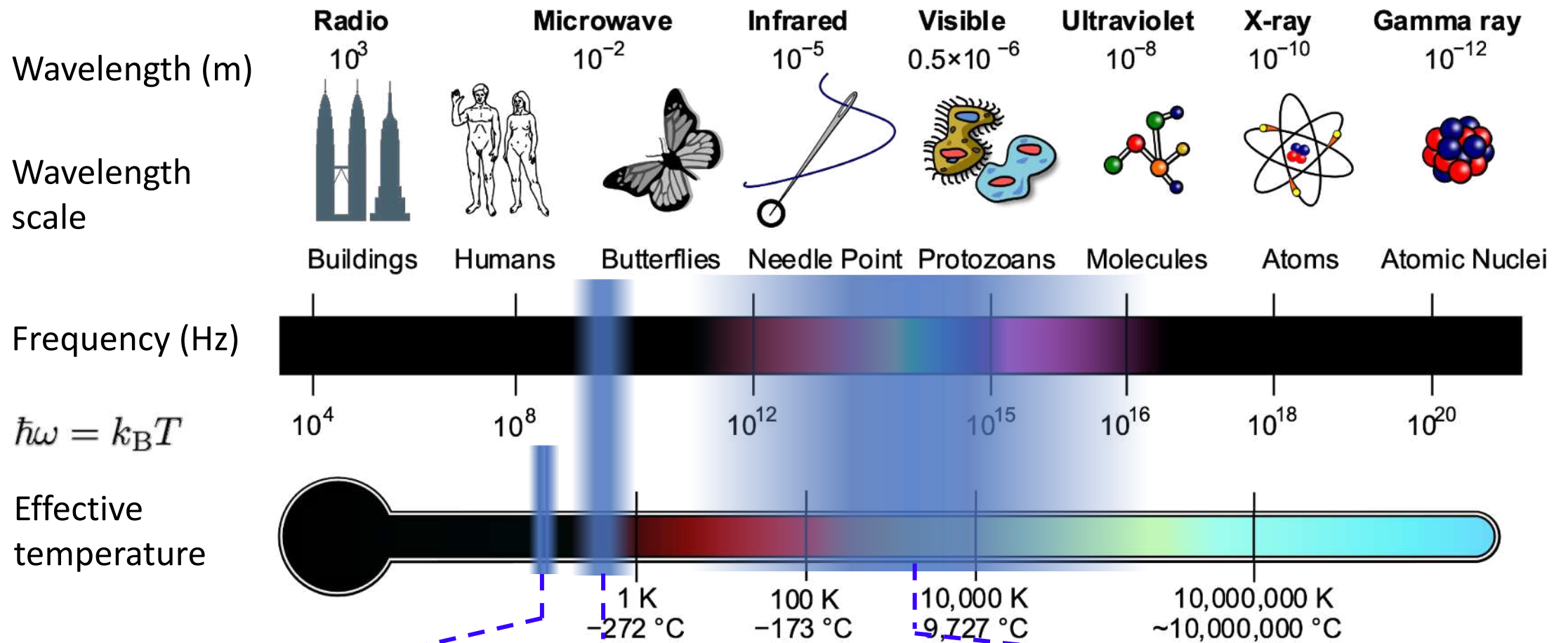
Nominally zero intrinsic dissipation and heat

Nominal temperature far below energy level splitting

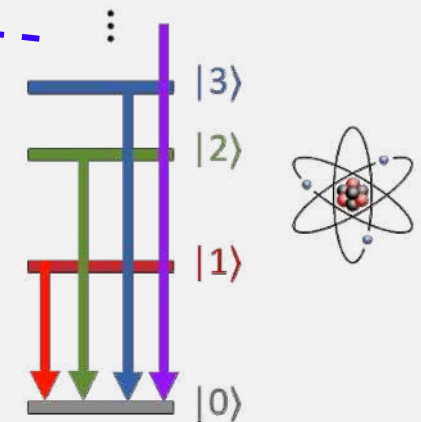
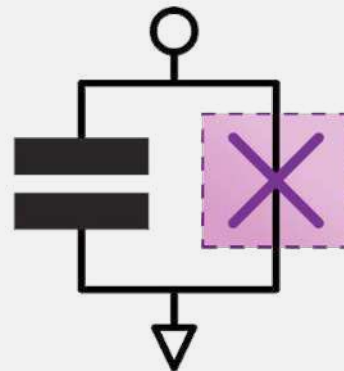
Non-linear, robust Josephson tunnel junction effect



* Low energy dynamics. Microwave image: vectorpocket; Nobel photograph: Jonathunder; Josephson photograph: Nobel Foundation archive.



Order of magnitude



Spectrum image: Inductiveload, NASA

A few introductory reviews

And many more... check online or ask us for specific topic

Qiskit Textbook (2020; more chapters coming)

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020)
Circuit Quantum Electrodynamics (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020)
Superconducting Qubits: Current State of Play
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019)
A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019)
Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017)
Quantum information processing with superconducting circuits. *Reports on Progress in Physics*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017)
Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

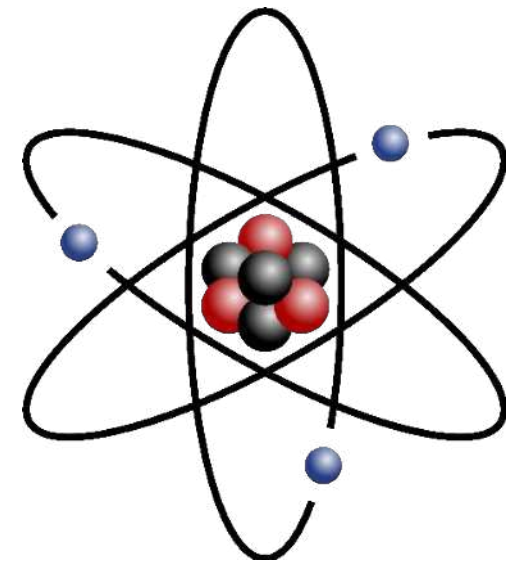
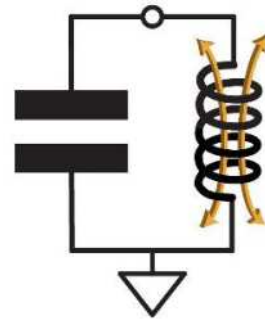
Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010)
Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

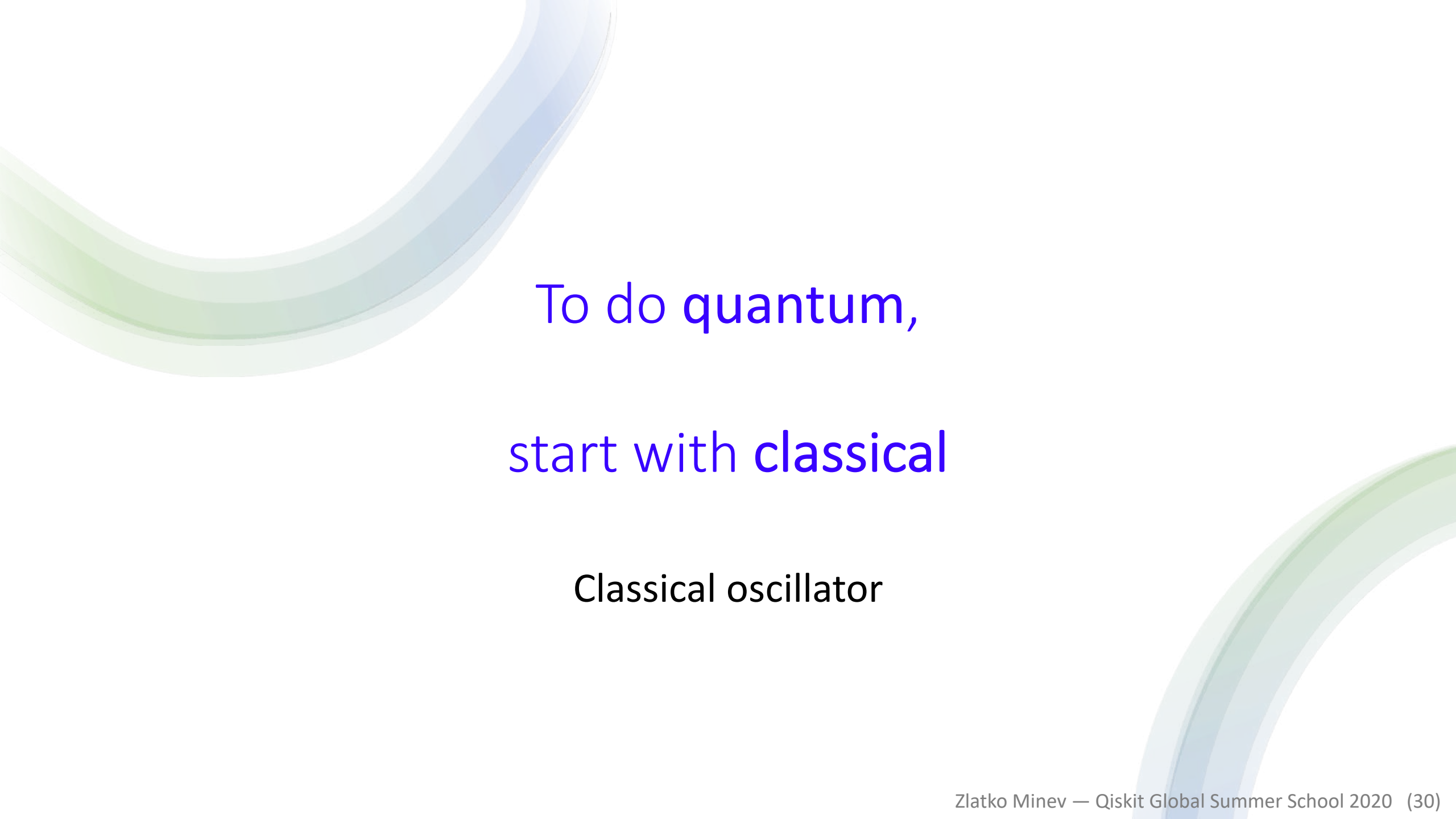
Clarke, J., & Wilhelm, F. K. (2008)
Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997)
Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

...

Circuit Quantum Electrodynamics

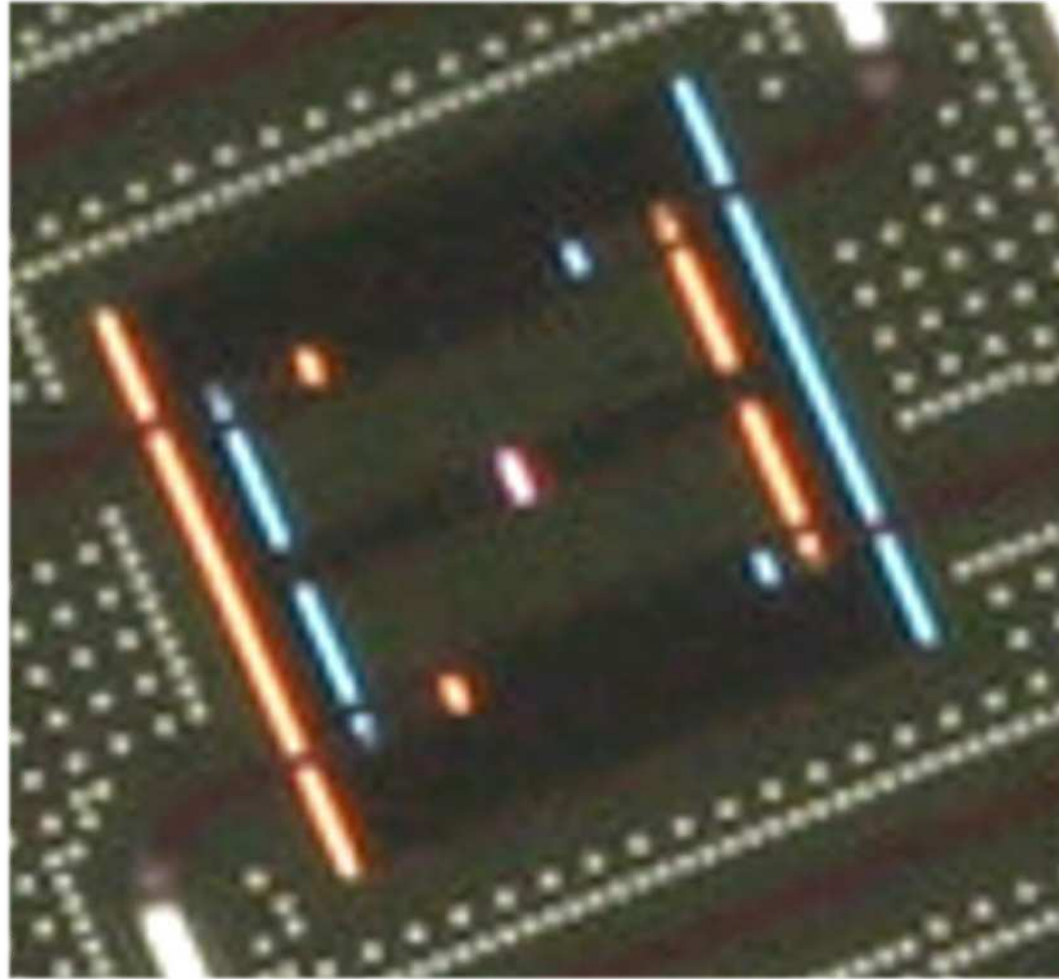




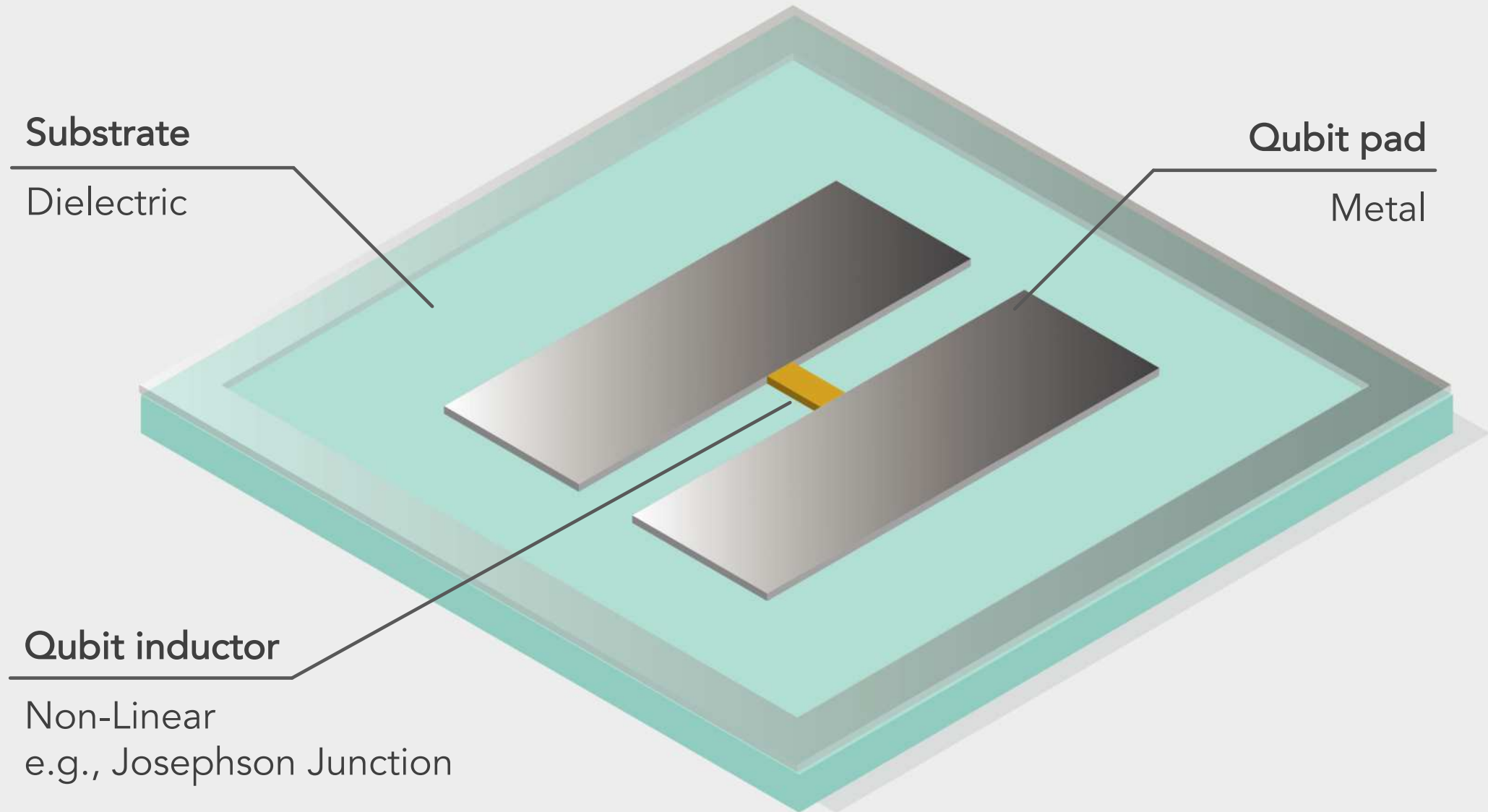
To do quantum,
start with classical

Classical oscillator

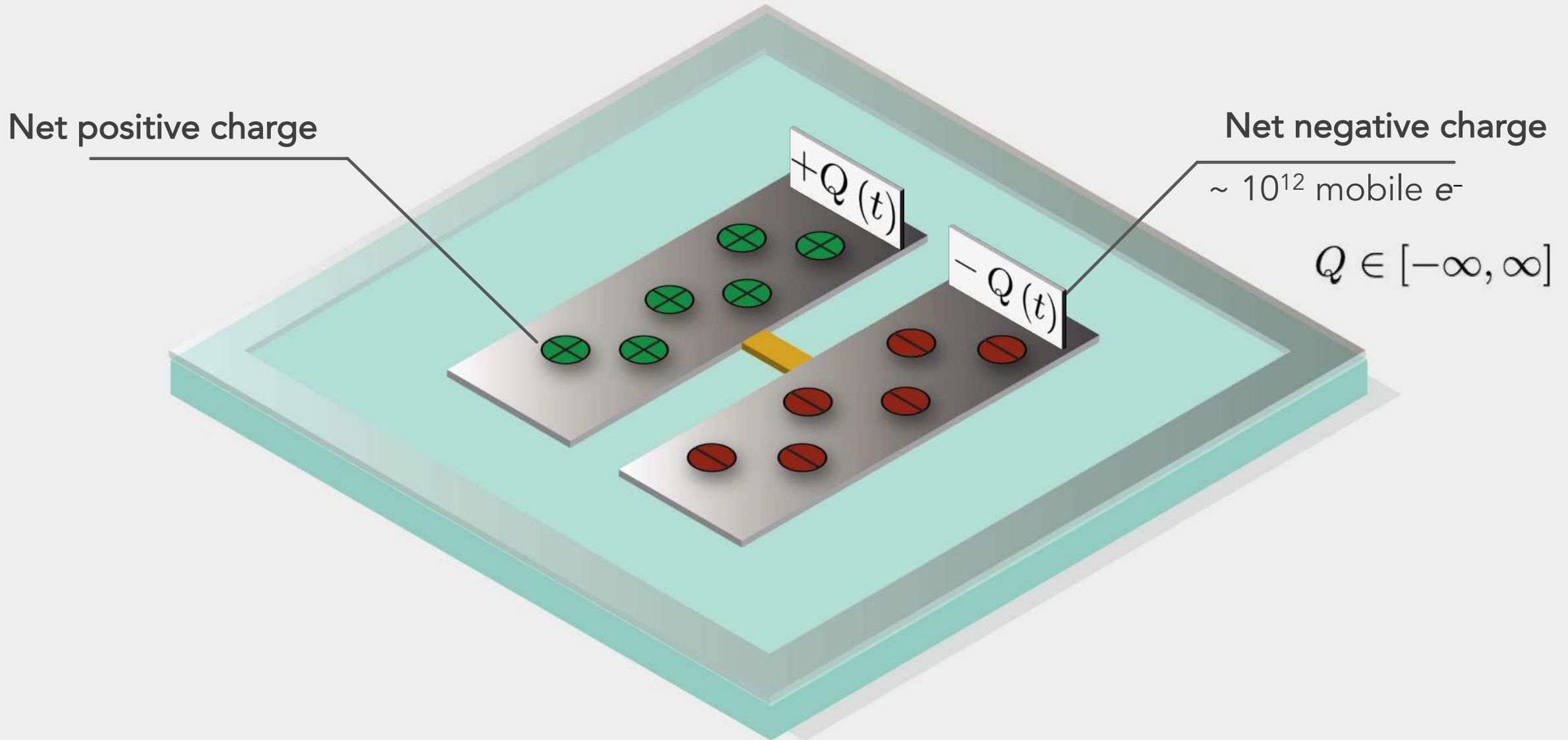
Transmon qubit



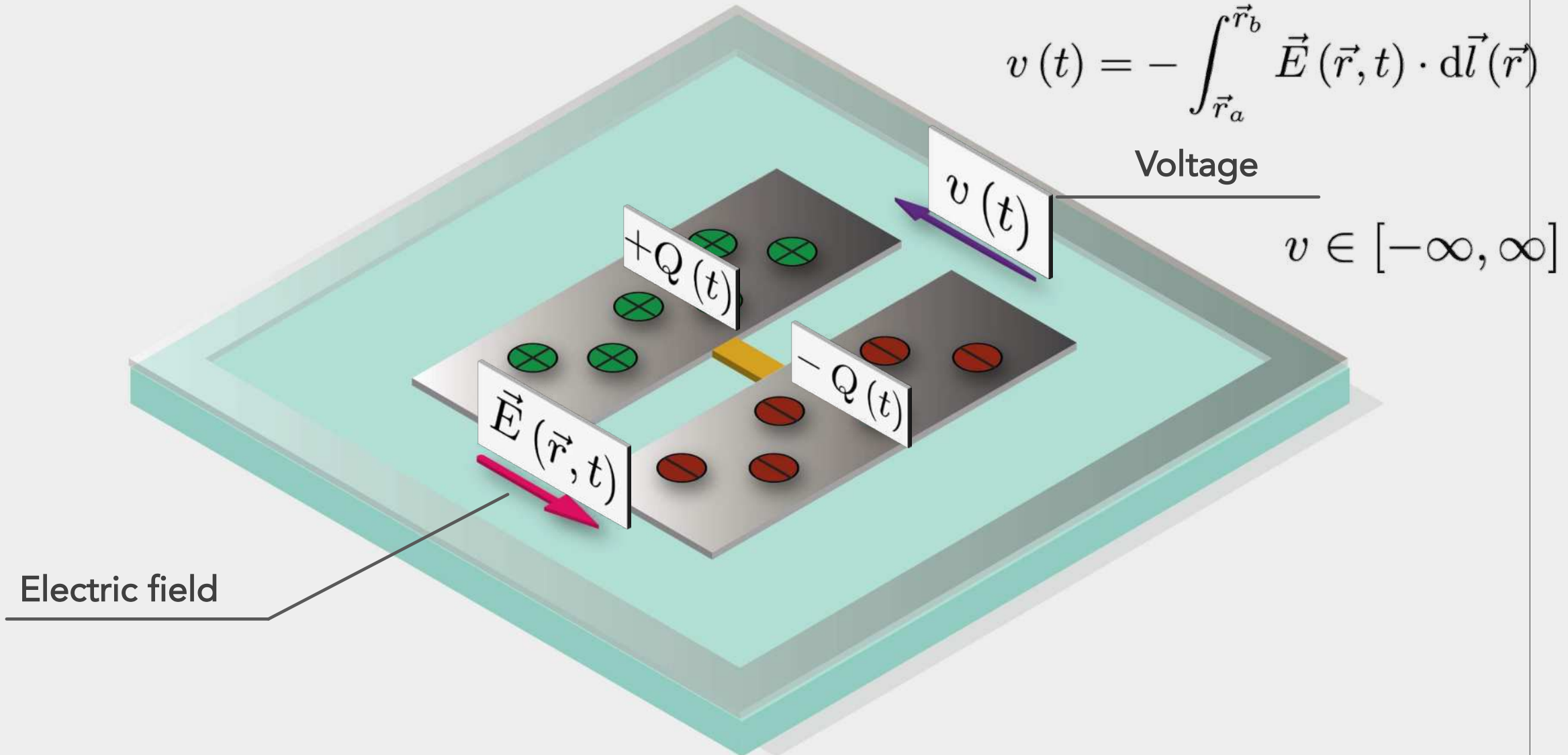
Transmon qubit



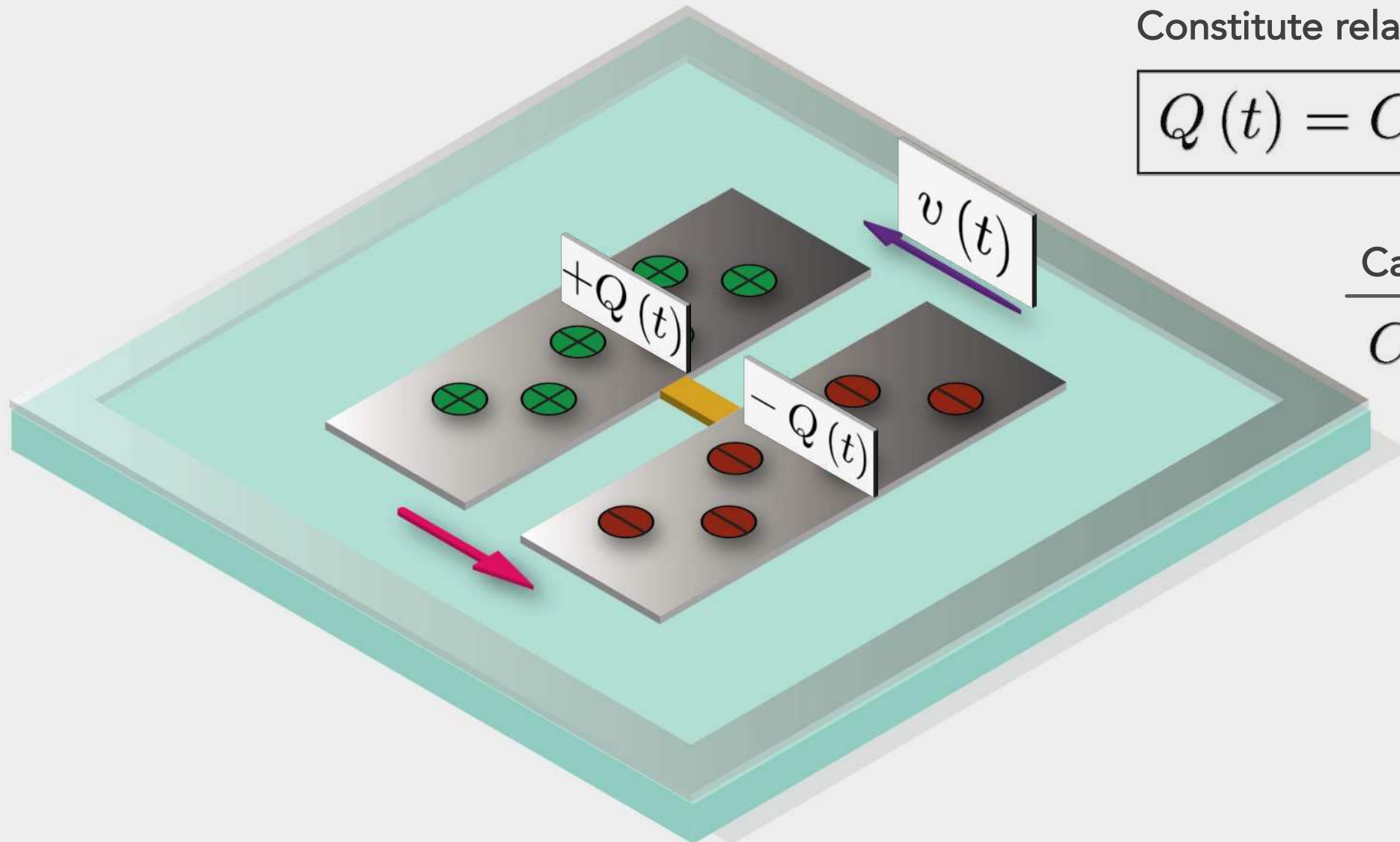
Transmon qubit: charge



Electric field and voltage



Charge and capacitance

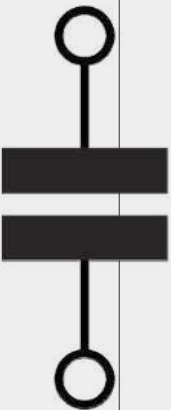


Constitute relationship

$$Q(t) = C v(t)$$

Capacitance

$$C \in (0, \infty)$$



For a good discussion, see "The Feynman Lectures on Physics Vol. II Ch. 22: AC Circuits." Caltech.

Charge and current

Conservation of charge

Universal relationship

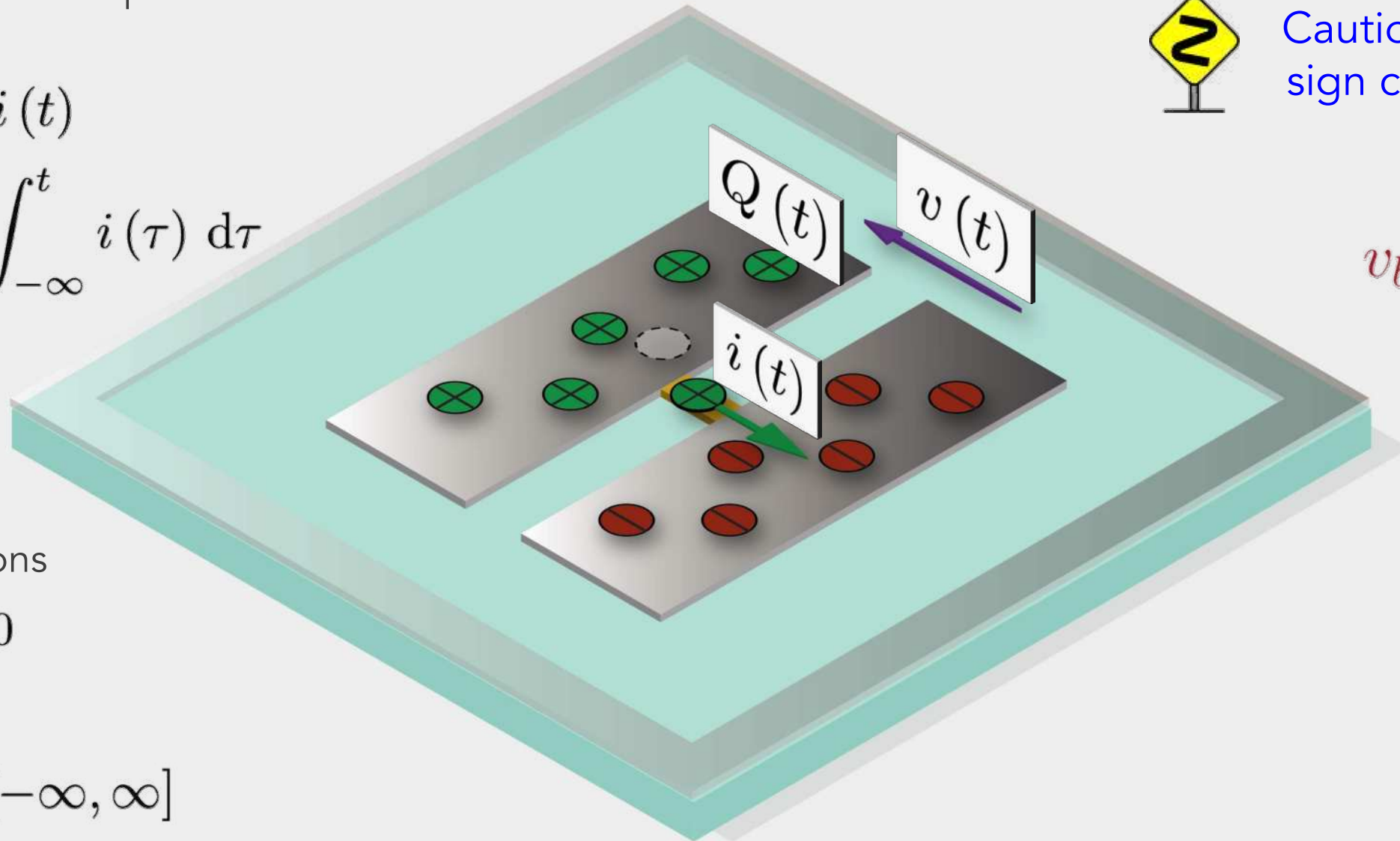
$$\frac{d}{dt}Q(t) = i(t)$$

$$Q(t) = \int_{-\infty}^t i(\tau) d\tau$$

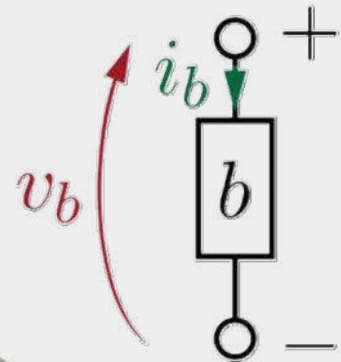
Initial conditions

$$Q(-\infty) = 0$$

$$i \in [-\infty, \infty]$$



Caution: Passive sign convention



Magnetic flux and inductance

Faraday's law of induction

Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$

$$\frac{d}{dt}\Phi(t) = v(t)$$

Initial conditions

$$\Phi(-\infty) = 0$$

Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic inductance

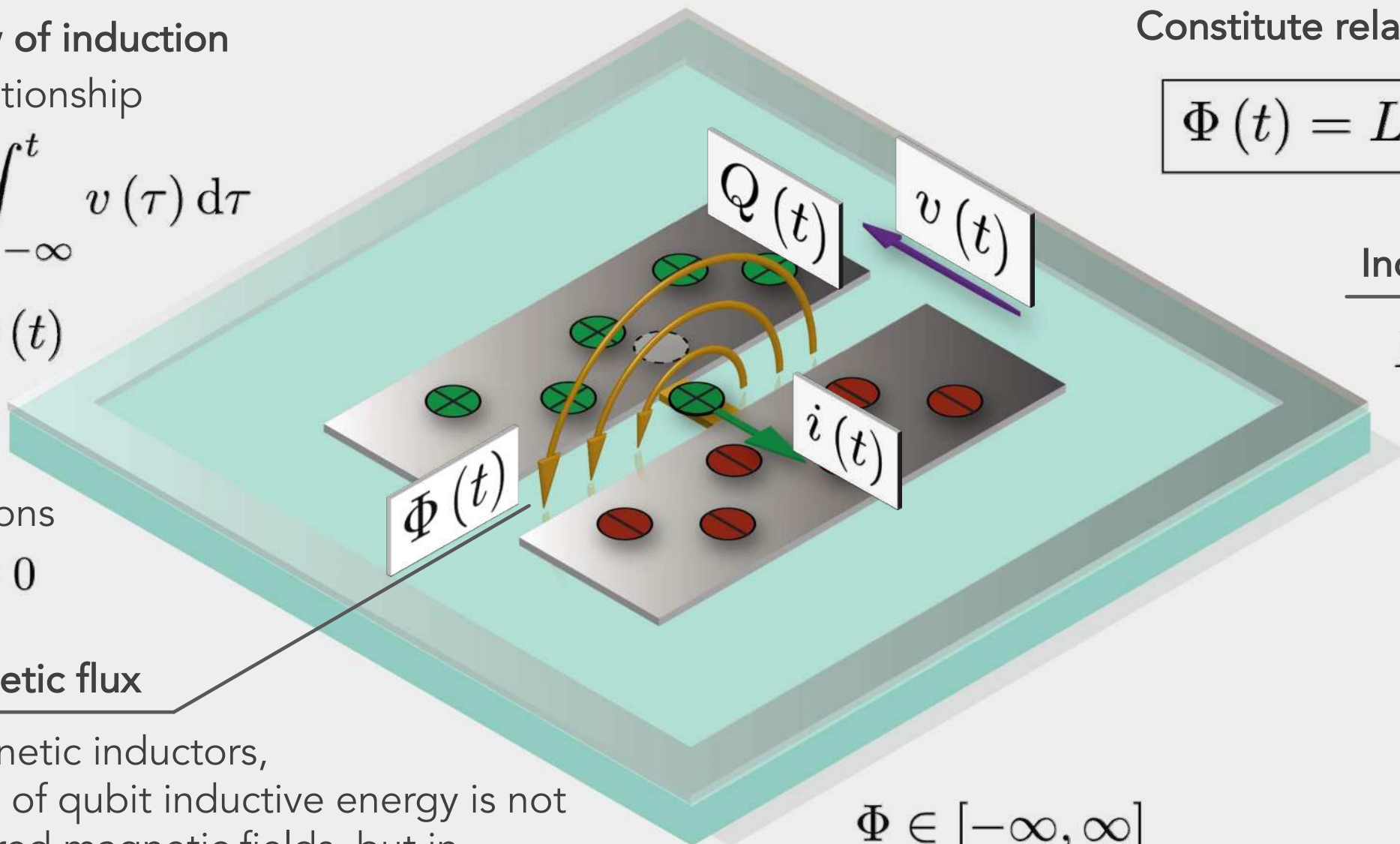
Constitute relationship

$$\Phi(t) = Li(t)$$

Inductance

$$L \in (0, \infty)$$

$$\Phi \in [-\infty, \infty]$$



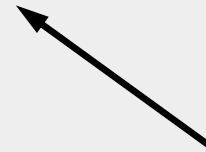
Power and energy

Universal

$$\frac{d}{dt}\mathcal{E}(t) = p(t) \equiv v(t)i(t)$$

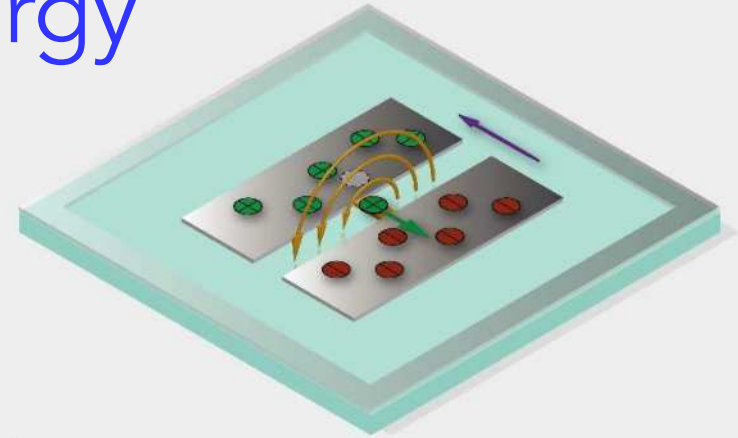


Energy stored *in*
(delivered *to*)
component



Instantaneous
power flowing
to component

$$\mathcal{E}(t) = \int_{-\infty}^t p(\tau) d\tau$$



$$\mathcal{E}_{\text{cap}}(\dot{\Phi}) = \frac{1}{2}C\dot{\Phi}^2$$

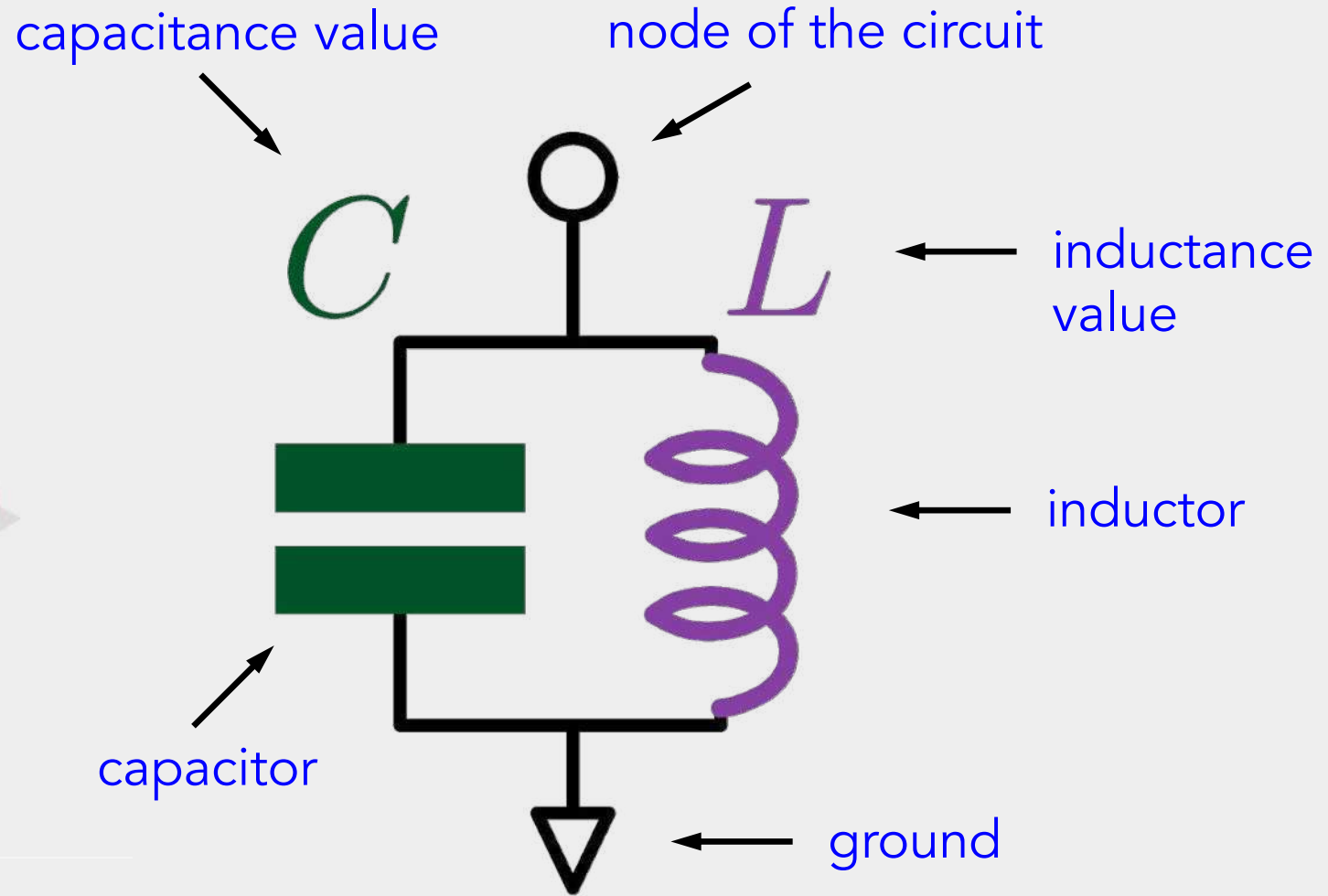
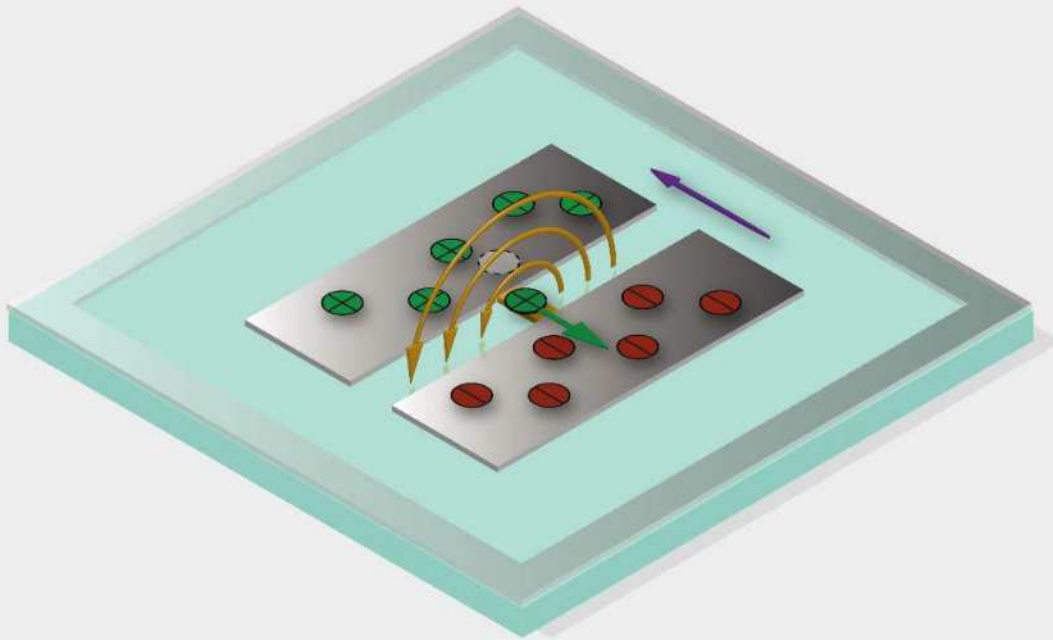
$$\mathcal{E}_{\text{cap}}(Q) = \frac{Q^2}{2C}$$



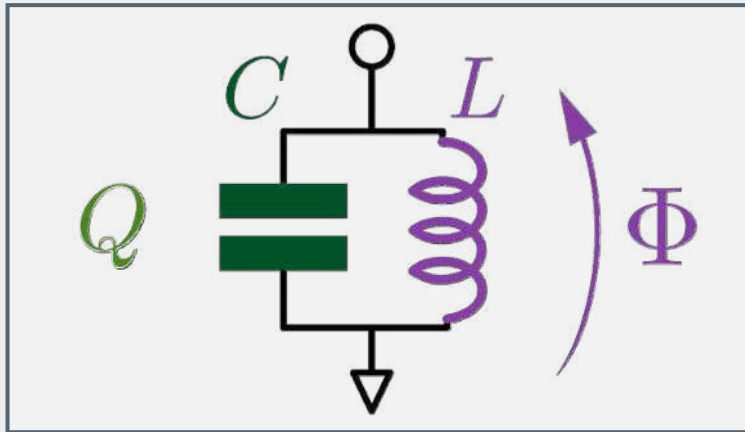
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\mathcal{E}_{\text{ind}}(\dot{Q}) = \frac{1}{2}L\dot{Q}^2$$

Electromagnetic oscillator



The LC classical harmonic oscillator



Position: $\Phi \mapsto x$

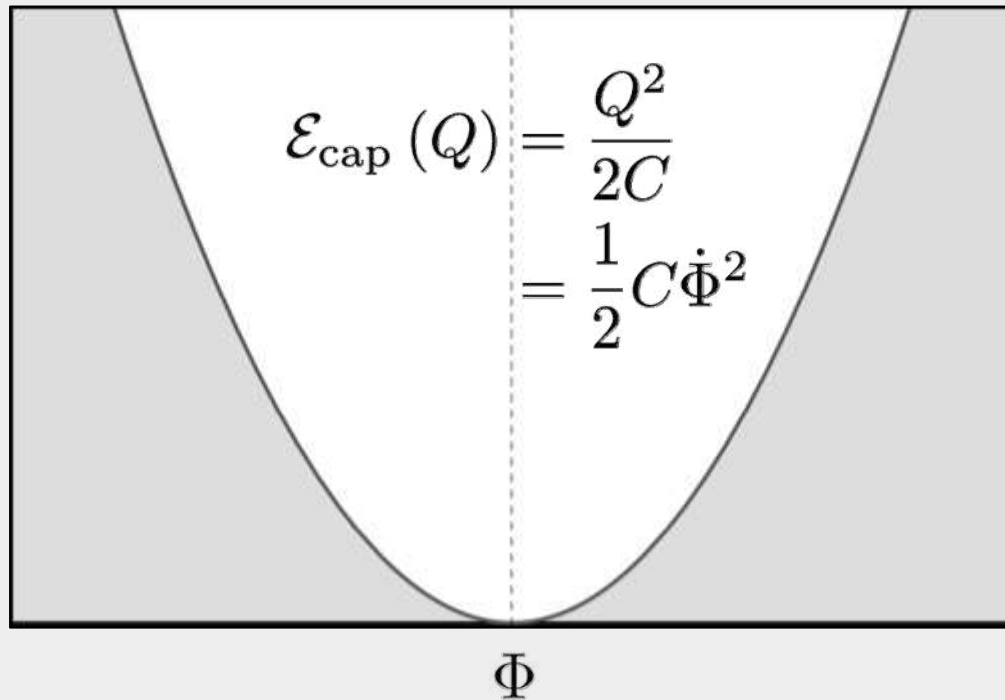
Mass: $C \mapsto m$

Momentum: $Q \mapsto p$

Spring constant: $L^{-1} \mapsto k$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

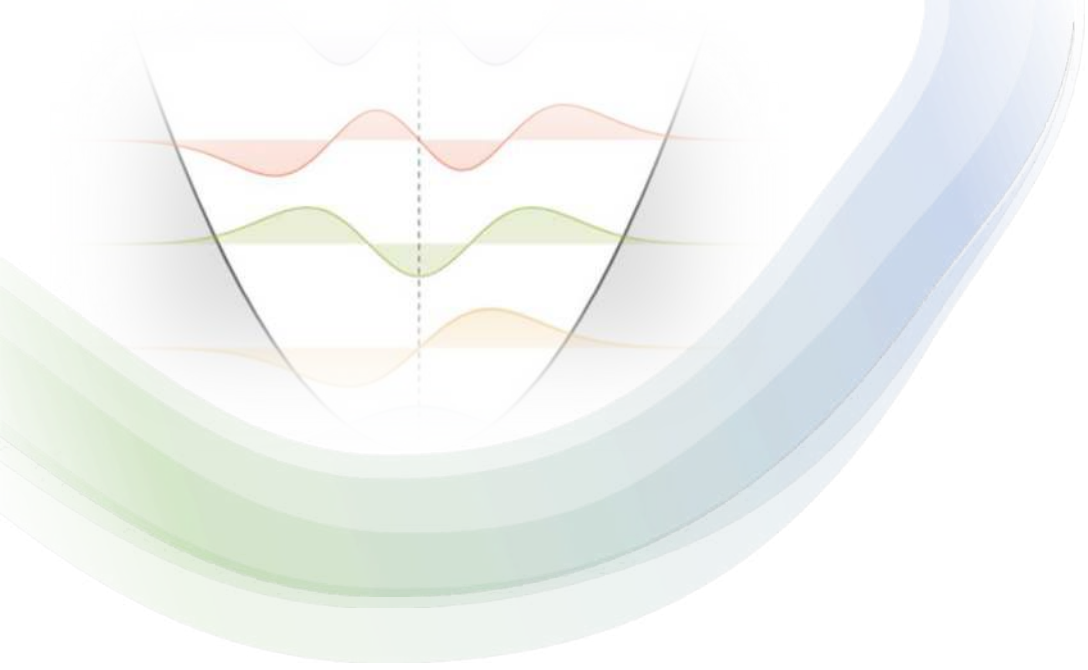
Energy



$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

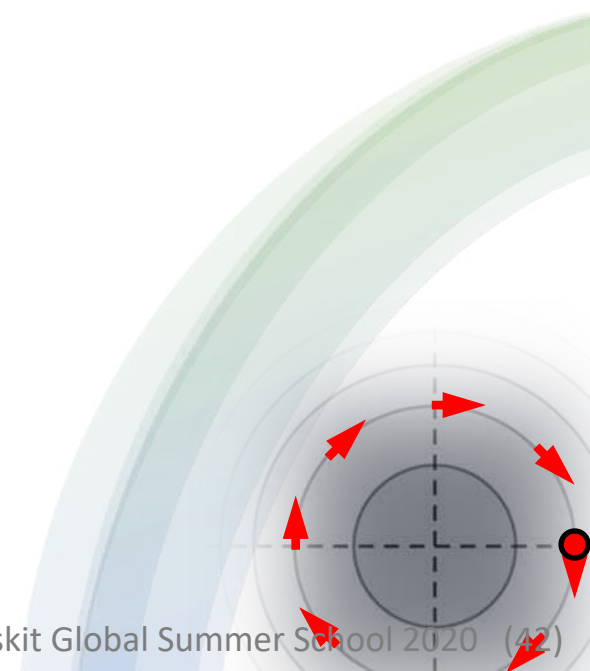
$$\omega_0^2 = \frac{1}{LC} \quad Z_0 = \frac{L}{C}$$

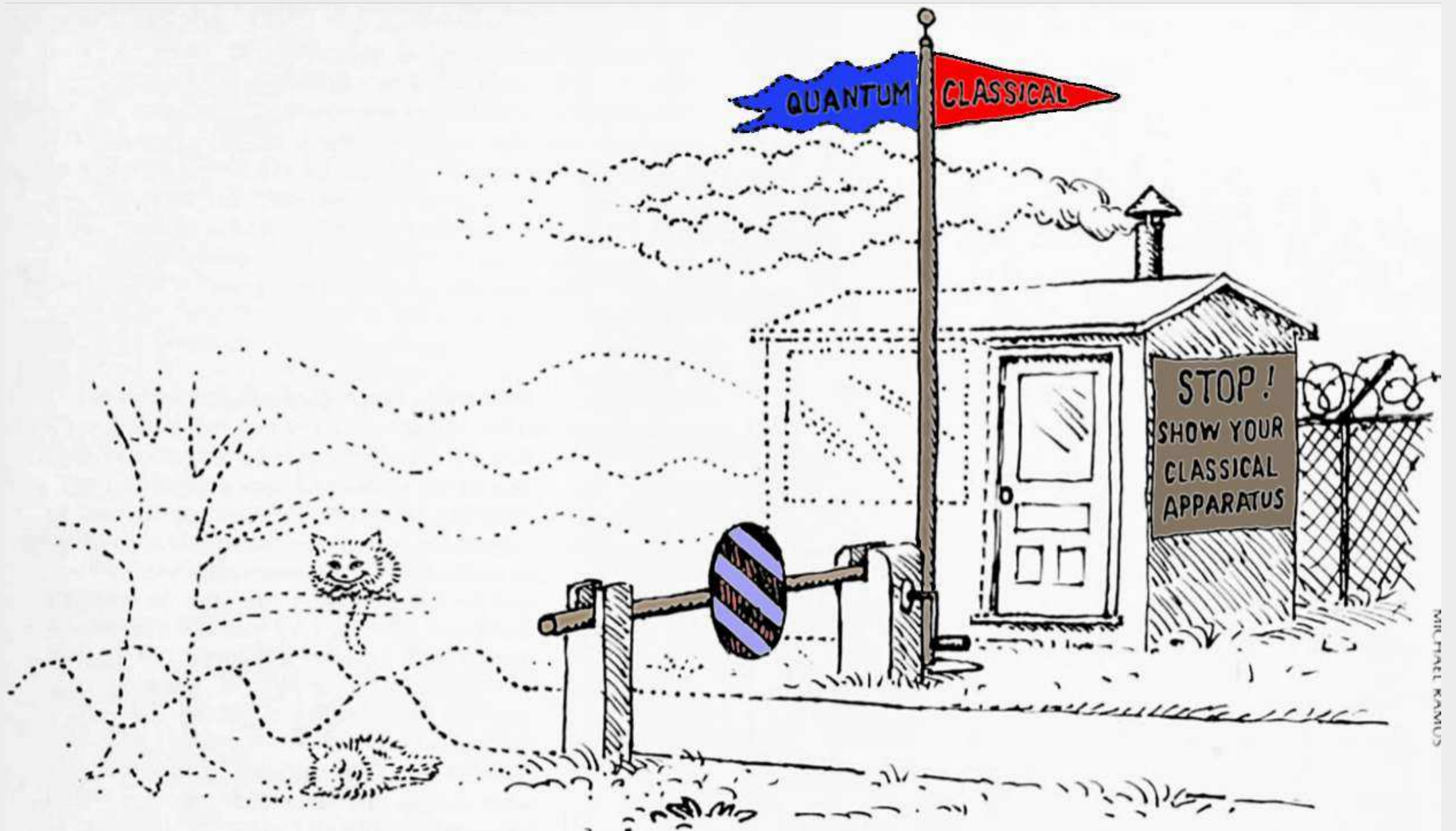
Graph: Minev; Spring-mass: Svjo



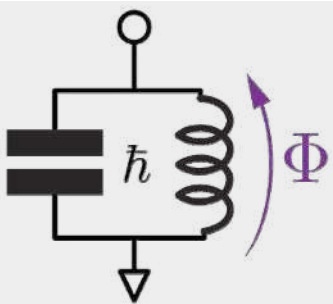
Unveiling the quantum

Quantum harmonic oscillator





Drawing: Zurek, Physics Today (1991)



The classical and quantum oscillator

Classical

Quantum

$$\Phi(t) \mapsto \hat{\Phi}$$

$$Q(t) \mapsto \hat{Q}$$

$$\mathcal{H}(\Phi, Q) \mapsto \hat{H}(\hat{\Phi}, \hat{Q})$$

$$\{\Phi, Q\} = 1 \mapsto [\hat{\Phi}, \hat{Q}] = i\hbar \hat{1}$$

$$\{\alpha, \alpha^*\} = 1/(i\hbar) \mapsto [\hat{a}, \hat{a}^\dagger] = \hat{1}$$

$$\{A, B\} = \frac{\partial A}{\partial \Phi} \frac{\partial B}{\partial Q} - \frac{\partial B}{\partial \Phi} \frac{\partial A}{\partial Q}$$

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

If I knew what I was doing, it
wouldn't be called research.

Albert Einstein
See Hawken *et al.* (2010)

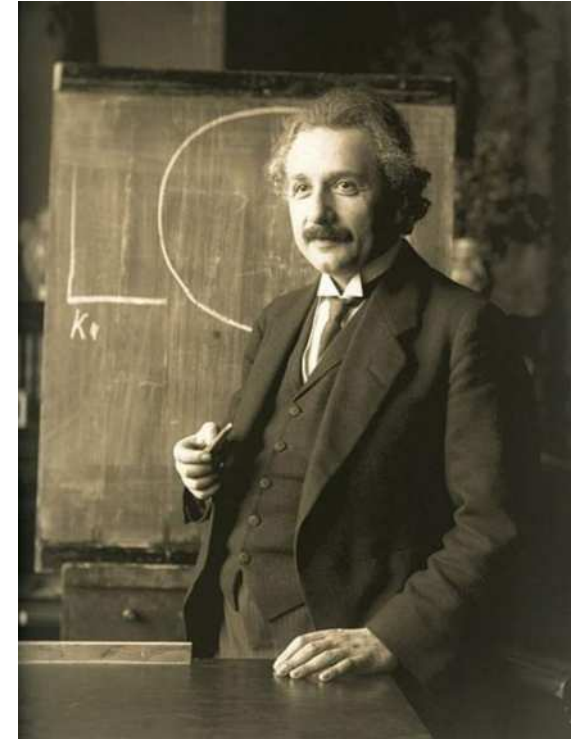
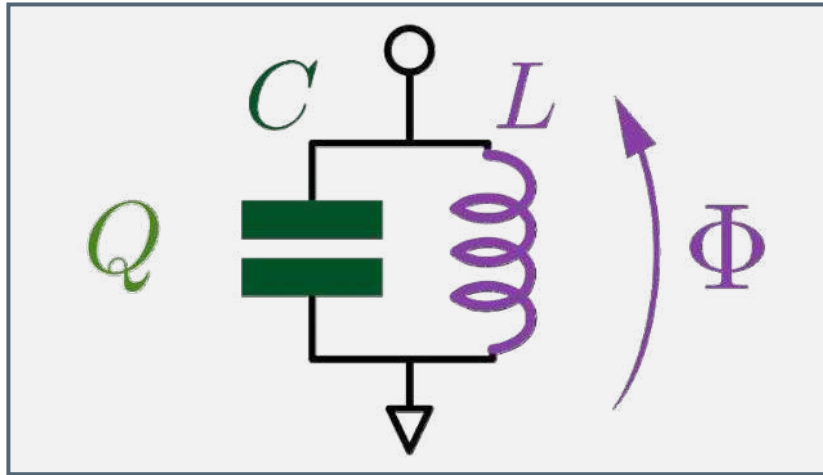


Photo: F. Schmutzer

The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$

Mass: $C \mapsto m$

Momentum: $Q \mapsto p$

Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

$$\hat{H}|n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad Z_0 = \frac{L}{C}$$

$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0 = \Phi_0 \sqrt{\frac{z_0}{2\pi}}, \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0^{-1} = (2e) \sqrt{\frac{1}{2\pi z_0}}, \end{aligned}$$

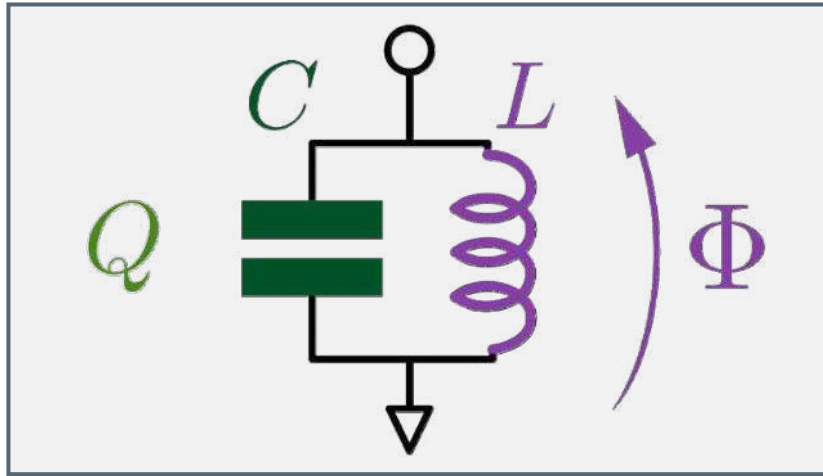
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\begin{aligned} \mathcal{E}_{\text{cap}}(Q) &= \frac{Q^2}{2C} \\ &= \frac{1}{2} C \dot{\Phi}^2 \end{aligned}$$

Energy

Φ/ϕ_0

The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$
Mass: $C \mapsto m$

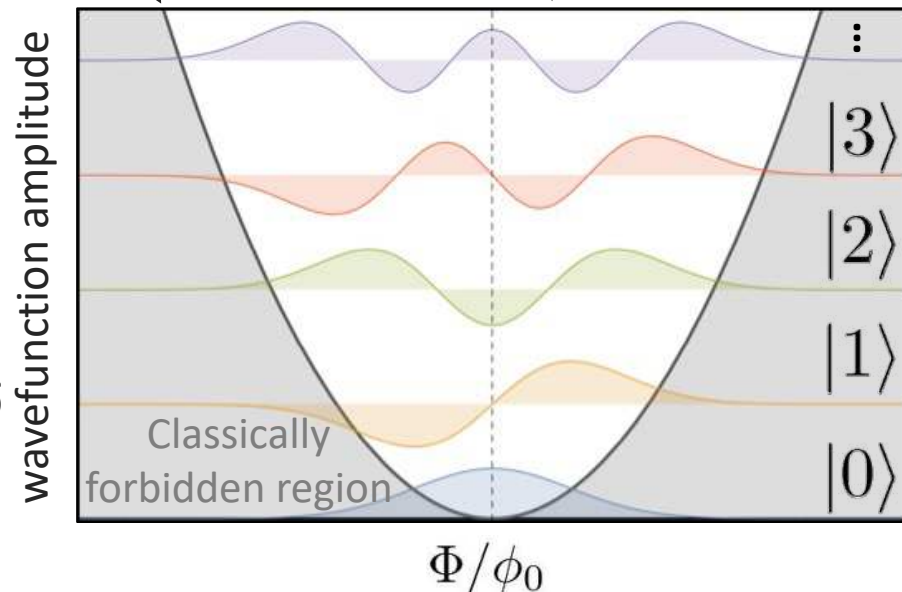
Momentum: $Q \mapsto p$
Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

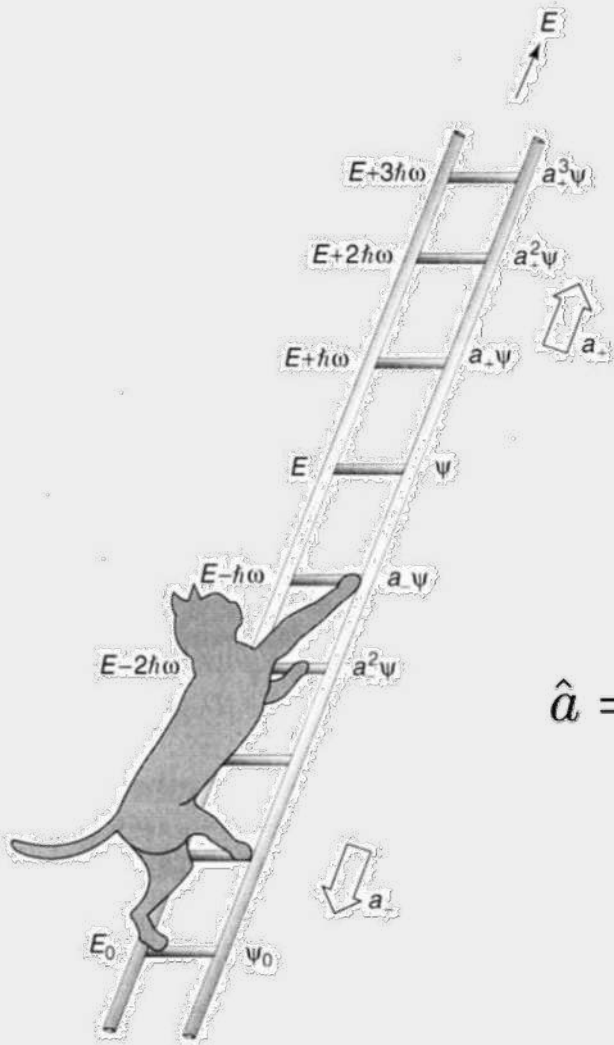
$$\hat{H}|n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad \Phi_{\text{ZPF}} Q_{\text{ZPF}} = \frac{\hbar}{2} \quad Z_0 = \frac{L}{C}$$

$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0} & \langle 0 | \hat{\Phi}^2 | 0 \rangle &= \Phi_{\text{ZPF}}^2 \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0^{-1}} & \langle 0 | \hat{Q}^2 | 0 \rangle &= Q_{\text{ZPF}}^2 \end{aligned}$$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L} \quad \psi_n(\Phi) \equiv \langle \Phi | n \rangle$$



Ladder operators and matrix representation



annihilation

$$\hat{a} |0\rangle = 0$$

$$\hat{a} |1\rangle = \sqrt{1} |0\rangle$$

creation

$$\hat{a}^\dagger |0\rangle = \sqrt{1} |1\rangle$$

$$\hat{a}^\dagger |1\rangle = \sqrt{2} |2\rangle$$

general hopping

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle$$

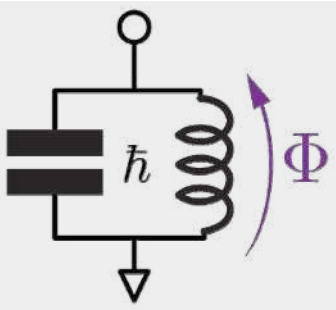
$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\hat{a} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

$$\hat{a}^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{3} & \ddots \end{pmatrix}$$

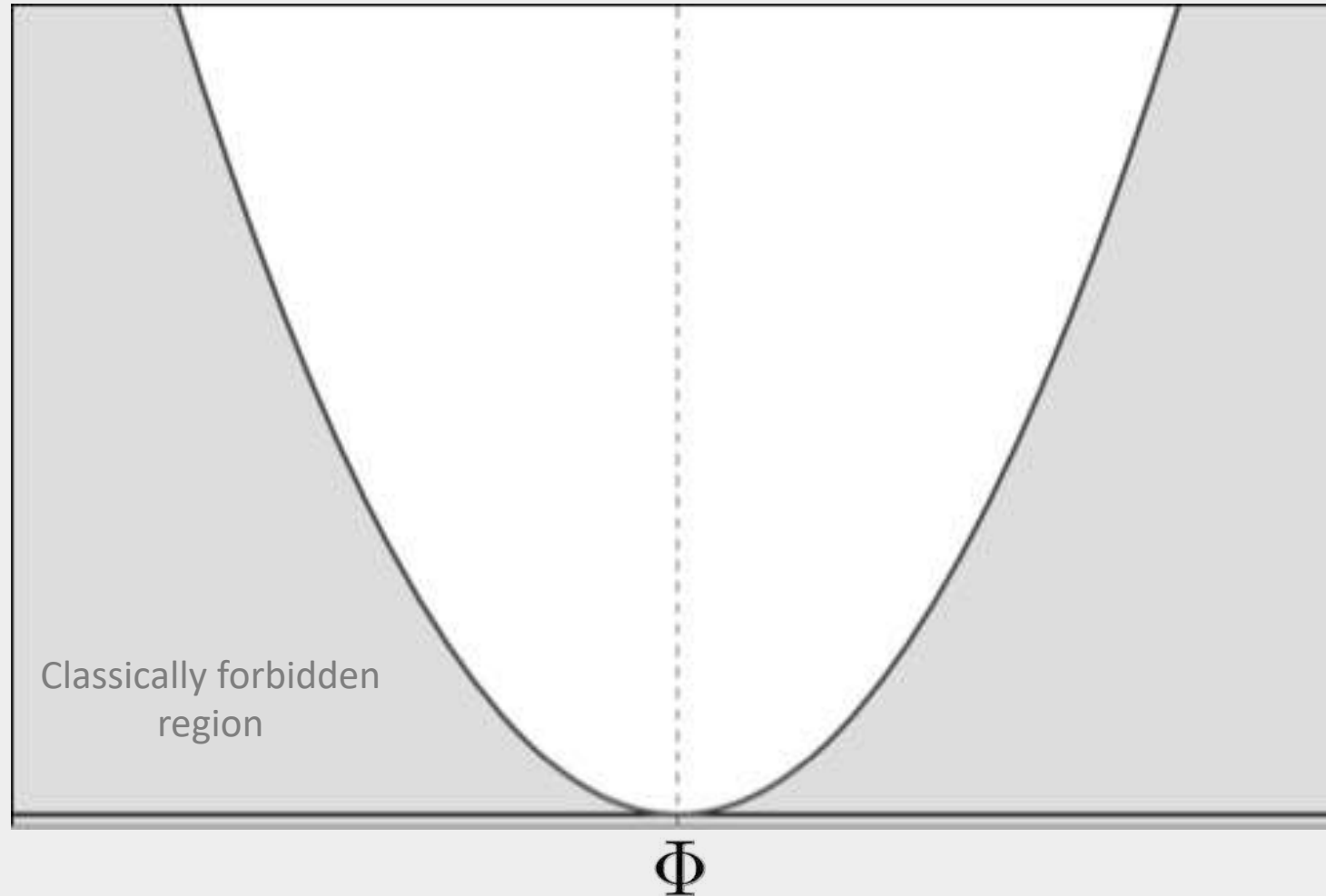
$$\hat{a}^\dagger \hat{a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

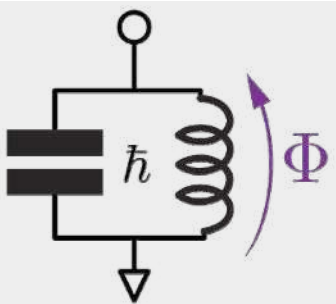
Wavefunctions of the quantum oscillator



$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



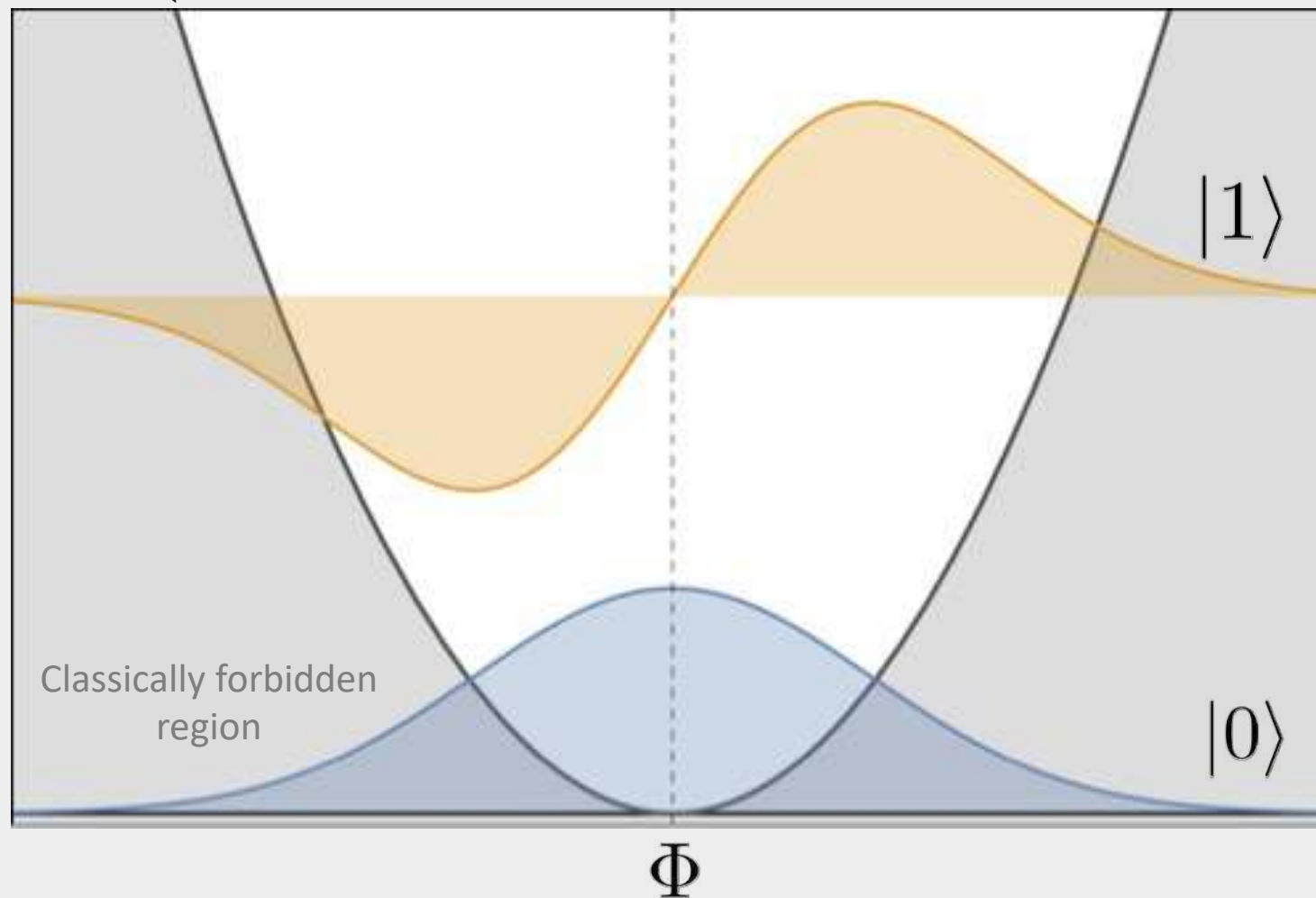


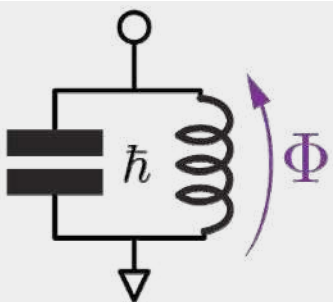
Wavefunctions of the quantum oscillator

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



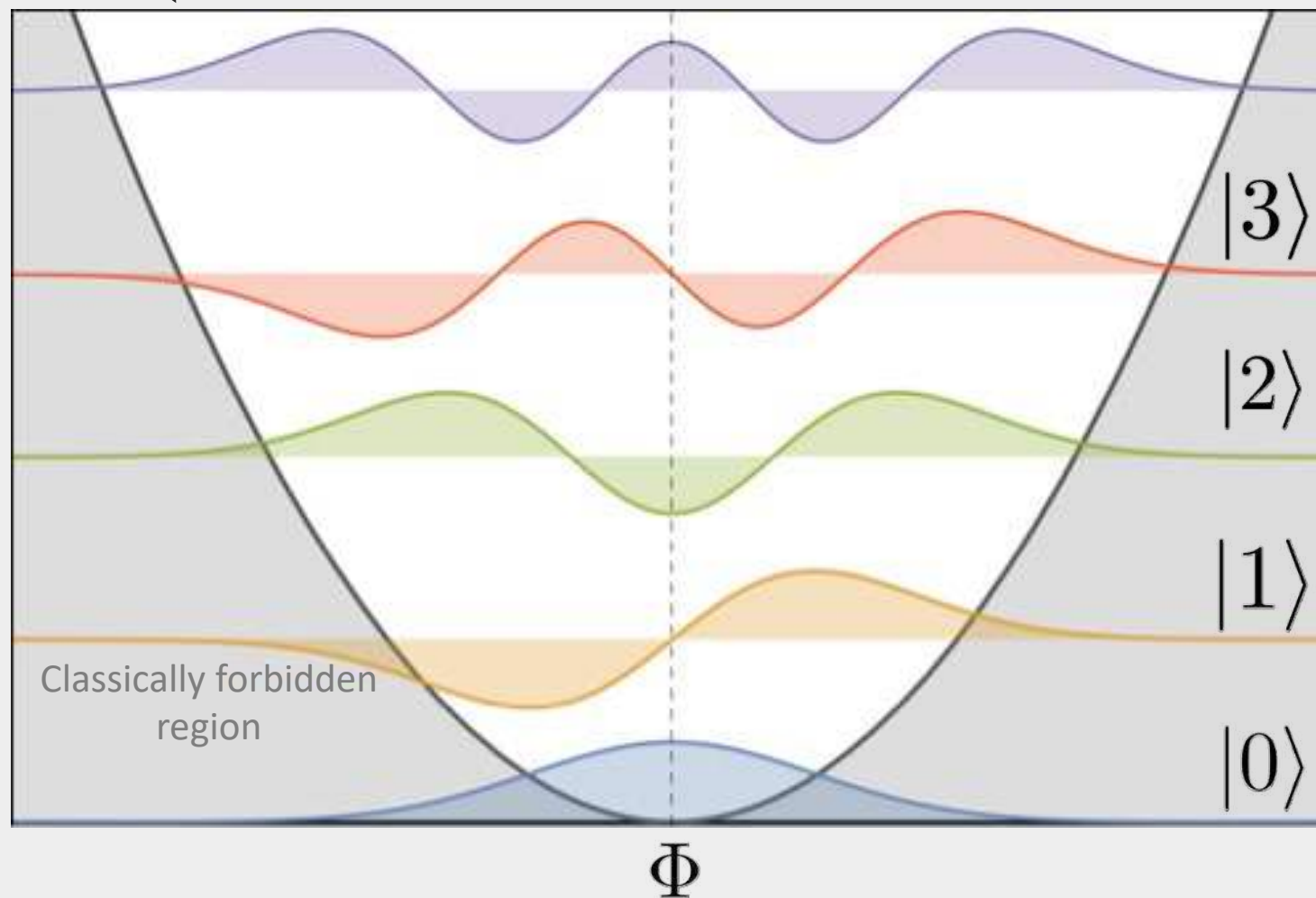


Wavefunctions of the quantum oscillator

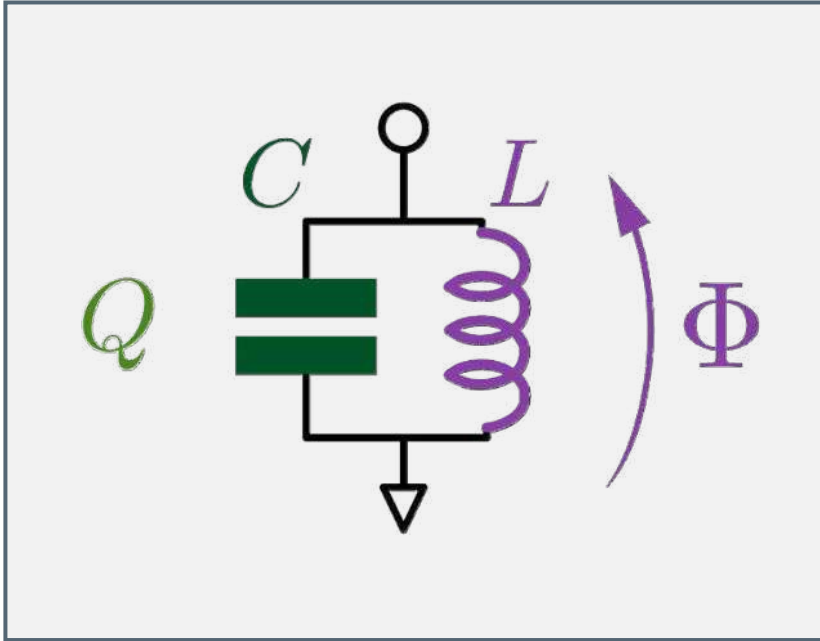
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



Pop-up question



The flux and charge operators are Hermitian observables.

How can some expectations, such as

$$\langle 0 | \hat{\Phi} \hat{Q} | 0 \rangle = \frac{1}{4} i ,$$

be imaginary?

Or, others be negative...?

Advanced questions



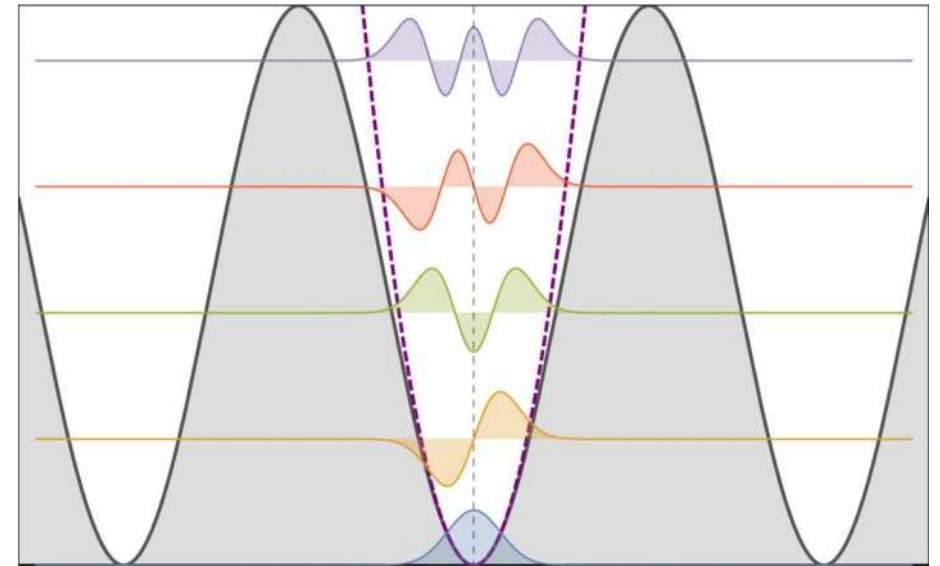
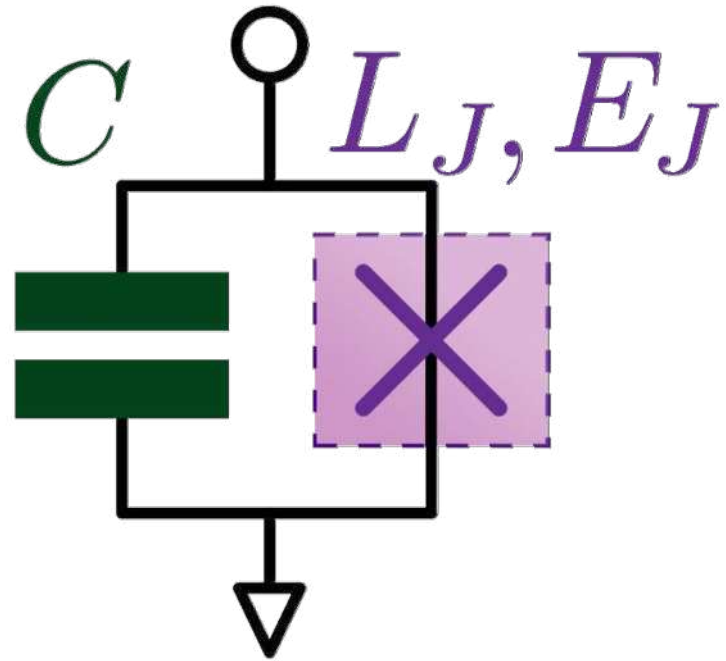
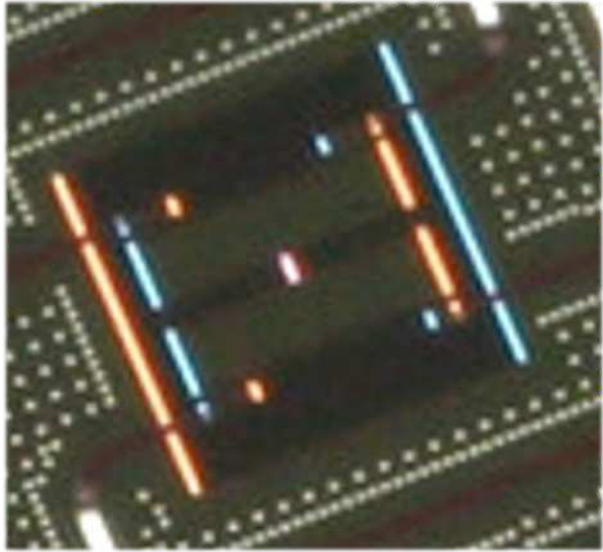
1. What is $\hat{\Phi}(t)$ in terms of $\hat{a}$ in the Heisenberg picture?
2. **Autocorrelation.** Find the autocorrelation operator $\hat{C}_{\Phi\Phi}(t) = \hat{\Phi}(t) \hat{\Phi}(0)$. What frequency components does it have?
 - (a) What is the flux autocorrelation expectation value $C_{\Phi\Phi}(t) = \langle n | \hat{\Phi}(t) \hat{\Phi}(0) | n \rangle$ for the Fock state $|n\rangle$? Is it real? Why not? What is the frequency spectrum?
 - (b) Repeat for a coherent state $|\alpha\rangle$.



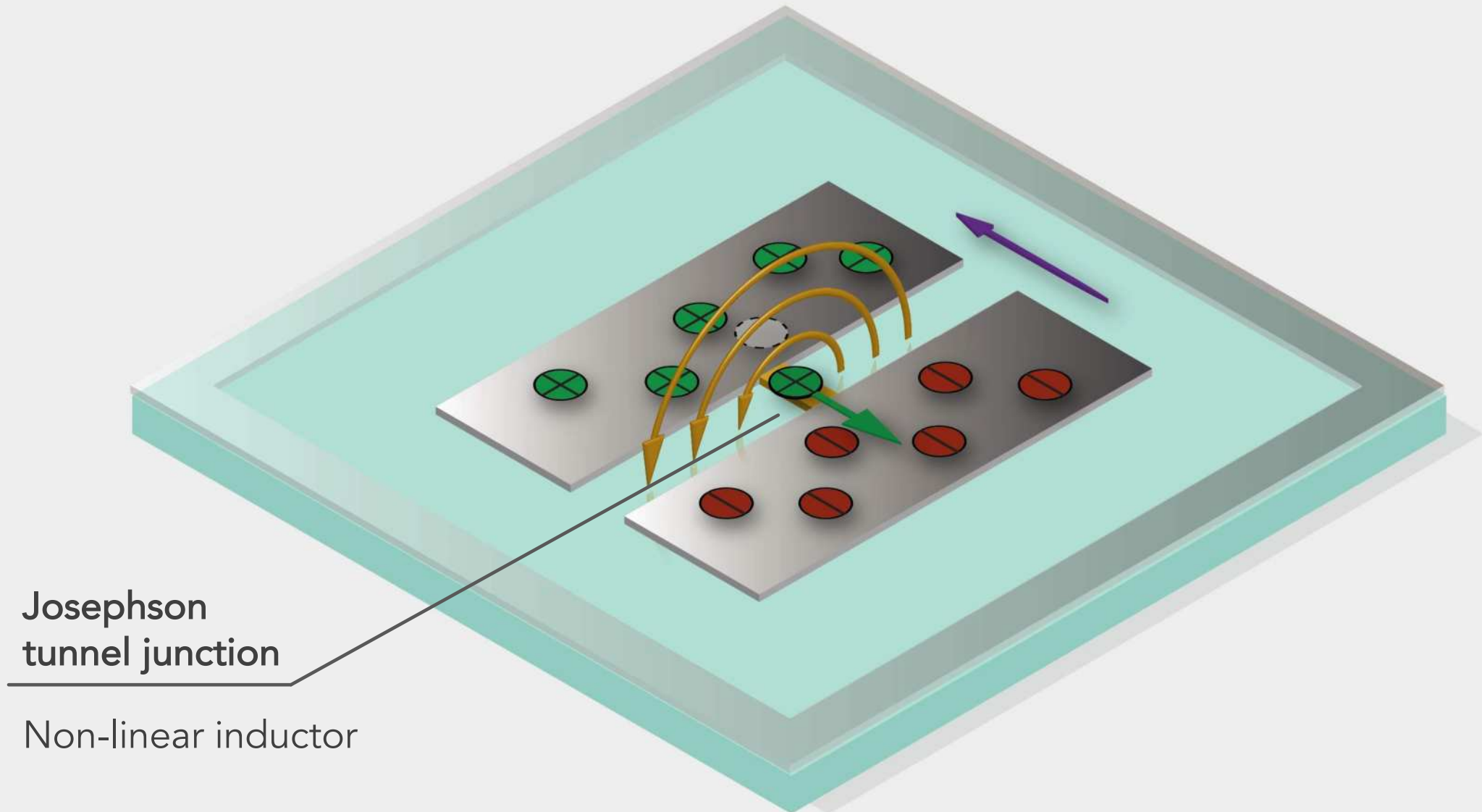
3. **Thermal state.** The thermal state is $\hat{\rho}_{\text{th}} = \exp[-\beta \hat{a}^\dagger \hat{a}] / \text{Tr}[\exp(-\beta \hat{a}^\dagger \hat{a})]$, where $\beta = \hbar\omega_0/k_B T$, and T is the temperature of the oscillator.
 - (a) What is the mean and variance of the flux $\hat{\Phi}$ and charge $\hat{Q}$ operators?
 - (b) How does the frequency spectrum of the autocorrelation $\langle \hat{C}_{\Phi\Phi}(t) \rangle = \text{Tr}[\hat{\rho}_{\text{th}} \hat{C}_{\Phi\Phi}(t)]$ change when with that for the state $|n\rangle$ and $|\alpha\rangle$?
 - (c) The spectrum is not symmetric in frequency. How can you interpret that positive and negative frequencies have different weights? How is this related to absorption and emission of the oscillator? (Consider Fermi's golden rule).
 - (d) What happens to the spectrum in the limit of high temperature, $k_B T \gg \hbar\omega_0$? How about low temperature, $k_B T \ll \hbar\omega_0$?
 - (e) How do the above conclusions change for charge; i.e., for $\langle \hat{C}_{QQ}(t) \rangle = \text{Tr}[\hat{\rho}_{\text{th}} \hat{C}_{QQ}(t)]$?

Will discuss
answers on my
blog sometime
soon

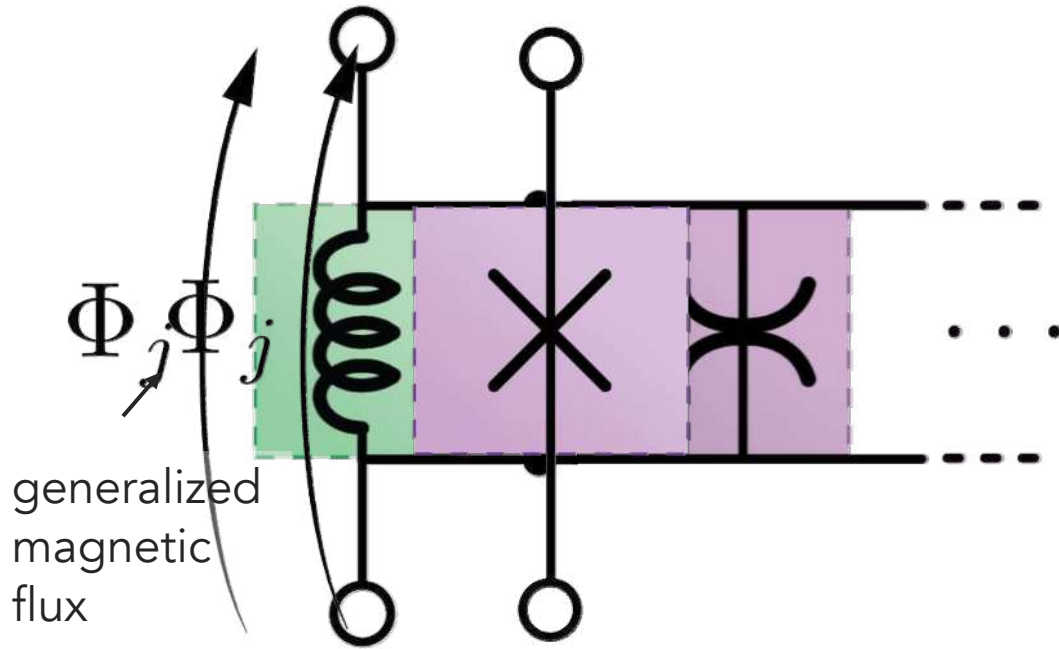
Transmon Qubit



The transmon as a non-linear oscillator



Josephson tunnel junction

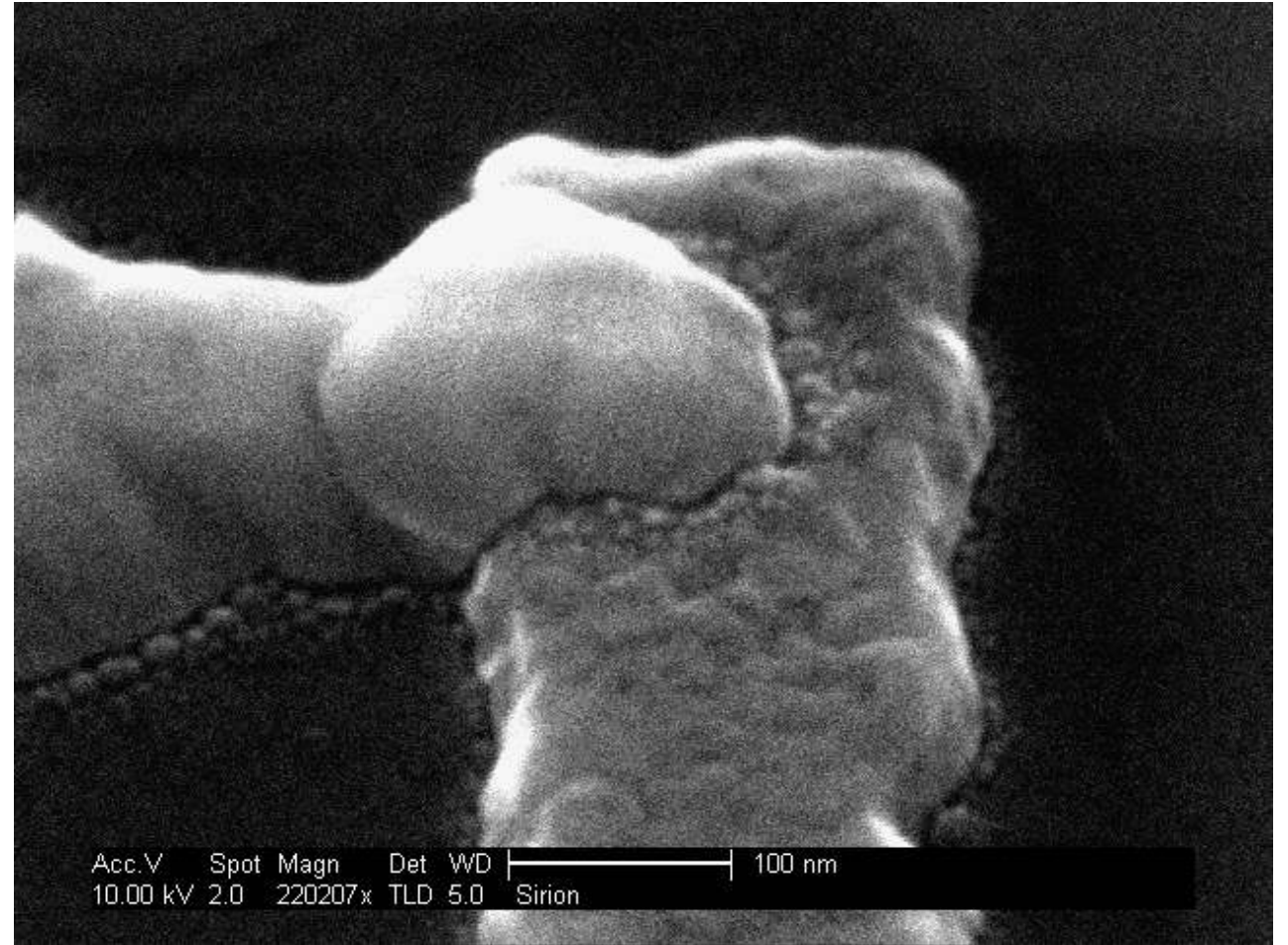


$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$= \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$\phi_0 \equiv \hbar/2e$$

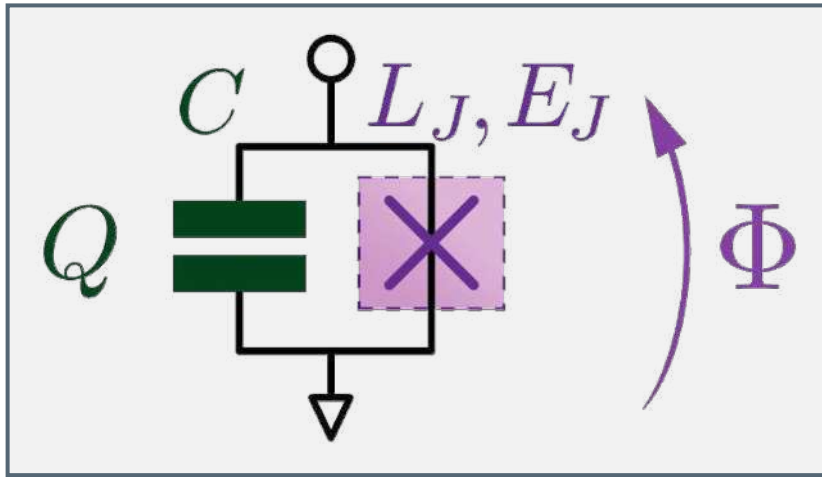
$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$



SEM image: L. Frunzio

Circuit image: Mineev et al., EPR to appear (2020)

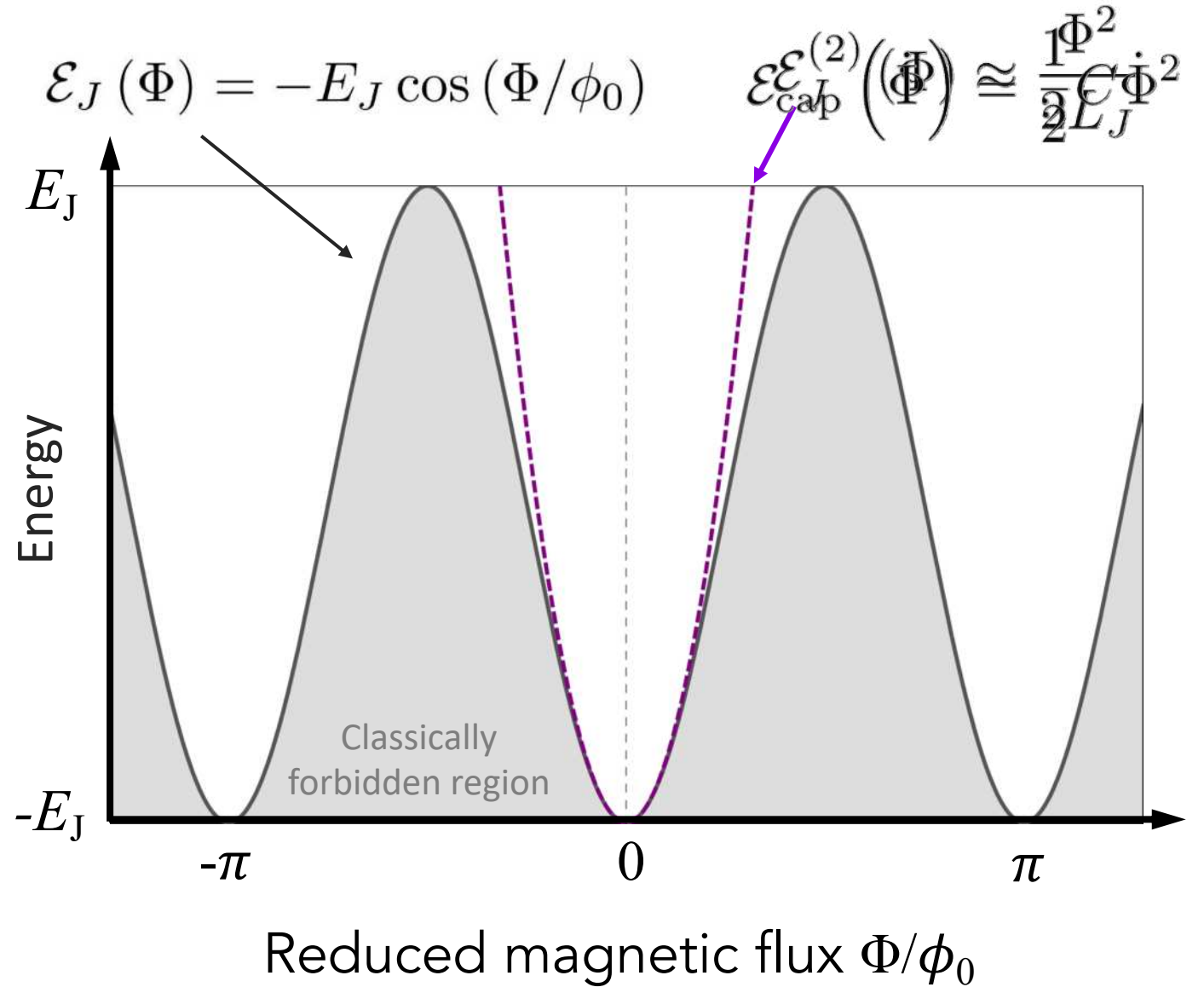
The Transmon qubit



$$E_J = \frac{\phi_0^2}{L_J}$$

$$\phi_0 = \frac{\hbar}{2e}$$

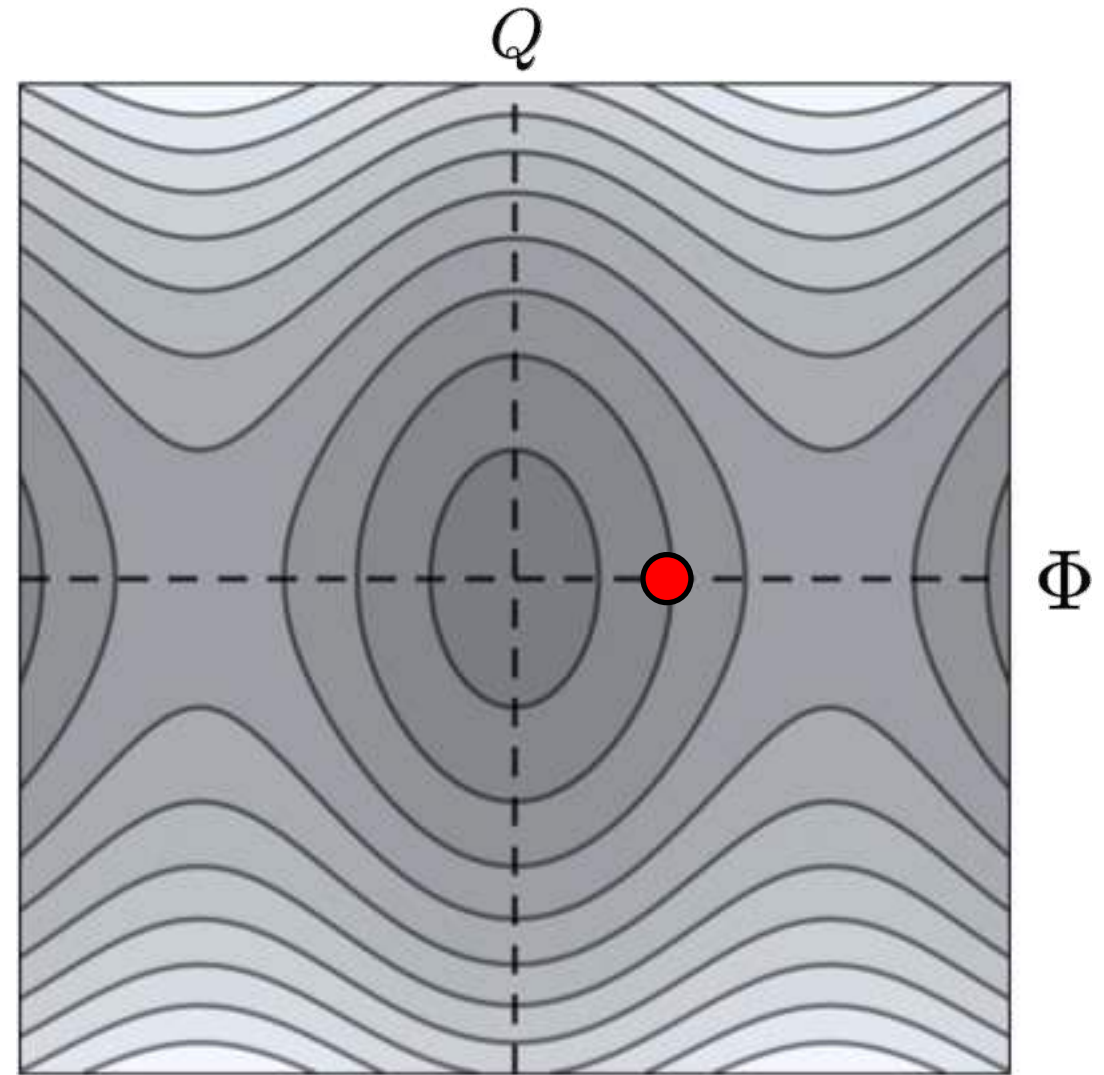
$$\approx 3.3 \times 10^{-16} \text{ Wb}$$



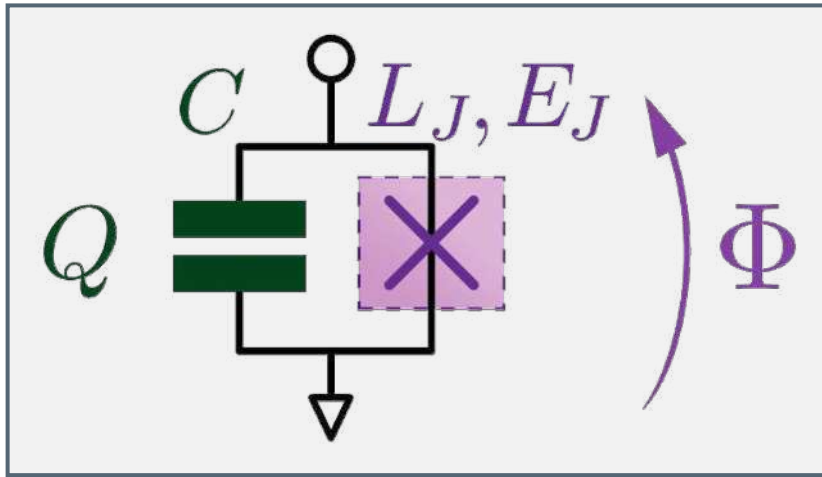
Semi-classical intuition: phase space picture

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0)$$

$$\begin{aligned} E &= \hbar\omega_0 \left(n + \frac{1}{2} \right) \\ &= \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0) \end{aligned}$$



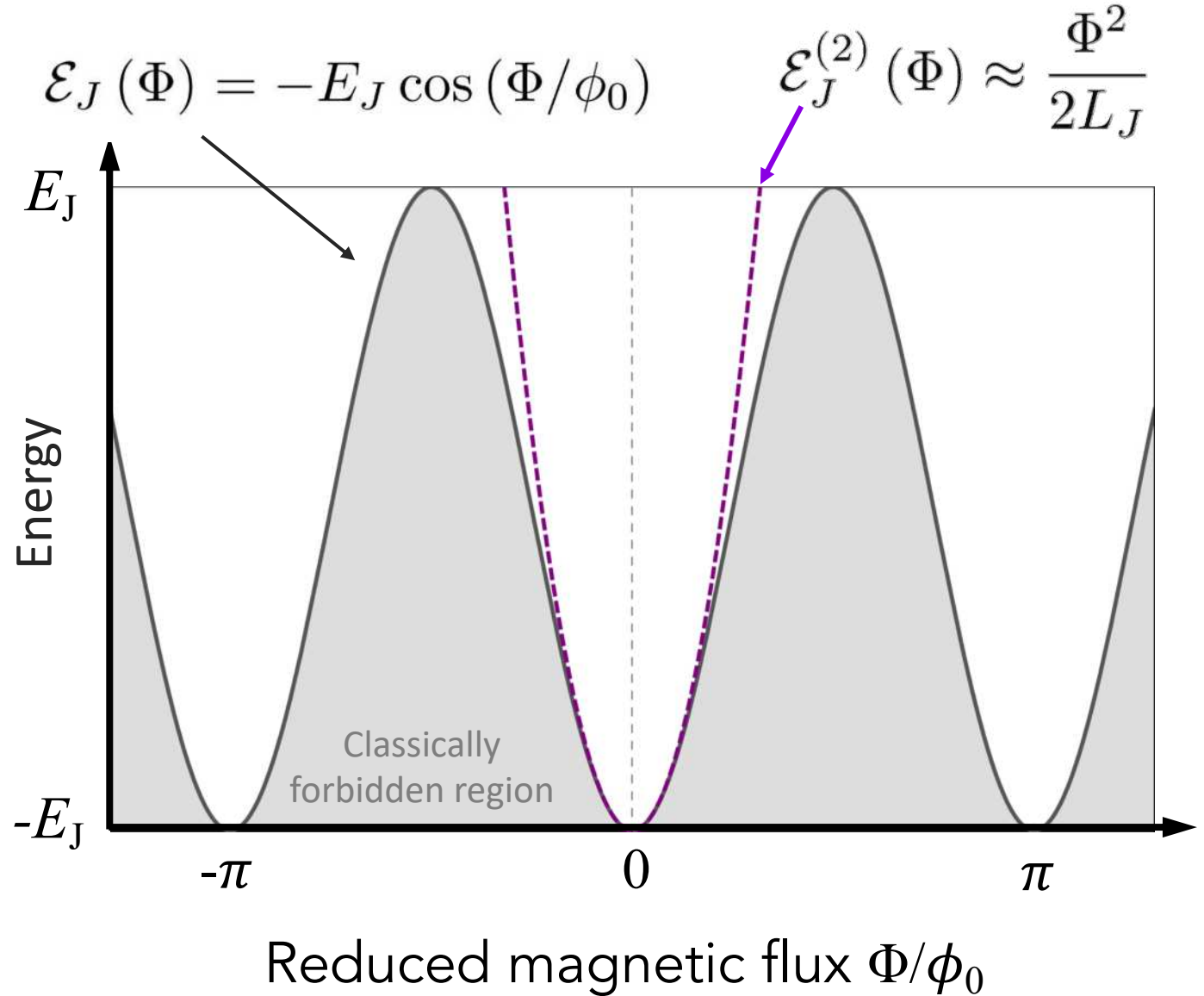
The Transmon qubit

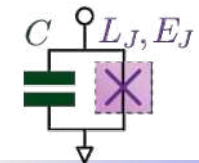


$$\hat{H} = \frac{\hat{Q}^2}{2C} - E_J \cos \left(\hat{\Phi} / \phi_0 \right)$$

$$\approx \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L_J} - \frac{E_J}{4!} \left(\frac{\hat{\Phi}}{\phi_0} \right)^4$$

$$= \hbar \omega_0 \hat{a}^\dagger \hat{a} - \frac{E_J \phi_{\text{ZPF}}^4}{4!} (\hat{a} + \hat{a}^\dagger)^4$$





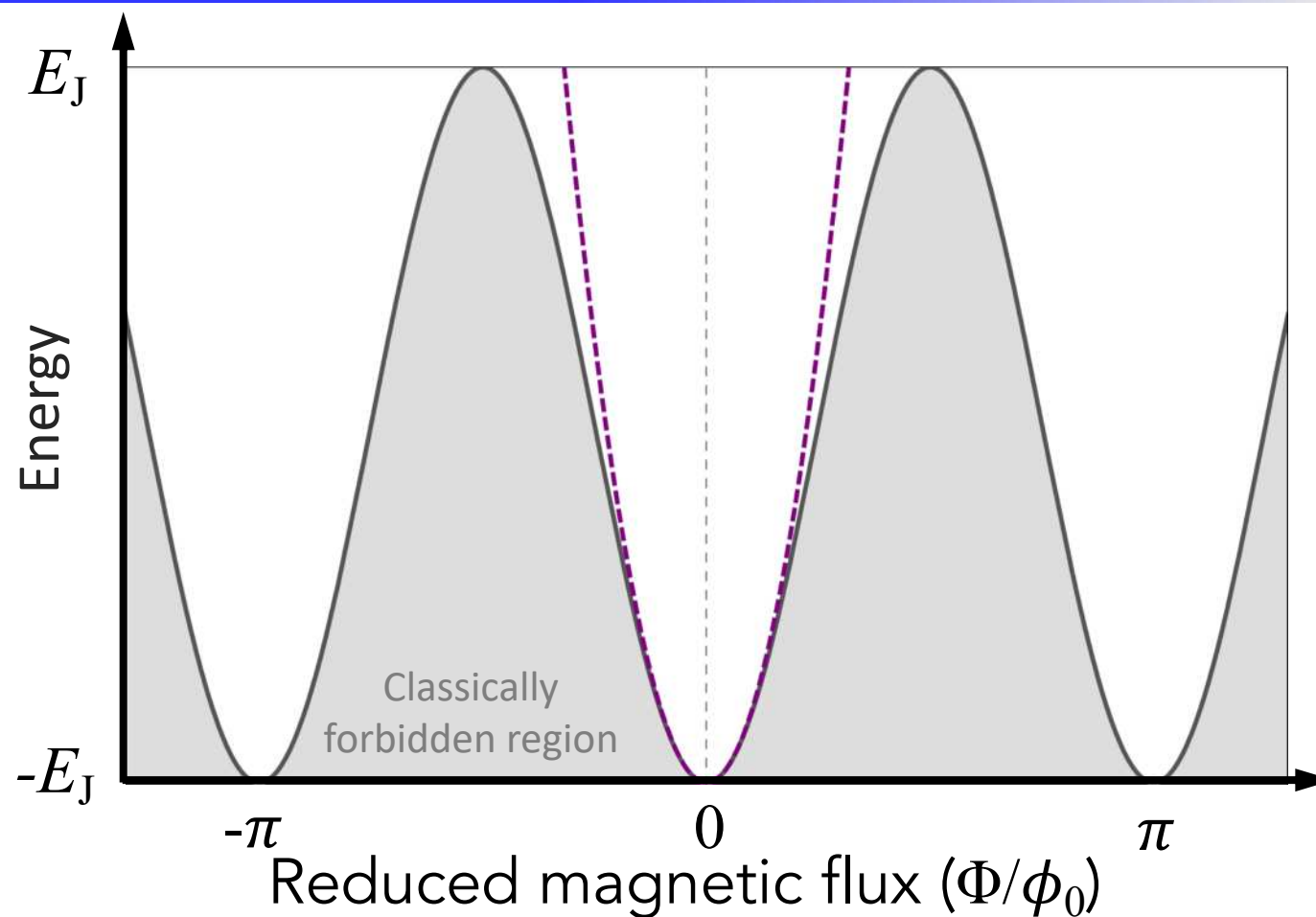
The Kerr Hamiltonian of the transmon

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

$$\hat{H}_4^{\text{RWA}} \approx \underbrace{\hbar\omega_q \hat{N}}_{\hat{H}_{\text{lin}}} - \underbrace{\frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)}_{\hat{H}_{\text{nl}}}$$

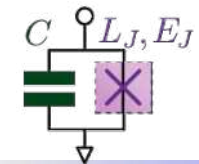
$\hat{H}_{\text{lin}}$ Solution known (SHO) $\hat{H}_{\text{nl}}$ New from nonlinearity

$$\hat{H}_4^{\text{RWA}} |n\rangle = \hbar\omega_q n \left(1 - \frac{\hbar\alpha_a}{2} (n - 1) \right) |n\rangle$$



1st order correction to energy: $E_n^{(1)} = \langle n^{(0)} | \hat{H}_{\text{nl}} | n^{(0)} \rangle$ and to eigenstates: $|n^{(1)}\rangle = \sum_{k \neq n} \frac{\langle k^{(0)} | \hat{H}_{\text{nl}} | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle.$

To first order perturbation theory the eigenstates do not change! Only the energy changes. Dispersive.



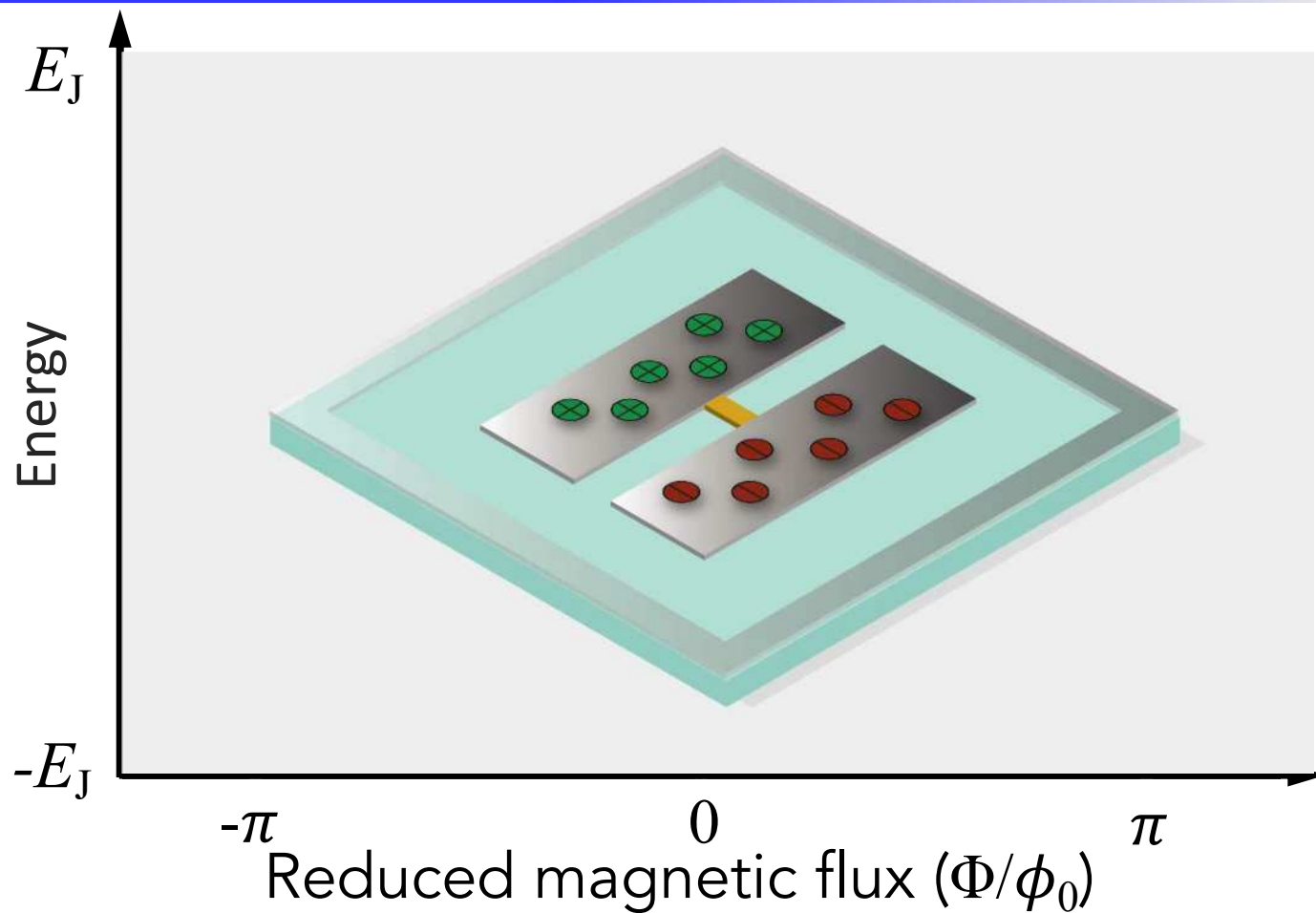
Exploring a real transmon qubit

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

$$\hat{H}_4^{\text{RWA}} \approx \underbrace{\hbar\omega_q \hat{N}}_{\hat{H}_{\text{lin}}} - \underbrace{\frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)}_{\hat{H}_{\text{nl}}}$$

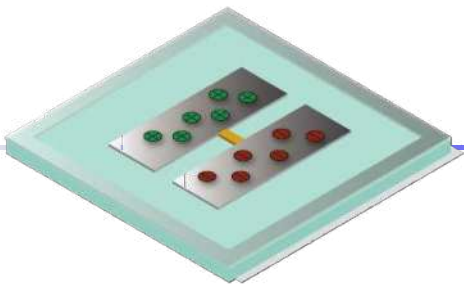
$\nearrow \hat{H}_{\text{lin}}$
 Solution known (SHO)

$\nwarrow \hat{H}_{\text{nl}}$
 New from nonlinearity



Experimental parameters

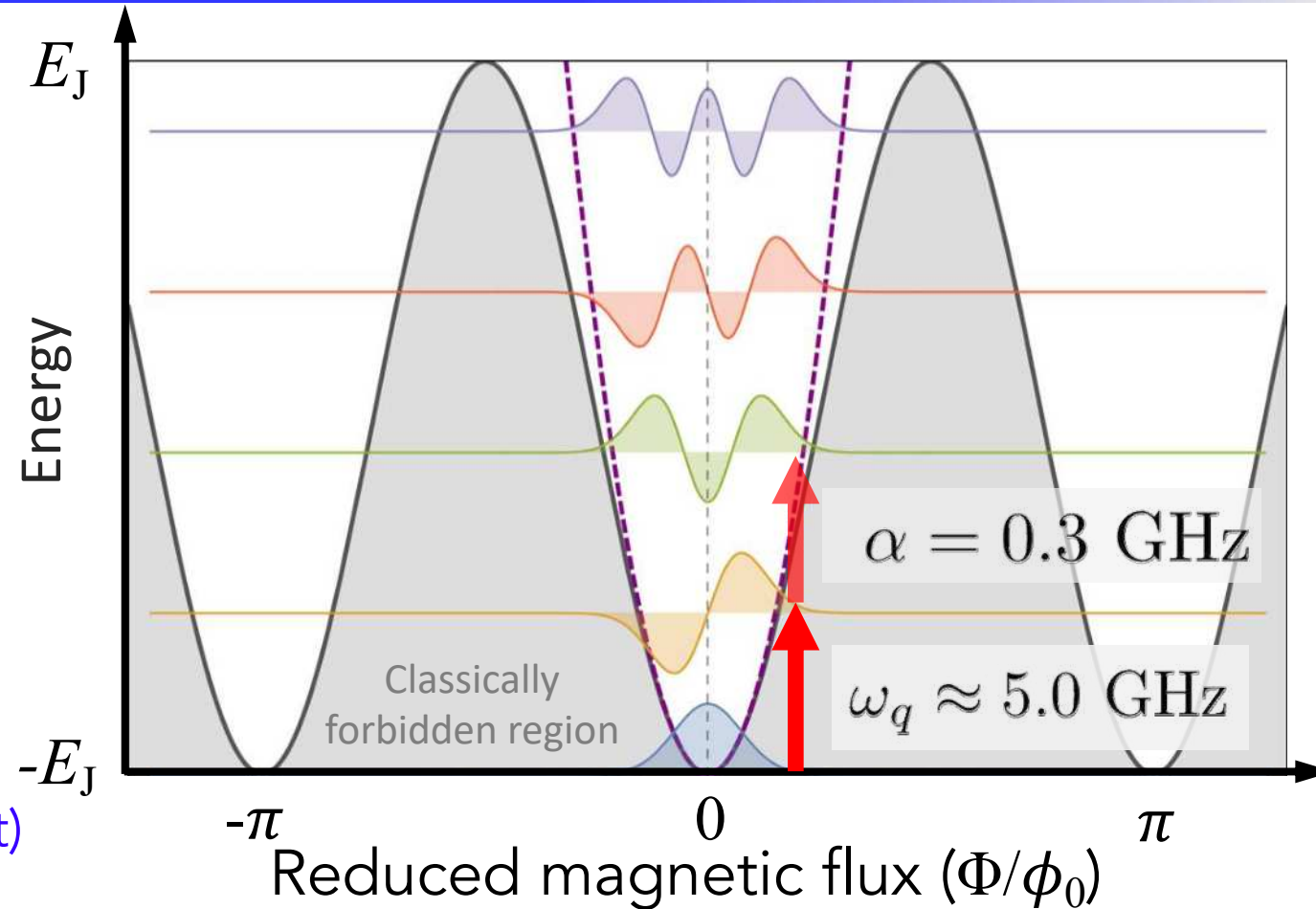
$$\begin{aligned}
 L_J &= 14 \text{ nH} & E_J &= \frac{\phi_0^2}{L_J} = 12 \text{ GHz} \\
 C_J &= 65 \text{ fF} & E_C &= \frac{e^2}{2C} = 0.3 \text{ GHz}
 \end{aligned}$$



Exploring a real transmon qubit

$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

Dispersive, states didn't change to 1st order
"Lamb shift" due to ZPF



Parameters used in figure (of a measured qubit)

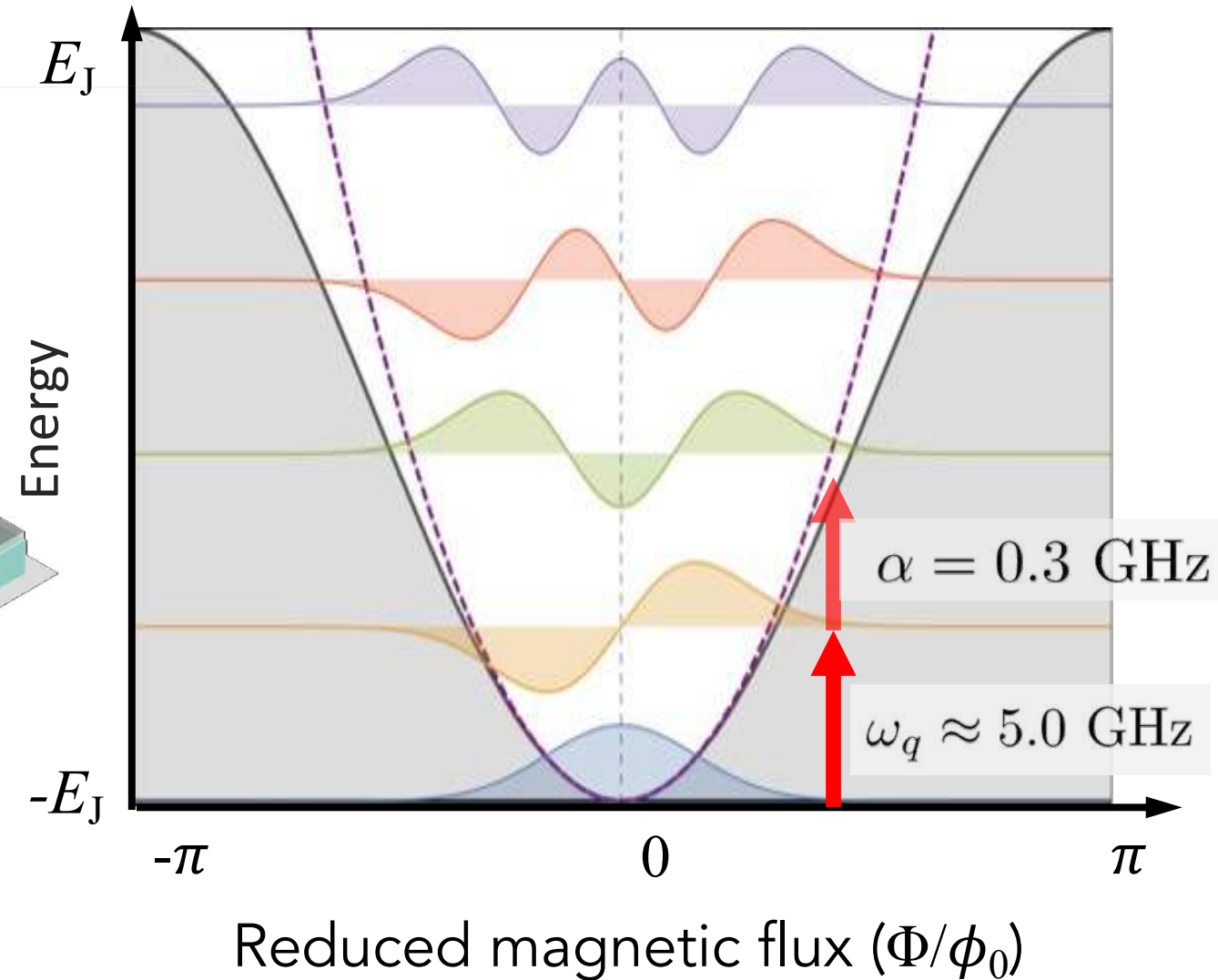
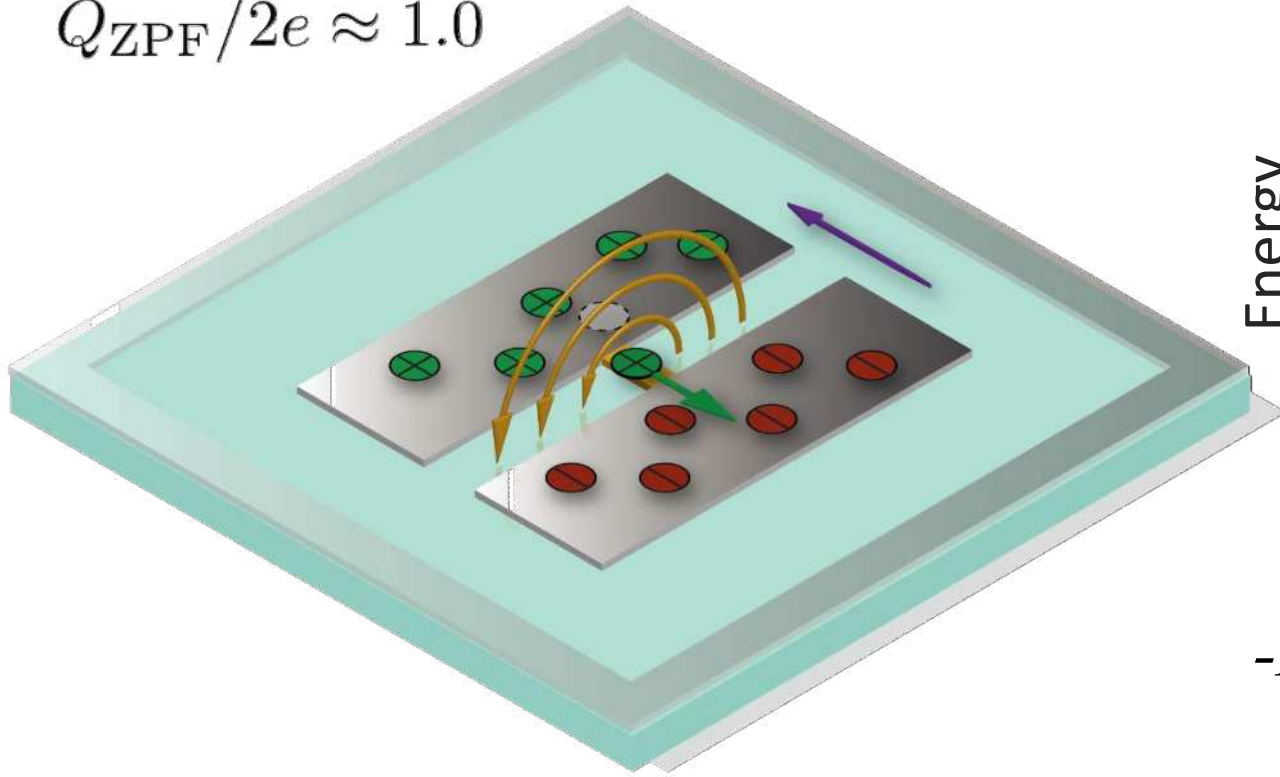
$$L_J = 14 \text{ nH} \quad E_J = \frac{\phi_0^2}{L_J} = 12 \text{ GHz} \quad \omega_0 = \sqrt{\frac{1}{LC}} = 5.3 \text{ GHz}$$

$$C_J = 65 \text{ fF} \quad E_C = \frac{e^2}{2C} = 0.3 \text{ GHz} \quad Z = \sqrt{\frac{L}{C}} \approx 450 \text{ } \Omega$$

Quantum fluctuations of the transmon qubit

$$\Phi_{\text{ZPF}}/\phi_0 \approx 0.5$$

$$Q_{\text{ZPF}}/2e \approx 1.0$$

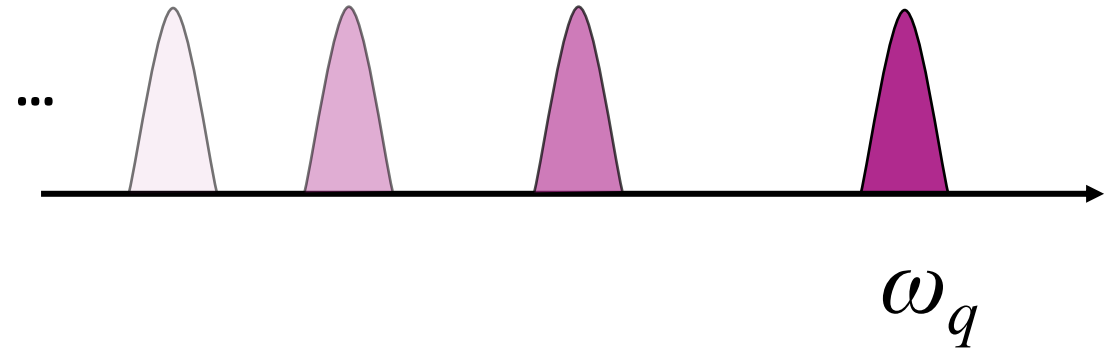


Energy diagram and transition spectrum

Energy levels

$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

Transition spectrum



The Transmon qubit: restricting Hilbert space

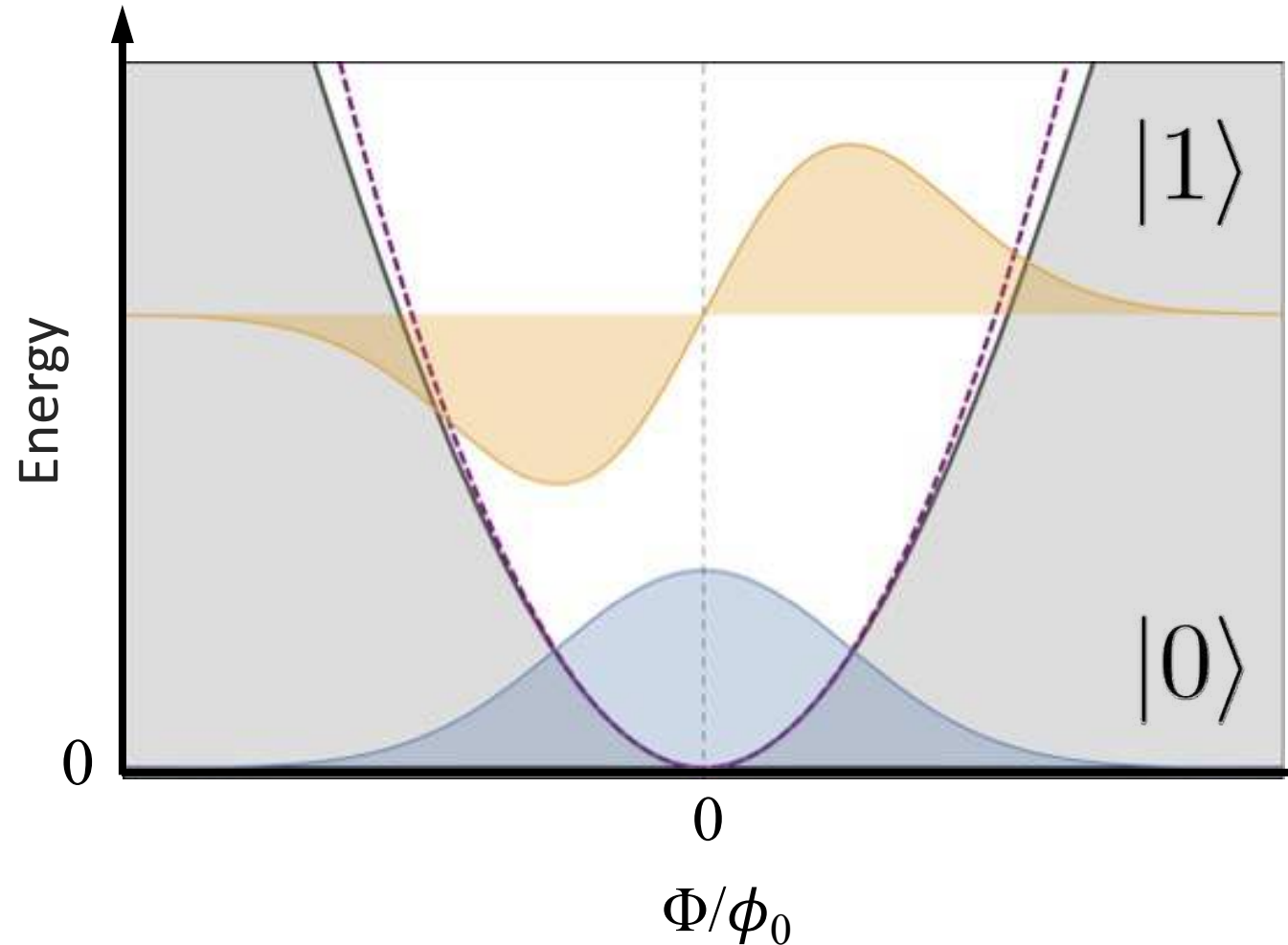
$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

Restrict to qubit subspace of $|0\rangle$ and $|1\rangle$

$$\hat{a}^\dagger \hat{a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

$$\hat{a} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$



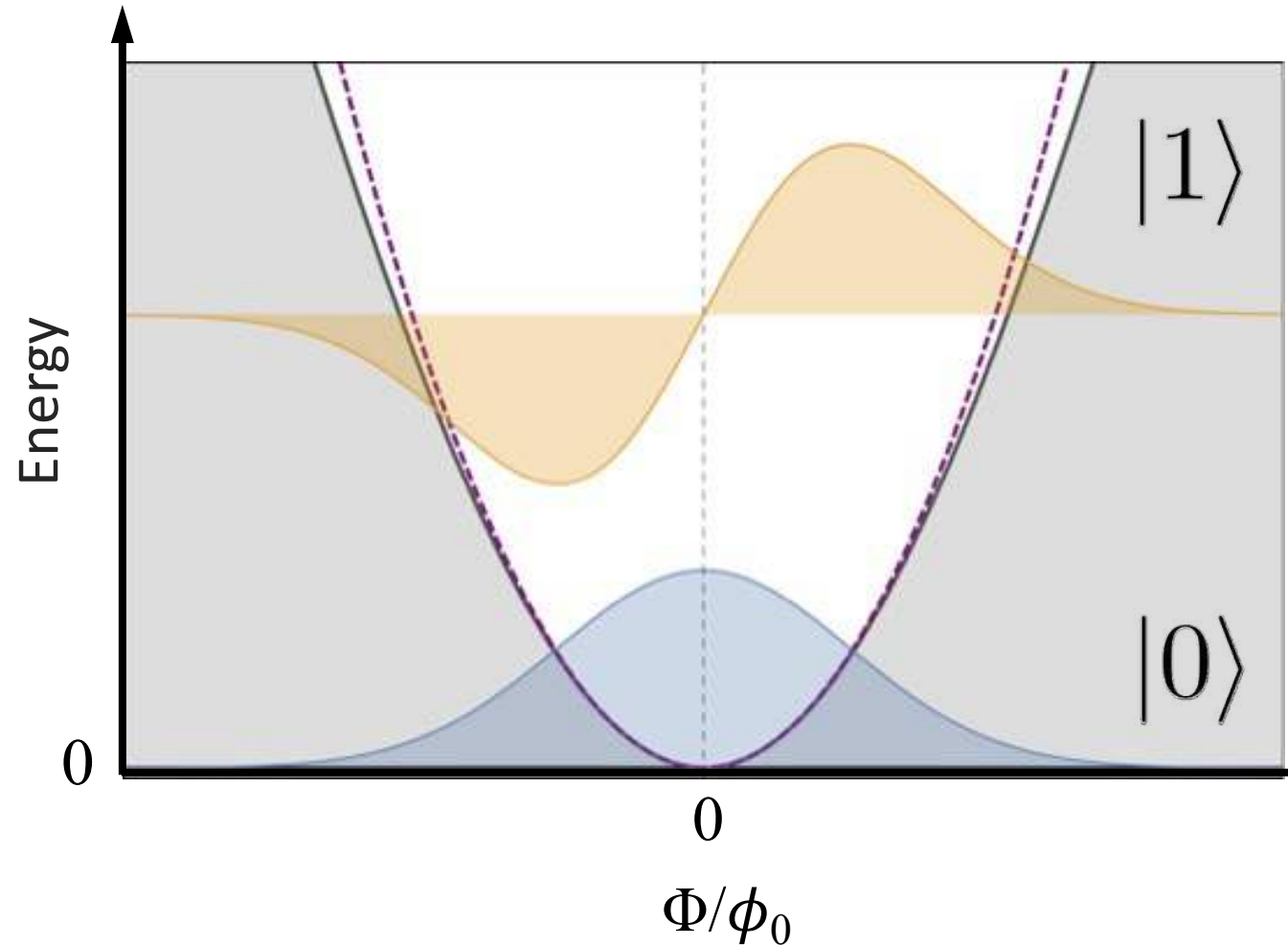
The Transmon qubit: restricting Hilbert space

$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

Restrict to qubit subspace of $|0\rangle$ and $|1\rangle$

$$\begin{array}{ccc} \left(\hat{N} - \frac{1}{2} \hat{I} \right) & \mapsto & -\frac{1}{2} \hat{Z} \quad \hat{a} \mapsto \hat{\sigma}_- = \frac{1}{2} (\hat{X} - i\hat{Y}) \\ \uparrow & & \uparrow \qquad \qquad \uparrow \qquad \uparrow \\ \text{Fock number} & \text{Qubit Pauli Z} & \text{Qubit Pauli X and Y} \\ \text{operator} & \text{operator} & \text{operators} \\ \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} & \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \end{array}$$



Transmon Qubit CPB Hamiltonian Charge Basis Demo

localhost:8888/notebooks/tutorials/Appendix/Quick Topic Tutorials Notebooks/Transmon Qubit CPB Hamiltonian Charge Basis Demo.ipynb#Experimental

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Trusted Python 3

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Modeling transmon qubit Cooper-pair box Hamiltonian in the charge basis

(Zlatko Mineev, Christopher Warren, Nick Lanzillo 2021)

This module models the transmon qubit in the cooper-pair charge basis, assuming wrapped junction phase variable. The Hamiltonian is given by:

$$\hat{H} = 4E_C (\hat{n}_I - n_g) - E_J \cos(\hat{\phi}),$$

where E_C is the charging energy, E_J is the Josephson energy, $\hat{n}$ is the number of Cooper pairs transferred between charge islands, $\hat{\phi}$ is the gauge-phase difference between charge islands, and n_g is effective offset charge of the device. Expressions for the charging energy, Josephson energy and charge can be written as:

$$E_C = \frac{e^2}{2C_\Sigma}, \quad n_g = -\frac{C_d \Phi_s(t)}{2e}, \quad E_J = \frac{\phi_0^2}{L_J},$$

where $C_\Sigma = C_J + C_B + C_g$ (the sum of the Josephson capacitance, shunting capacitance and gate capacitance), L_J is the inductance of the Jose junction, and ϕ is the magnetic flux.

The variables are

$$\hat{\phi} \equiv \frac{\hat{\Phi}}{\phi_0}, \quad \hat{n} \equiv \frac{\hat{Q}}{2e},$$

Observe that $\hat{\phi}$ and $\hat{n}$ are both dimensiunless, and they obey the commutation relationship:

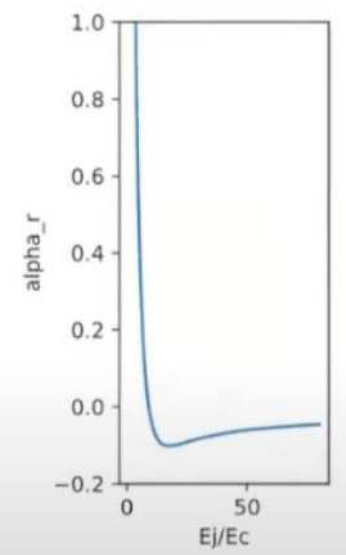
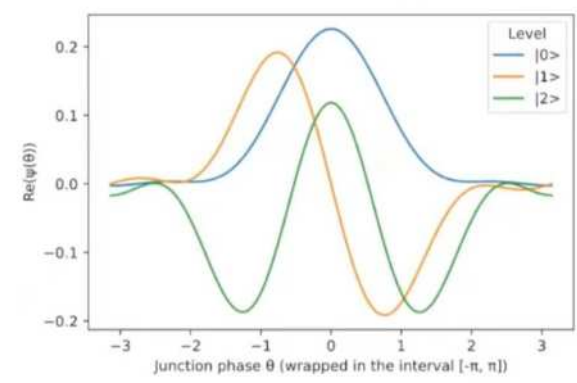
$$[\hat{\phi}, \hat{n}] = i$$

The Hamiltonian can be written in the charge ($\hat{n}$) basis as:

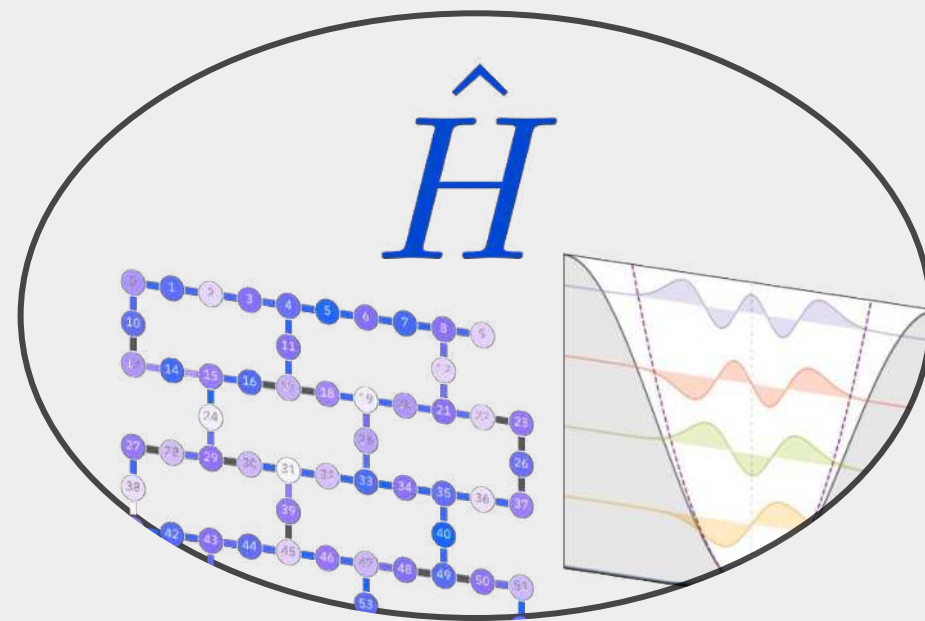
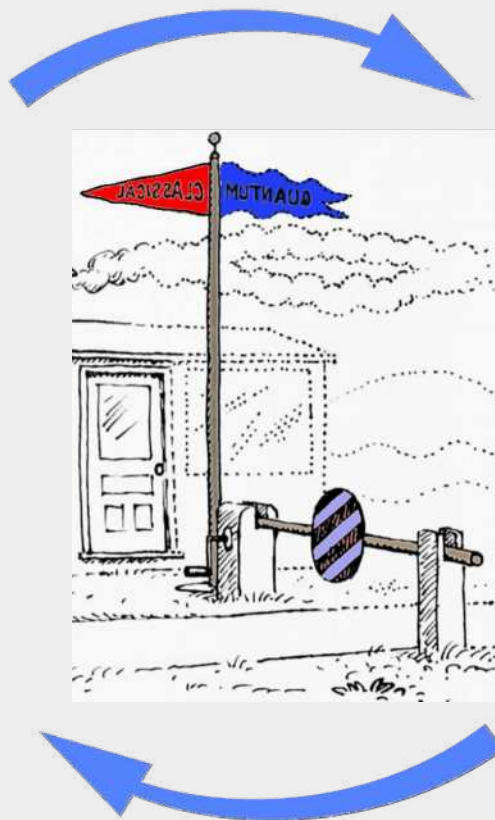
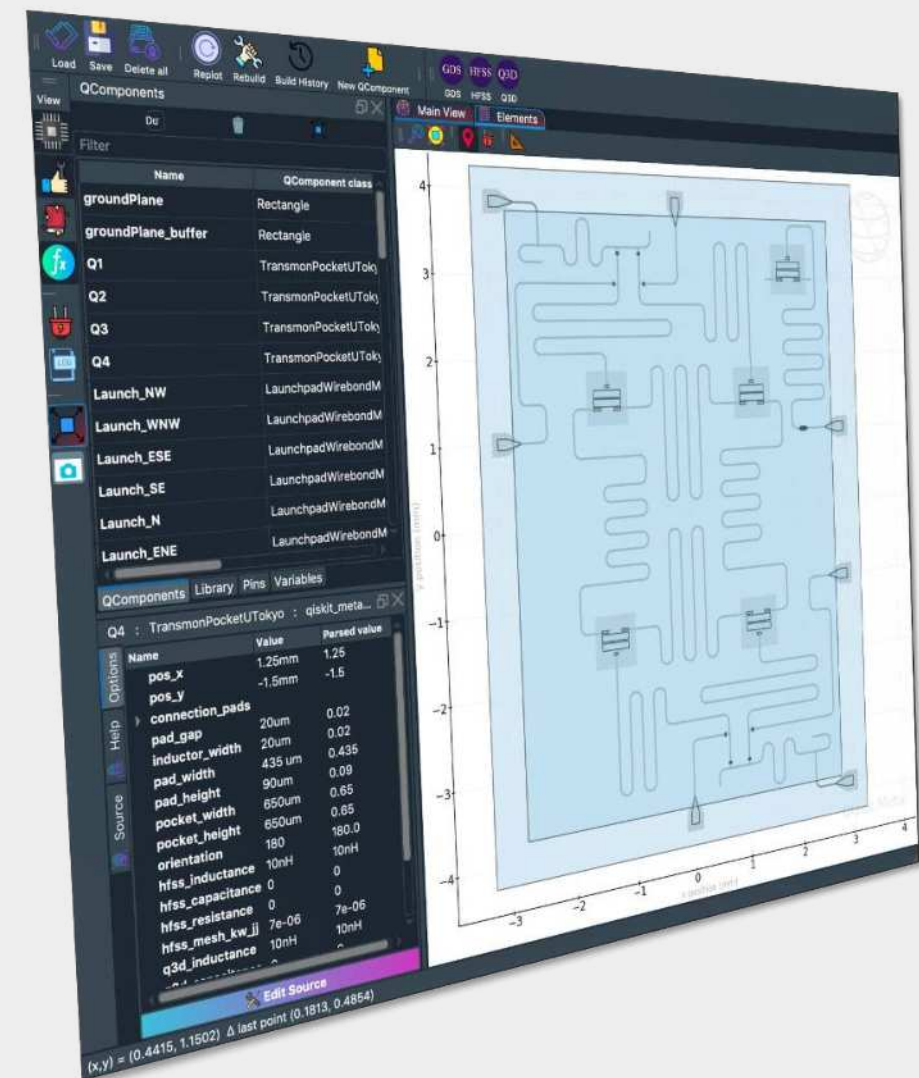
$$H = 4E_C (\hat{n} - n_g)^2 - \frac{1}{2} E_J \sum_n (|n\rangle \langle n+1| + \text{h.c.}),$$

Where $\hat{n} = \sum_n |n\rangle \langle n|$

```
import matplotlib.pyplot as plt
for k in range(3):
    psi, theta = H.psi_k(k,101)
    plt.plot(theta,psi.real+psi.imag, label=f'|{k}>')
plt.xlabel("Junction phase theta (wrapped in the interval [-pi, pi])")
plt.ylabel("Re(psi(theta))")
plt.legend(title="Level")
```



Make easy?



Docs & tutorials
qiskit.org/documentation/metal

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YouTube – see docs

Slack
[#metal](#) (qiskit workspace)

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You are here 😊 - or check slack

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Qiskit Metal | Quantum Device Design & Analysis (Q-EDA) 0.0.3

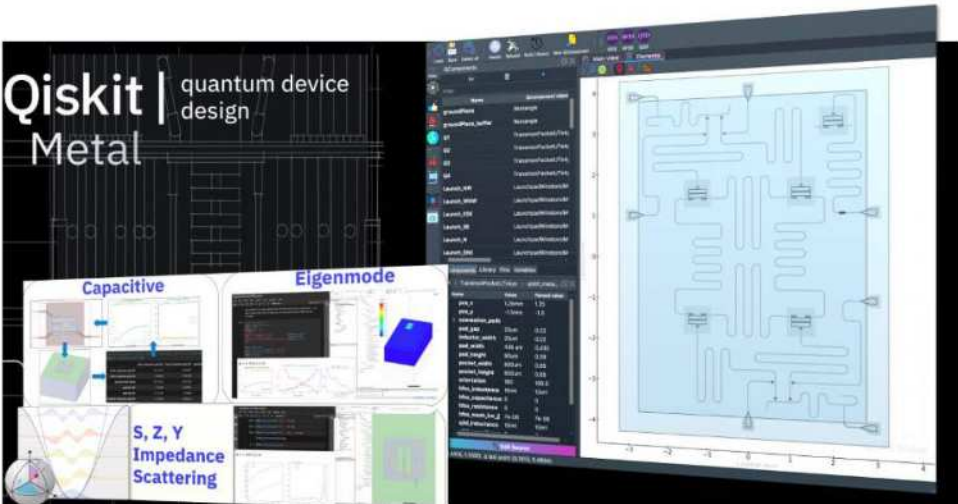
ATTENTION

This is an alpha version of Qiskit Metal, the code is still under development. Please let us know about anything you might want us to add or elaborate upon in the Slack channel [#metal](#) in the [qiskit workspace](#).

HINT

You can open this documentation using

```
import qiskit_metal
qiskit_metal.open_docs()
```



About

Qiskit Metal | for quantum device design & analysis ('Qiskit Metal') is an open-source framework (and library) for the design of superconducting quantum chips and devices. Call it quantum EDA (Q-EDA) and analysis. Qiskit Metal is:

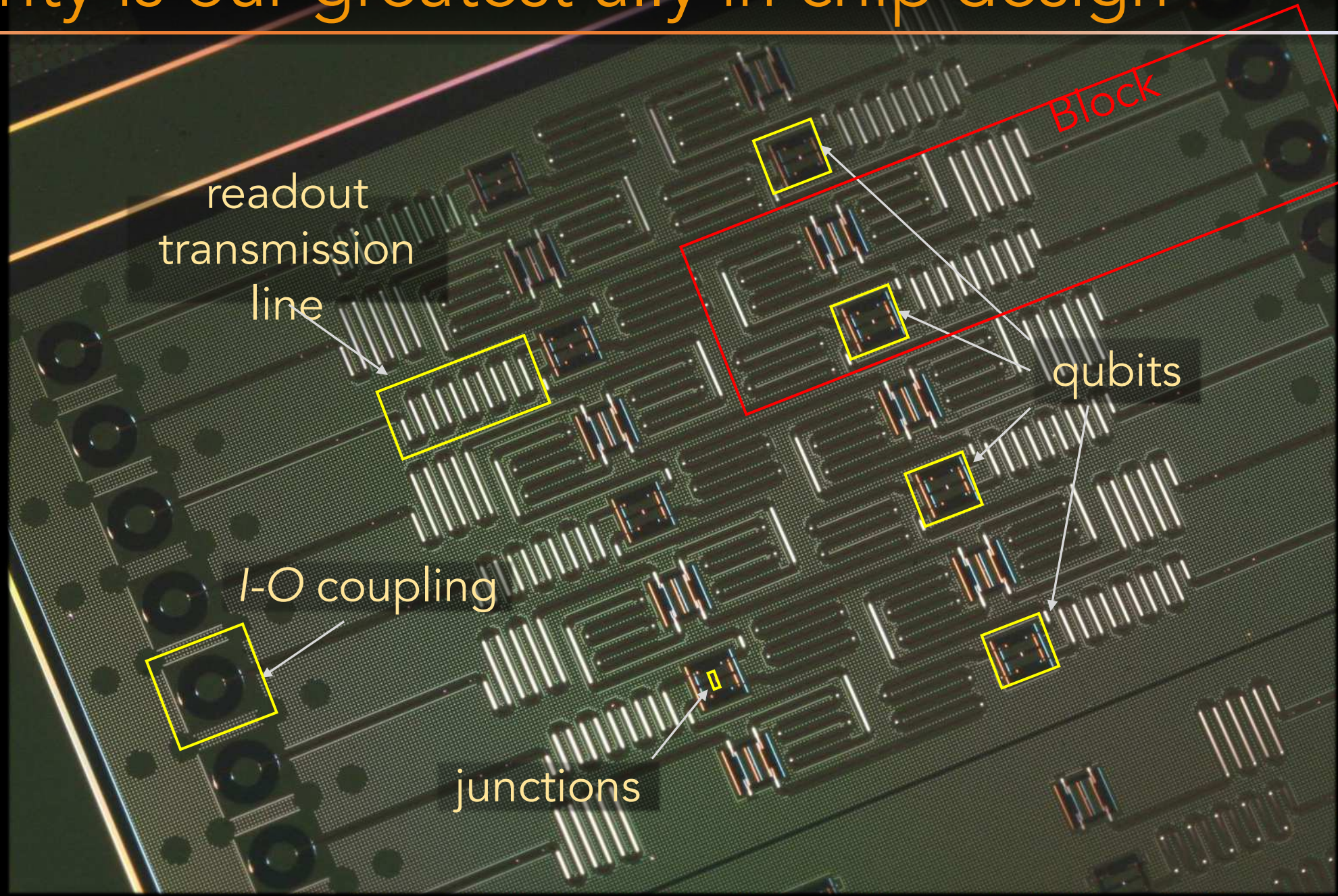
- Open source
- Community-driven
- A python API and a front-end visual GUI interface

Let's layout, analyze, and
optimize a four-qubit chip...

Open source
qiskit.org/metal

Regularity is our greatest ally in chip design

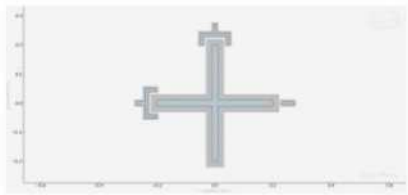
Reuse
Fine-tune
Automate
Extend



Device example library

Resonators

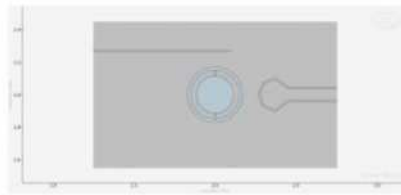
Qubits



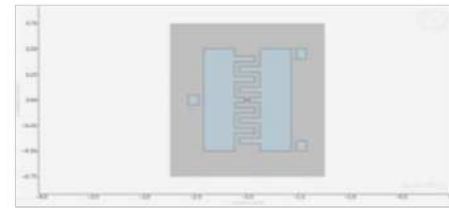
Single Transmon - Grounded (xmon)



Single Transmon - Floating



Concentric Transmon



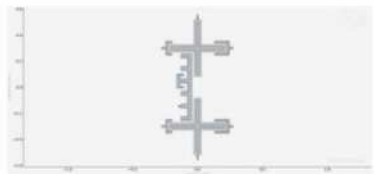
Interdigitated Transmon Qubits



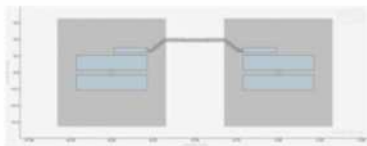
Simple CPW Meander

...

Composite systems / small chips / ...



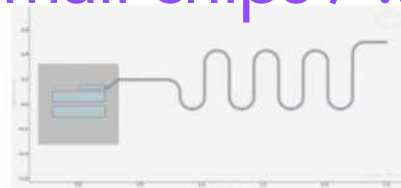
Tunable Coupler (MIT)



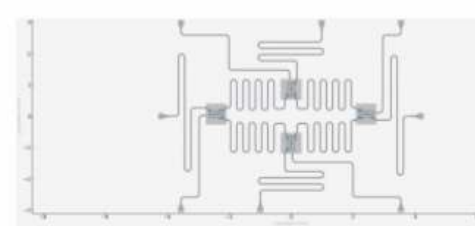
Direct Coupler (transmon-transmon)



Bus Resonator Coupler (transmon-transmon)



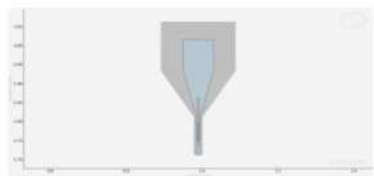
Transmon coupled to CPW resonator



Four Qubit Chip Design

...

Design flow examples



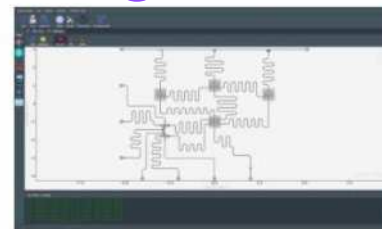
CPW Launch Pad



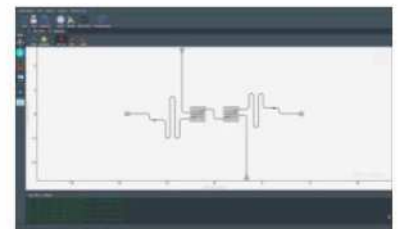
Readout line



Charge Line



Full Chip Design



South Korea Qiskit Hackathon



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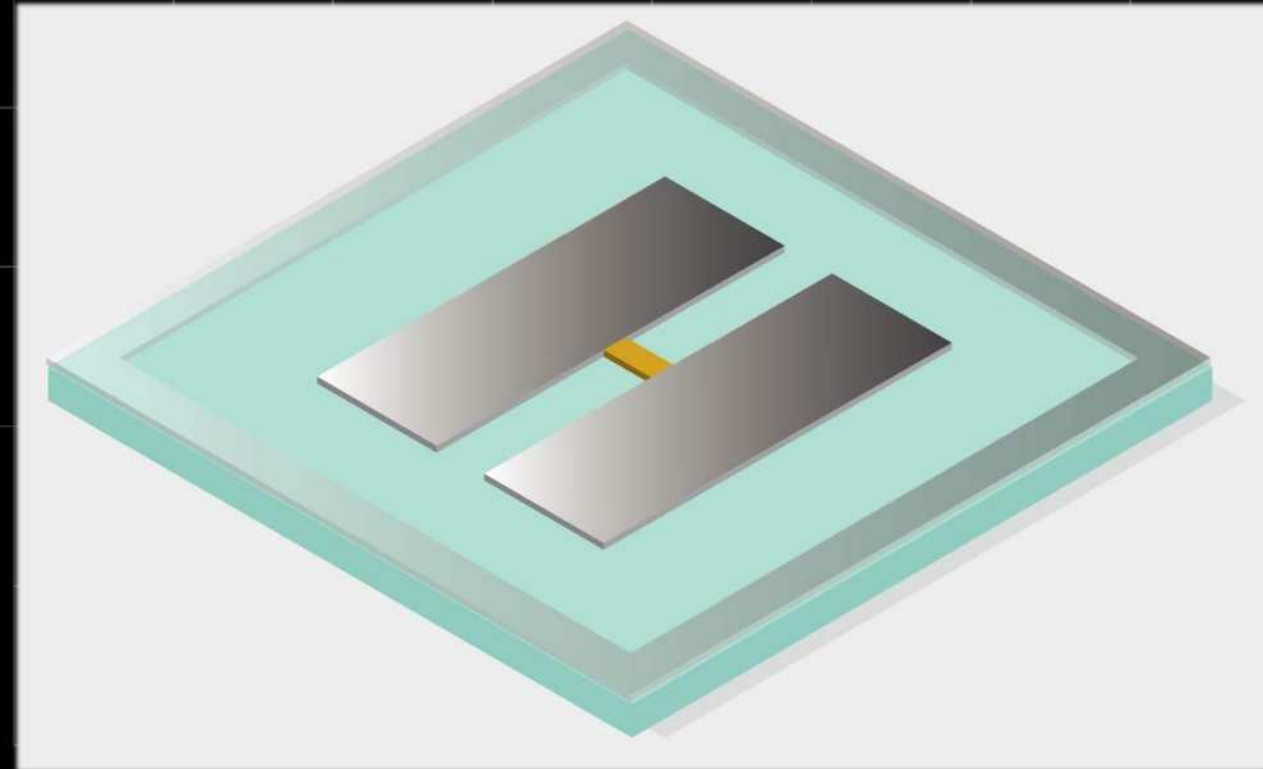
Quantum hardware design and analysis

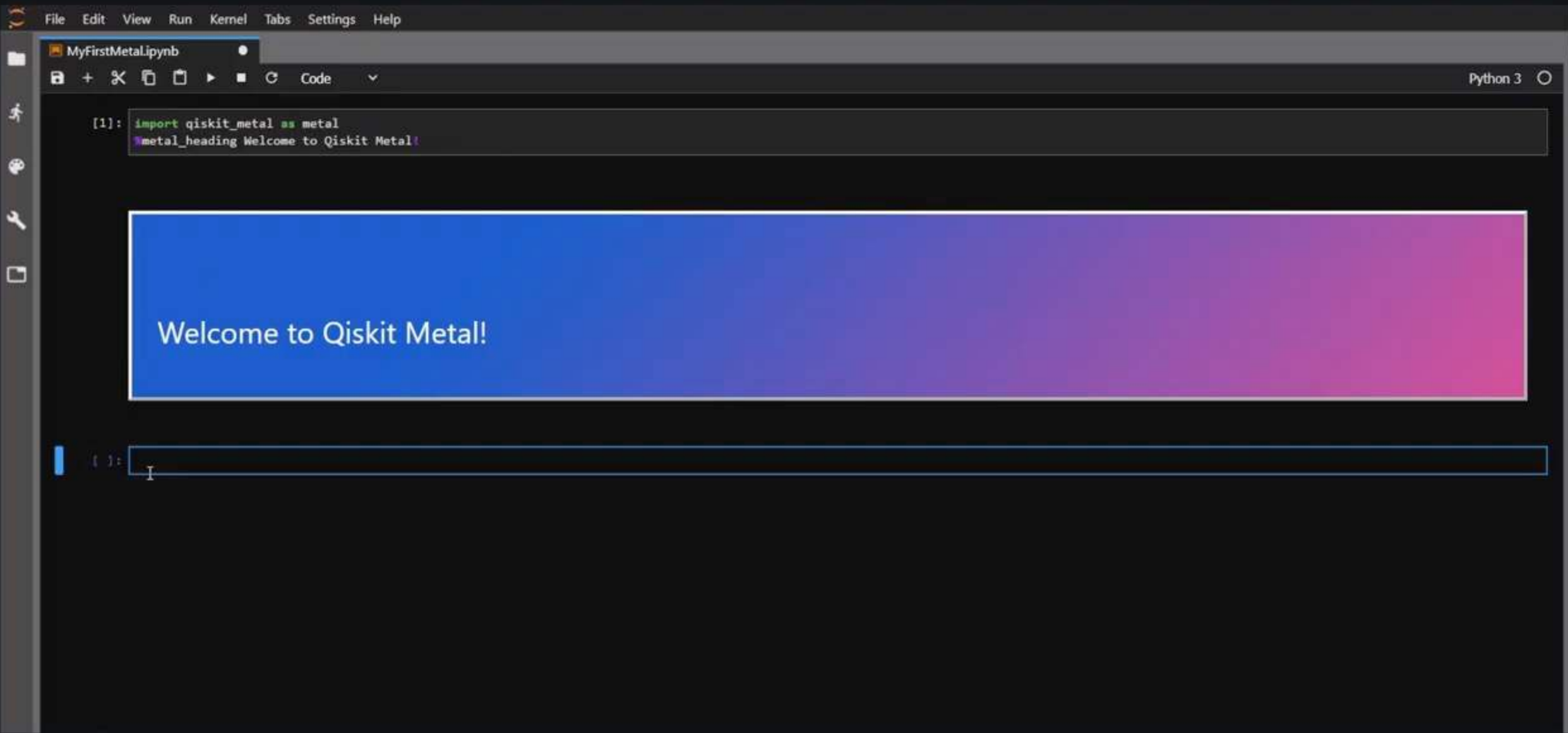


Create a transmon qubit

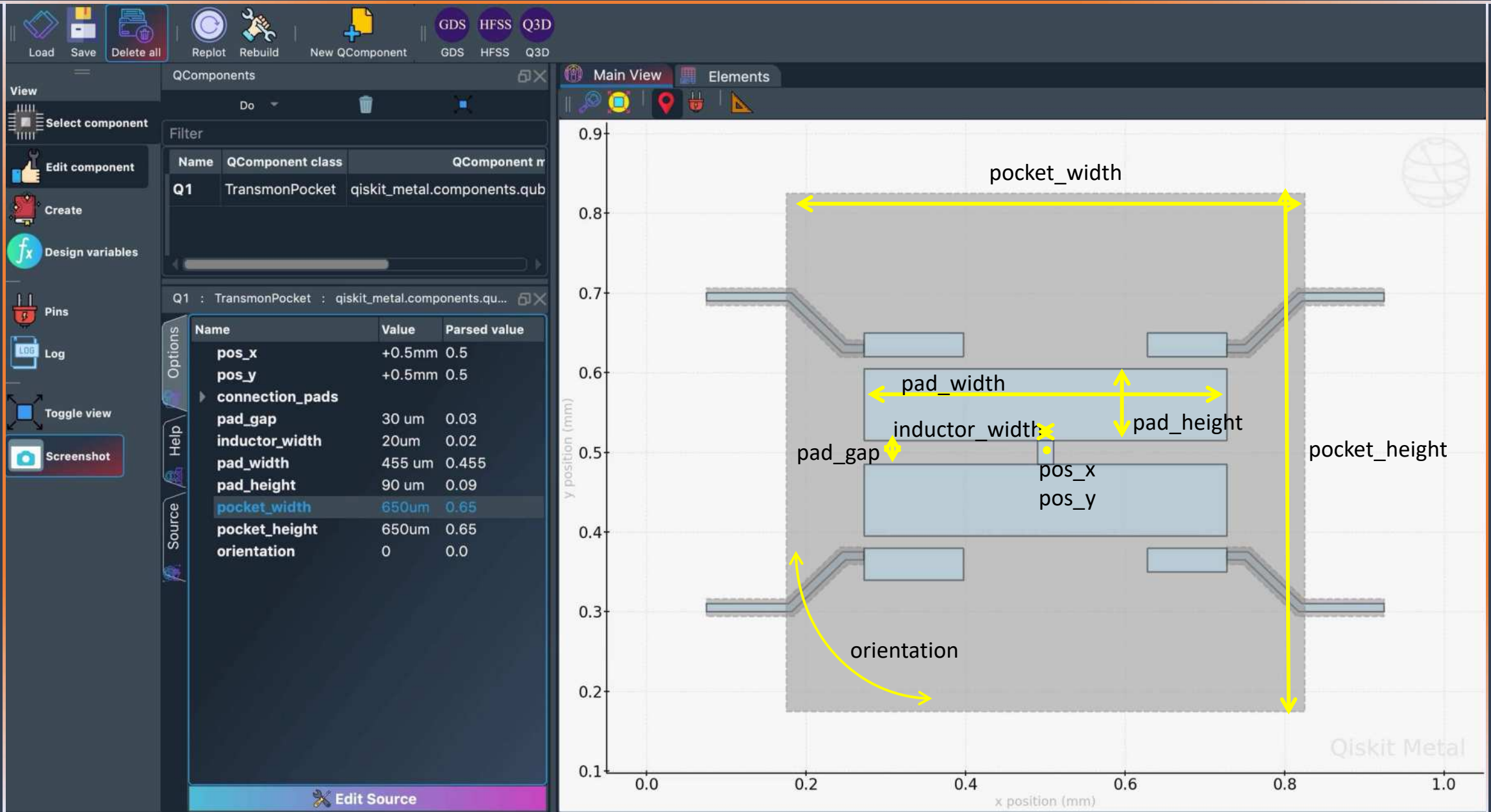
```
from qiskit_metal.qlibrary import qubits  
q1 = qubits.TransmonPocket('Q1', options=dict(...))
```

```
from qiskit_metal import MetalGUI  
MetalGUI()
```

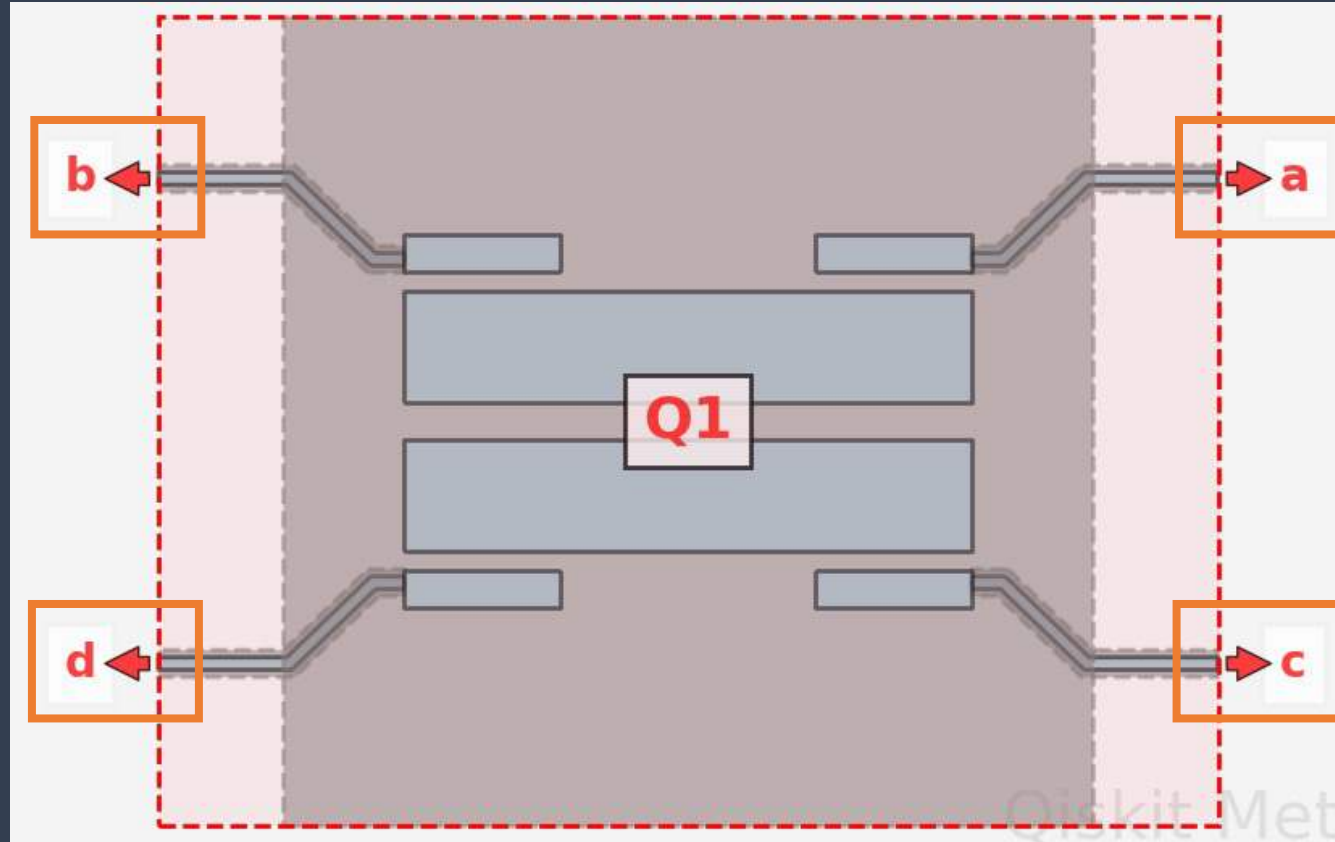


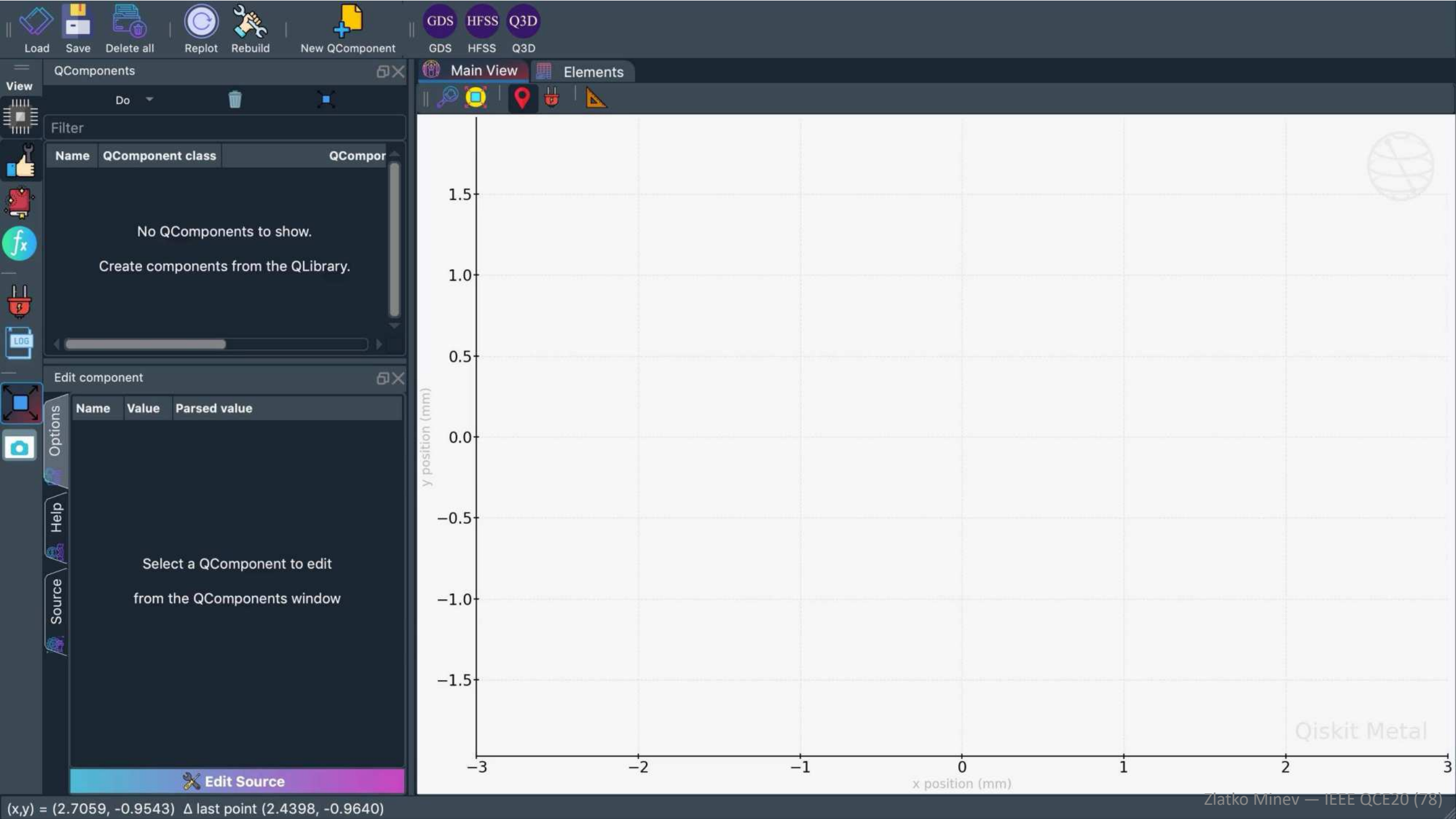


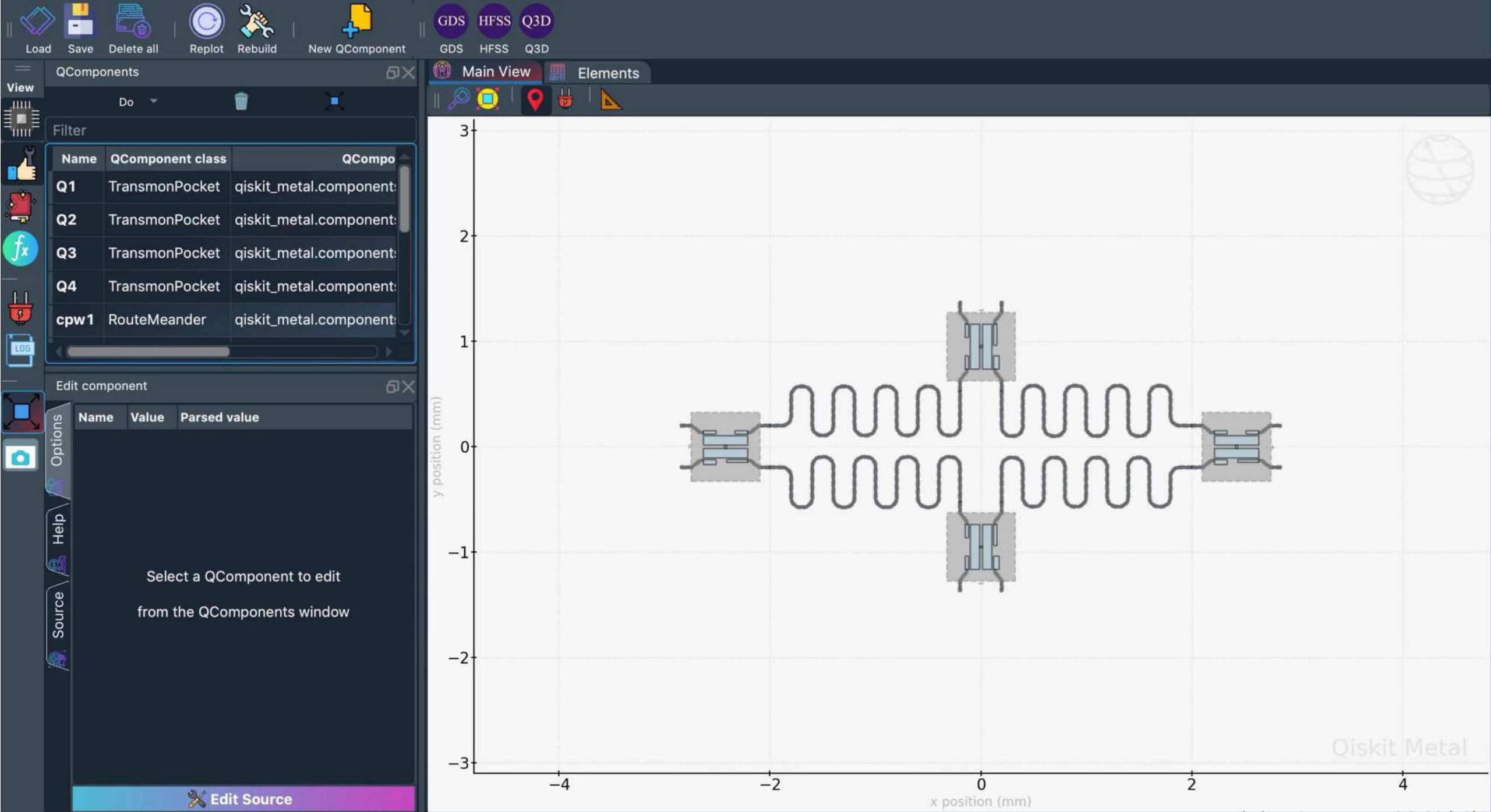
Fine-tune and automate parameters



Dynamically –connected quantum devices







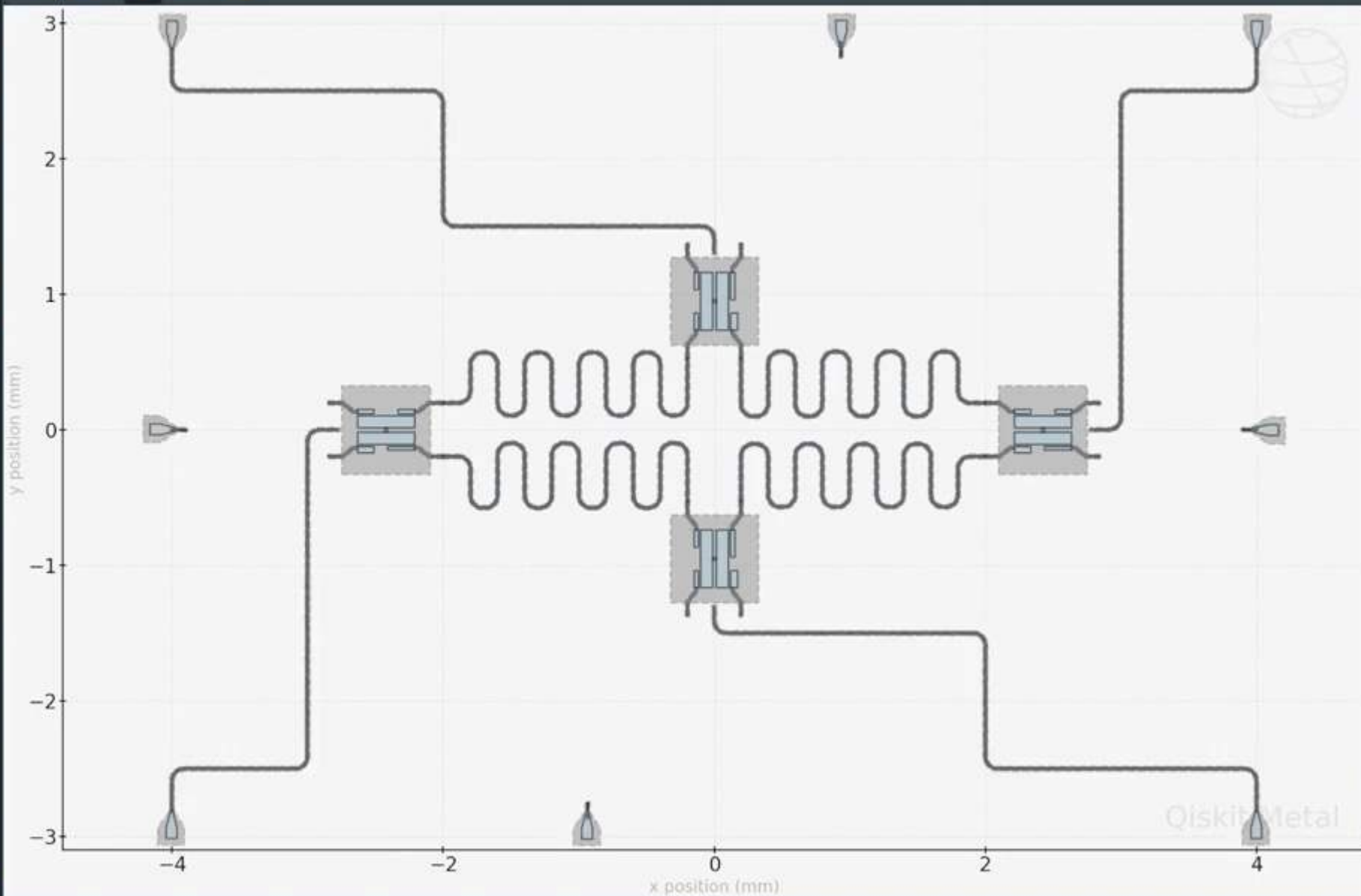
Filter

Name	QComponent class	QCompo
Q1	TransmonPocket	qiskit_metal.component:
Q2	TransmonPocket	qiskit_metal.component:
Q3	TransmonPocket	qiskit_metal.component:
Q4	TransmonPocket	qiskit_metal.component:
cpw1	RouteMeander	qiskit_metal.component:

Edit component

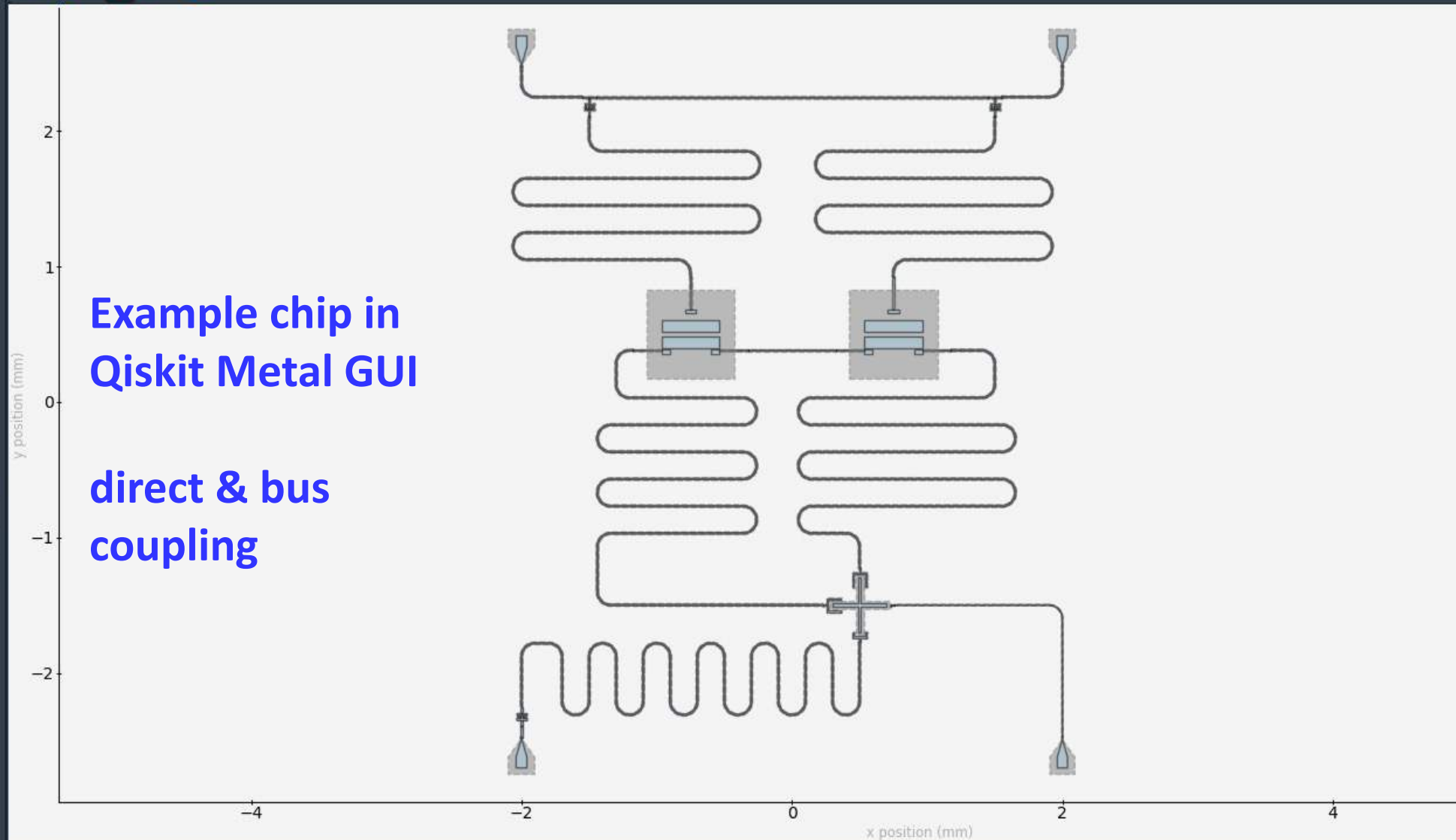
Name	Value	Parsed value
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Edit Source



Example chip in
Qiskit Metal GUI

direct & bus
coupling



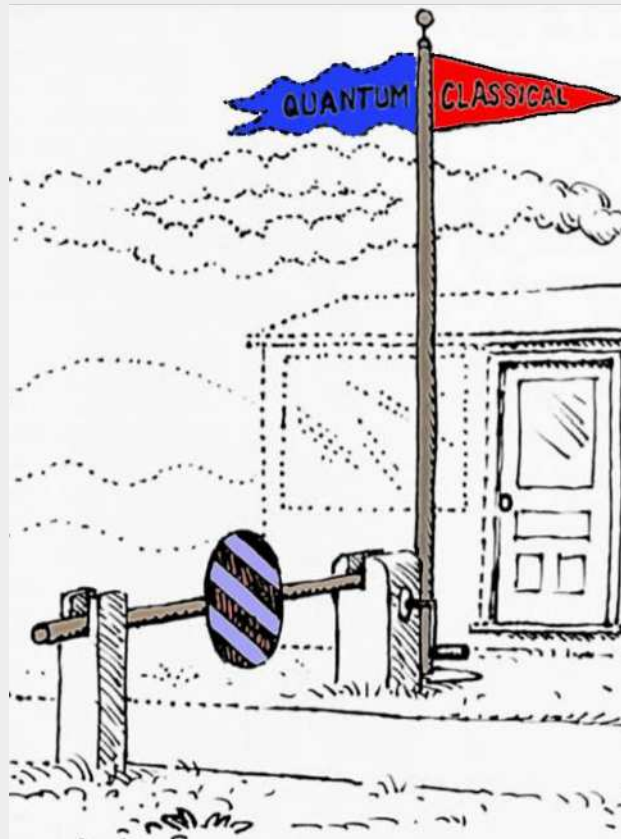
Can I make my own qubit?

```
smile = draw.shapely.geometry.Point(0, 0).buffer(0.8)
cut_sq = draw.shapely.geometry.box(-1, -0.3, 1, 1)
smile = draw.subtract(smile, cut_sq)
face = draw.subtract(face, smile)
face = draw.subtract(face, eye_r)
face = draw.subtract(face, eye_l)
face
```

[7]:



Analysis



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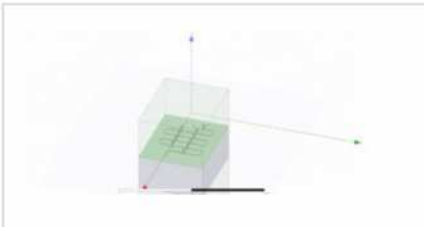
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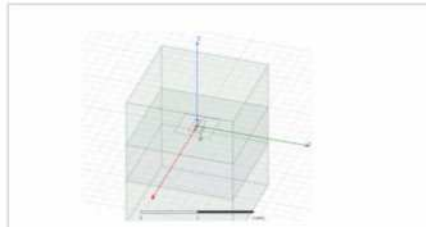
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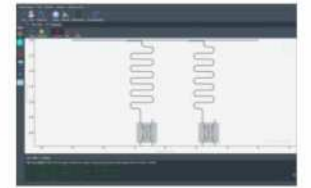
Eigenmode and EPR



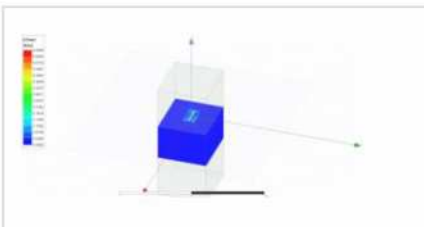
Impedance Analysis



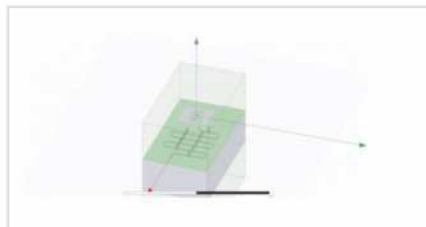
Capacitance Matrix and LOM Analysis



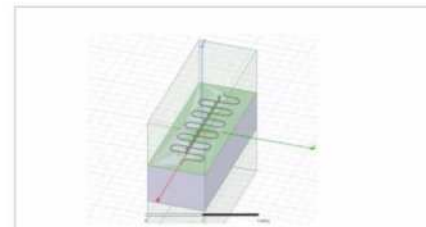
Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



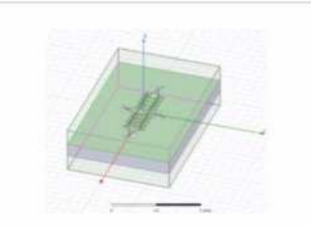
Analyzing and Tuning a Transmon Qubit



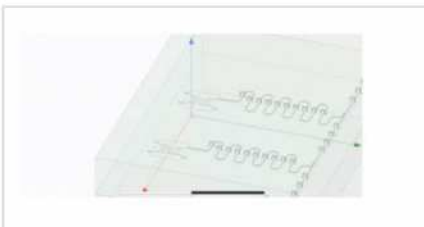
Analyzing a transmon and a resonator



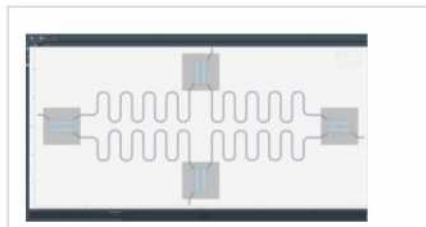
Analyzing a resonator



How to Render a Metal Design into Ansys



Driven Modal Simulation and S-Parameters



Sweep an option in QComponent, get capacitance matrix from Ansys Q3D and pyEPR



Sweep an option in QComponent, get Ansys HFSS eigenmode solution matrix.



How to Render a Metal Design into GDS

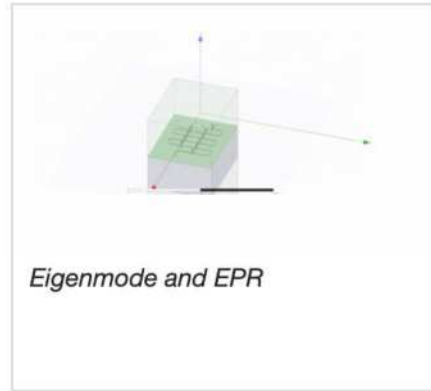
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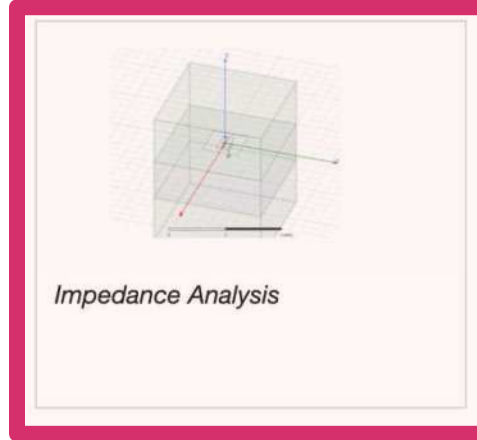
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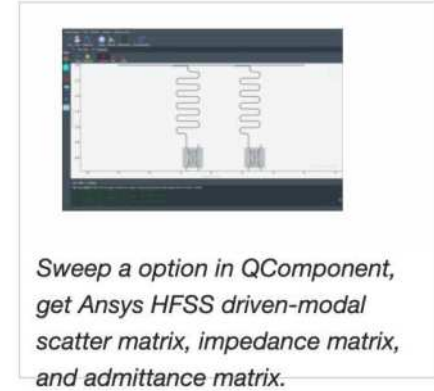
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Impedance Analysis



Capacitance Matrix and LOM Analysis

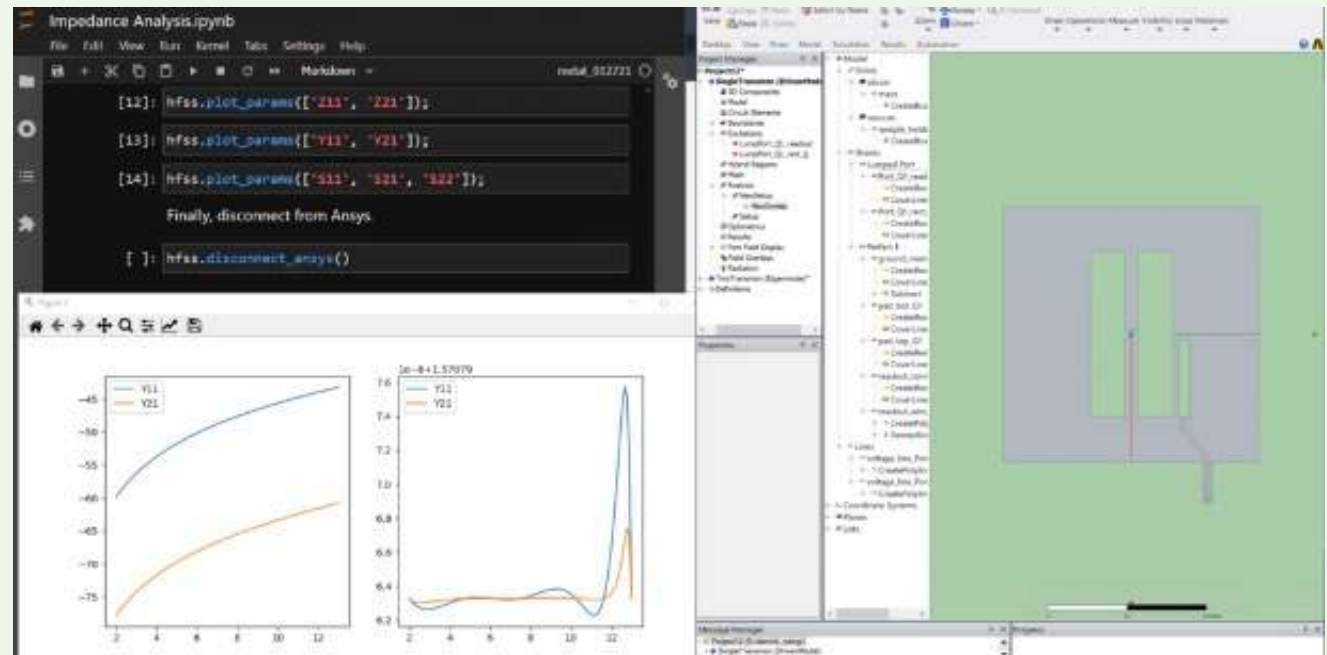


Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



S, Z, Y Impedance Scattering

arXiv:1204.0587 ...
quantum analysis part WIP



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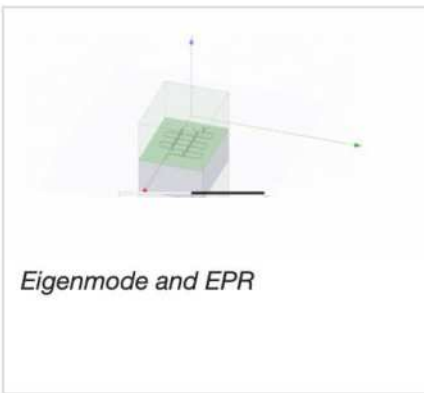
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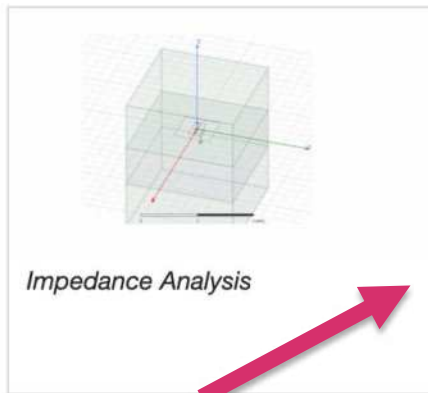
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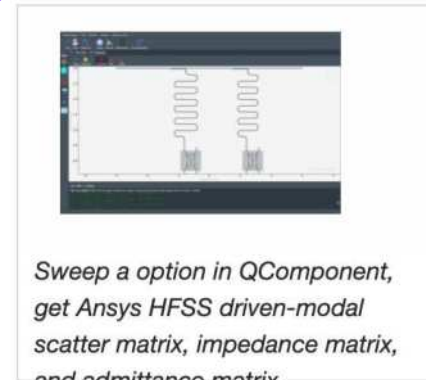
Eigenmode and EPR



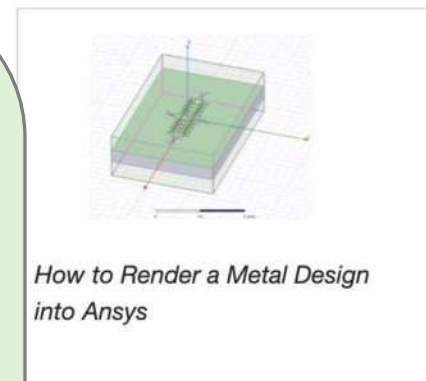
Impedance Analysis



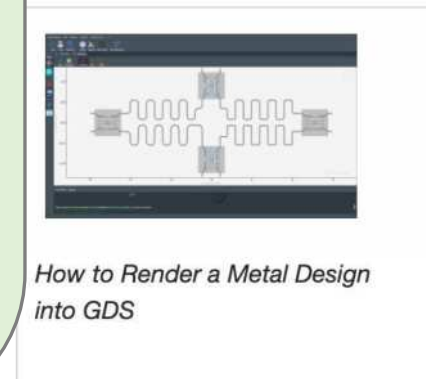
Capacitance Matrix and LOM Analysis



Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



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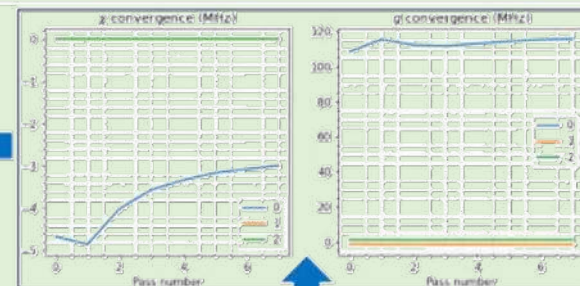
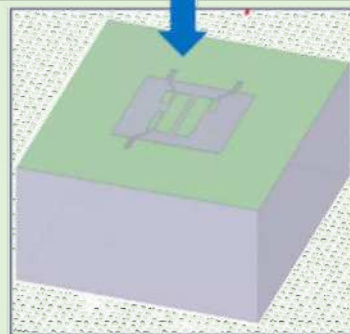
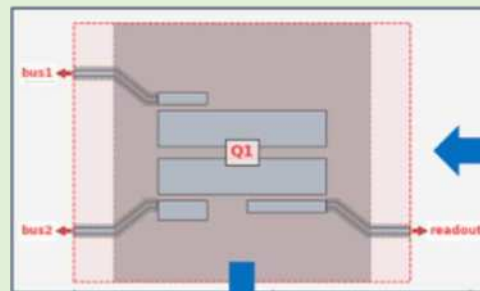


How to Render a Metal Design into GDS

Capacitive

Jay's code

arXiv:2103.10344 ...



```
# Using the analysis results, get capacitance matrix as a dataframe
q3d.get_capacitance_matrix()
```

	bus1_connector_pad_Q1	bus2_connector_pad_Q1	ground
bus1_connector_pad_Q1	47.71247	-0.38203	
bus2_connector_pad_Q1	-0.38203	51.80766	
ground_main_plane	-33.11632	-35.62303	
pad_bot_Q1	-1.34246	-12.58801	
pad_top_Q1	-12.12145	-1.59640	
readout_connector_pad_Q1	-0.17280	-0.96276	

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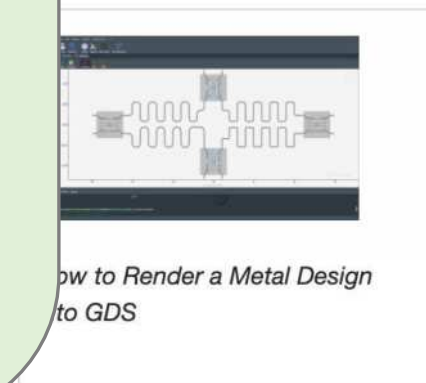
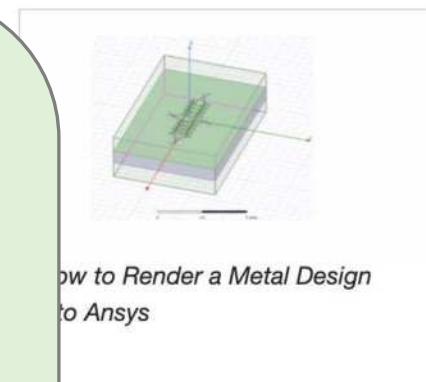
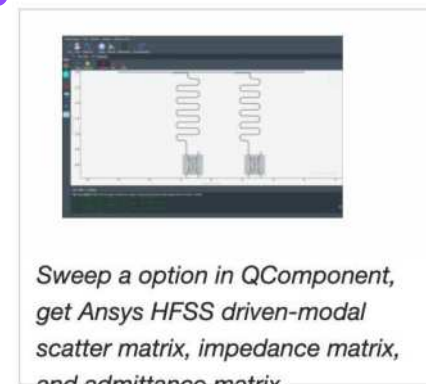
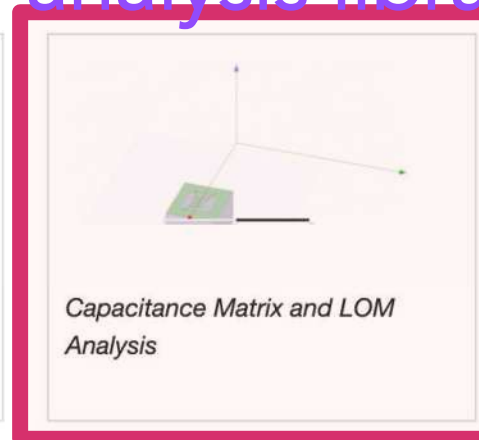
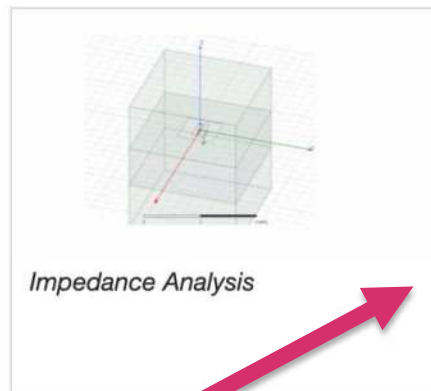
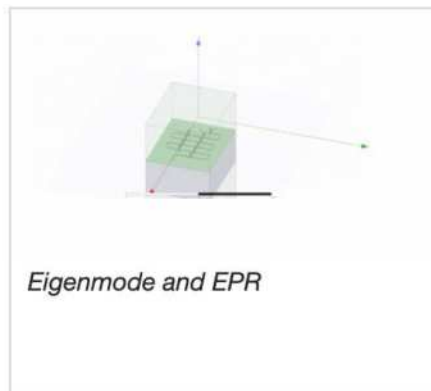
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Tutorials: Quantum analysis library



Cornell University

arXiv.org > quant-ph > arXiv:2103.10344

Quantum Physics

Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Minev,^{*} Thomas G. McConkey, Maika Takita, Antonio Corcoles, and Jay M. Gambetta
IBM Quantum, IBM T.J. Watson Research Center, Yorktown Heights, US

WIP: General capacitive analysis code
map to known building blocks: e.g.,
transmon, fluxonium, zero-pi, etc.

Resonators
Composite Bi-Partite Systems
Qubit Couplers
Input-Output Coupling
Small Quantum Chips
Design Flow

Libraries
All Quantum Devices

API References

Overview

QD

Tutorials: Quantum analysis library

Eigenmode and EPR

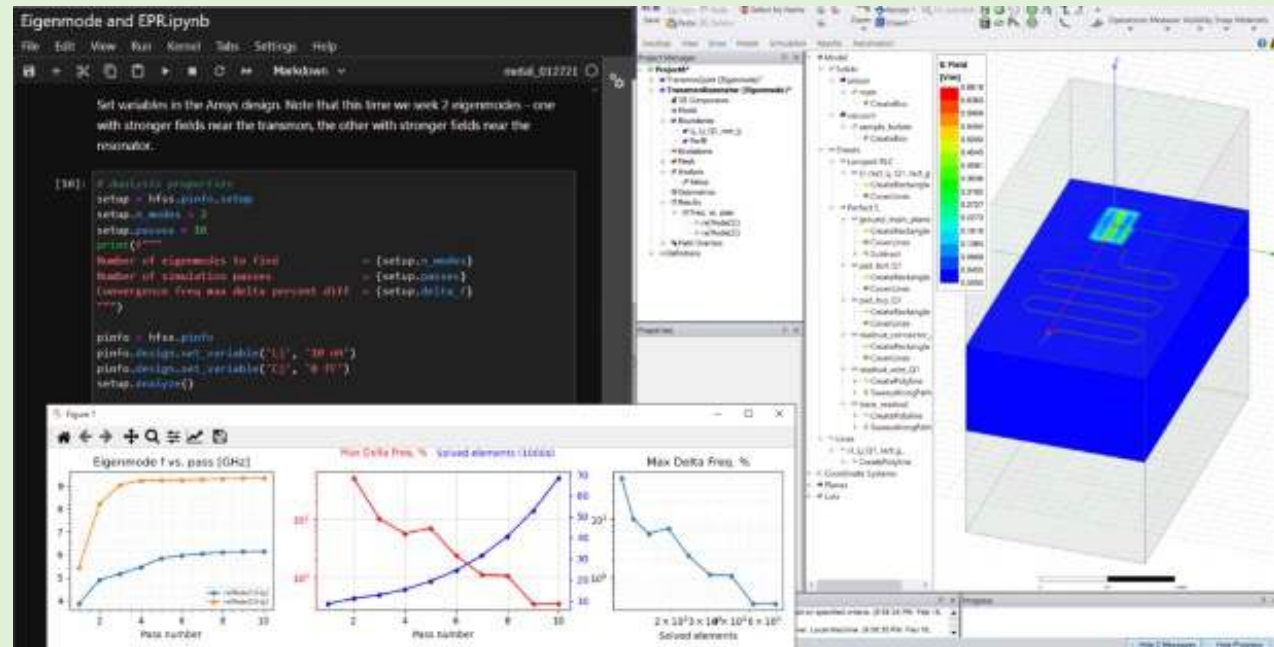
Impedance Analysis

Capacitance Matrix and LOM
Analysis

Sweep a option in QComponent,
get Ansys HFSS driven-modal
scatter matrix, impedance matrix,
and admittance matrix.

Eigenmode

arXiv: 2010.00620
arXiv: 1902.10355

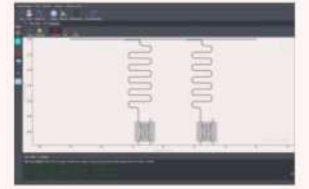


Tutorials: Quantum analysis library

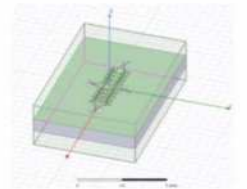
Sweeps

with

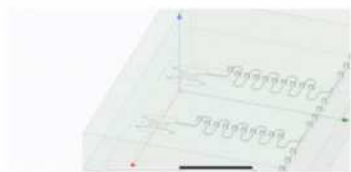
Z S Y impedance, scattering params
eigenmode / EPR
capacitive / lumped model



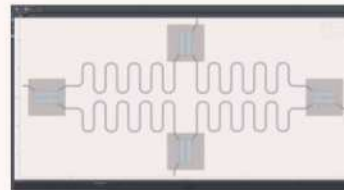
Sweep a option in QComponent,
get Ansys HFSS driven-modal
scatter matrix, impedance matrix,
and admittance matrix.



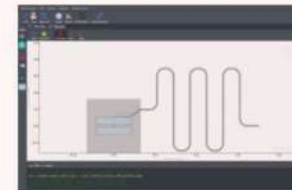
How to Render a Metal Design
into Ansys



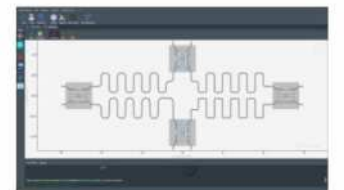
Driven Modal Simulation and S-
Parameters



Sweep an option in QComponent,
get capacitance matrix from Ansys
Q3D and pyEPR

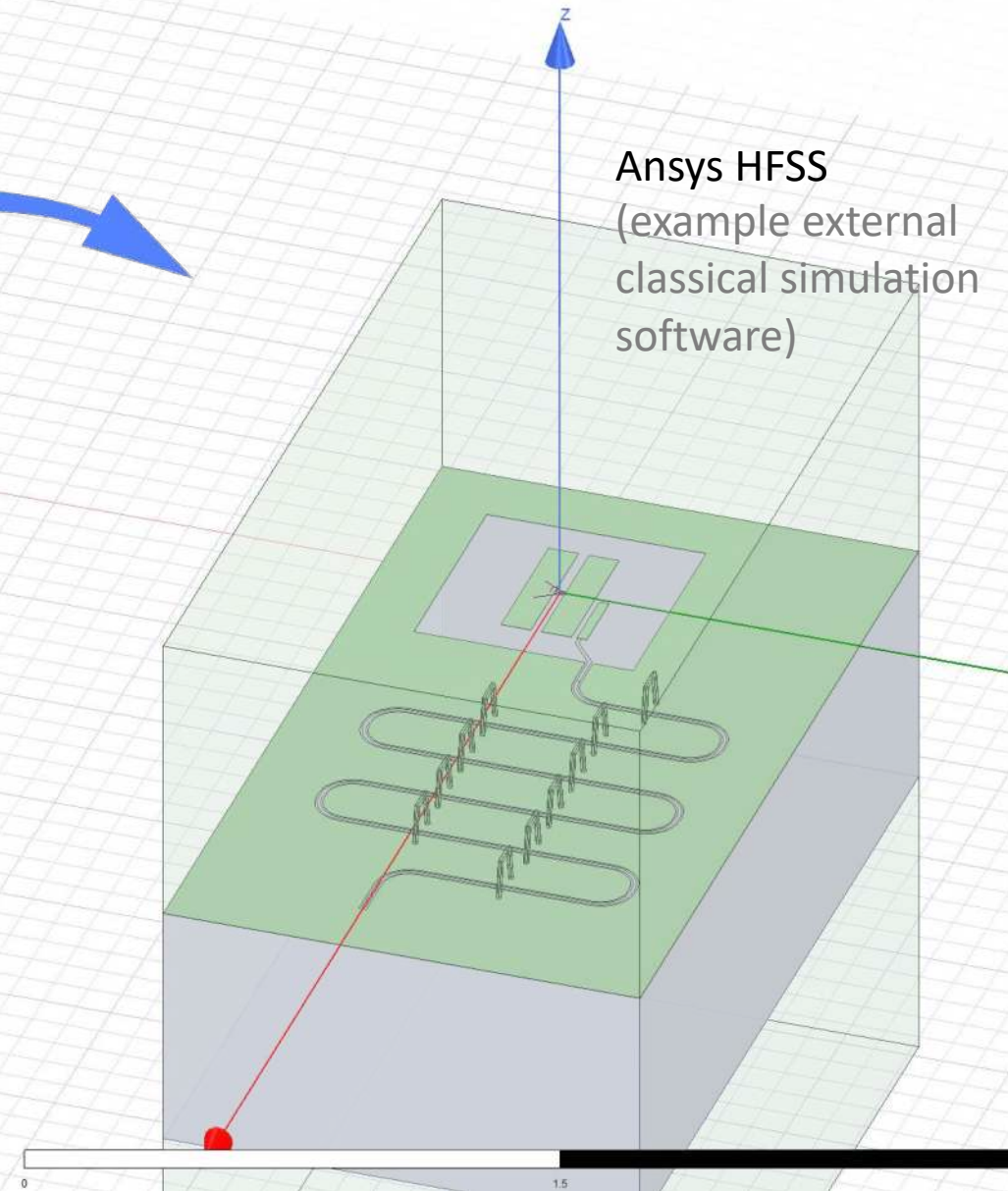
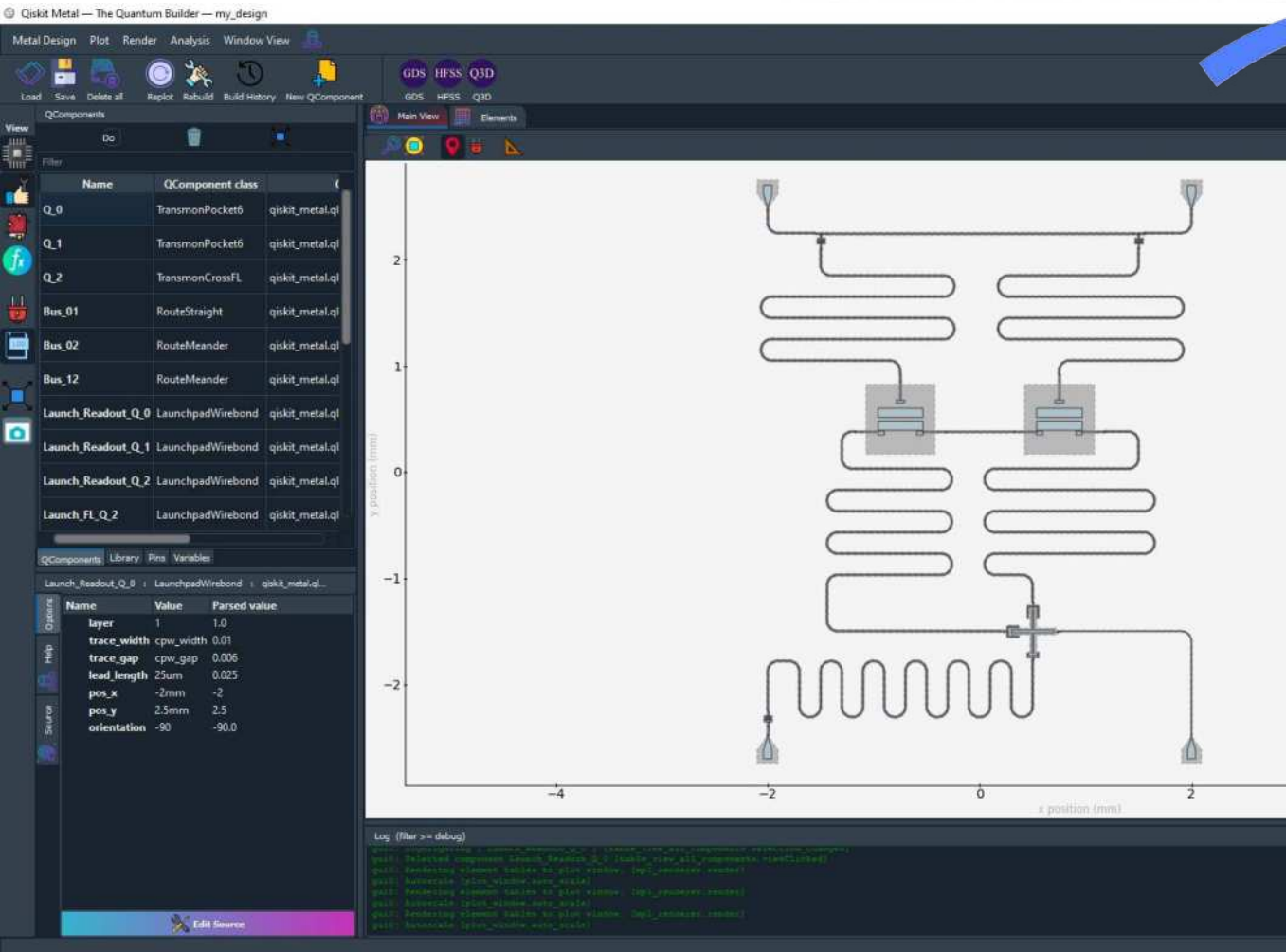


Sweep an option in QComponent,
get Ansys HFSS eigenmode
solution matrix.

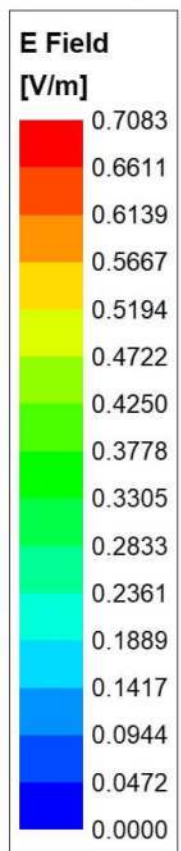


How to Render a Metal Design
into GDS

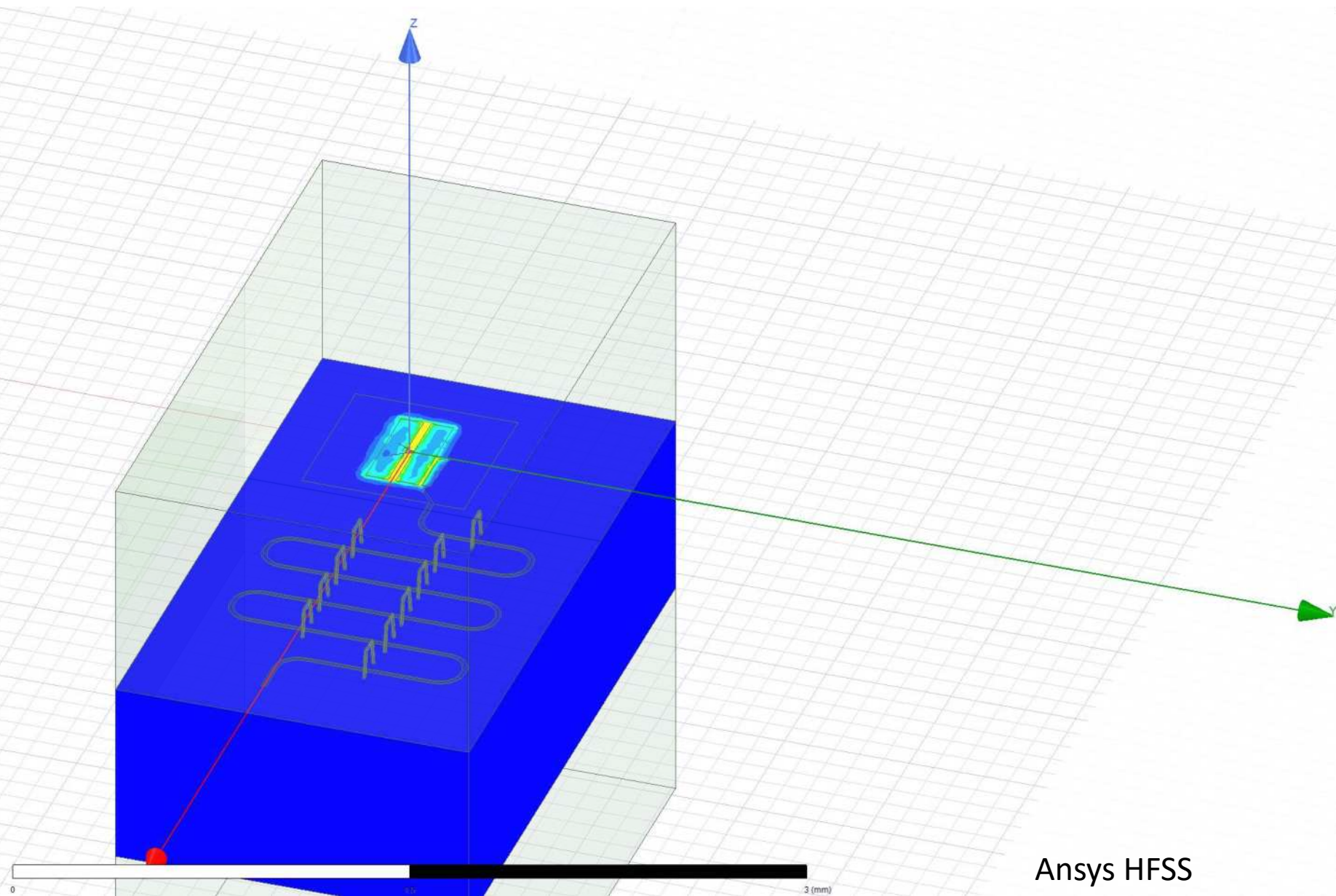
Automated with
Qiskit | quantum device design



Ansyes HFSS
(example external
classical simulation
software)

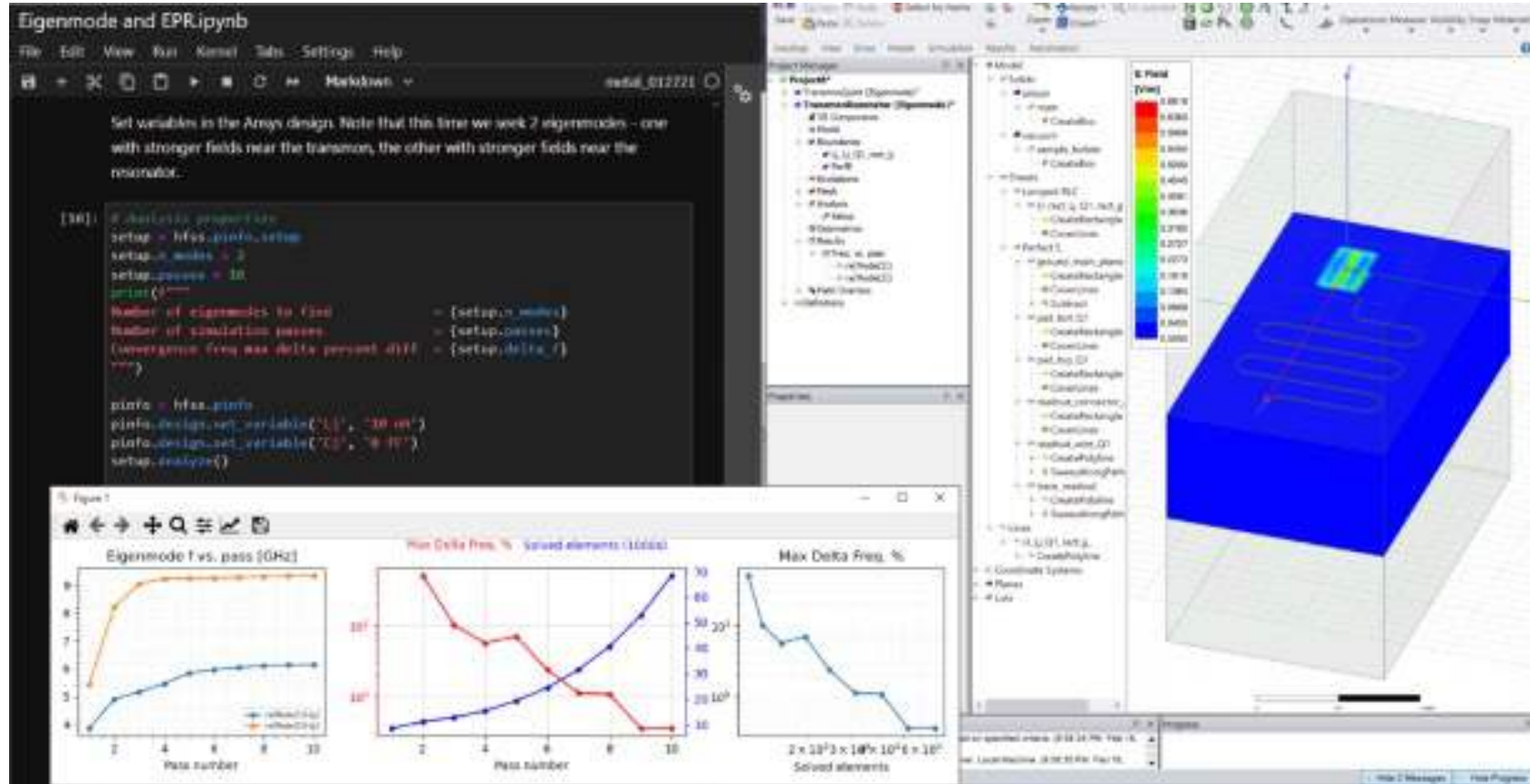


`.analyze('Q1')`



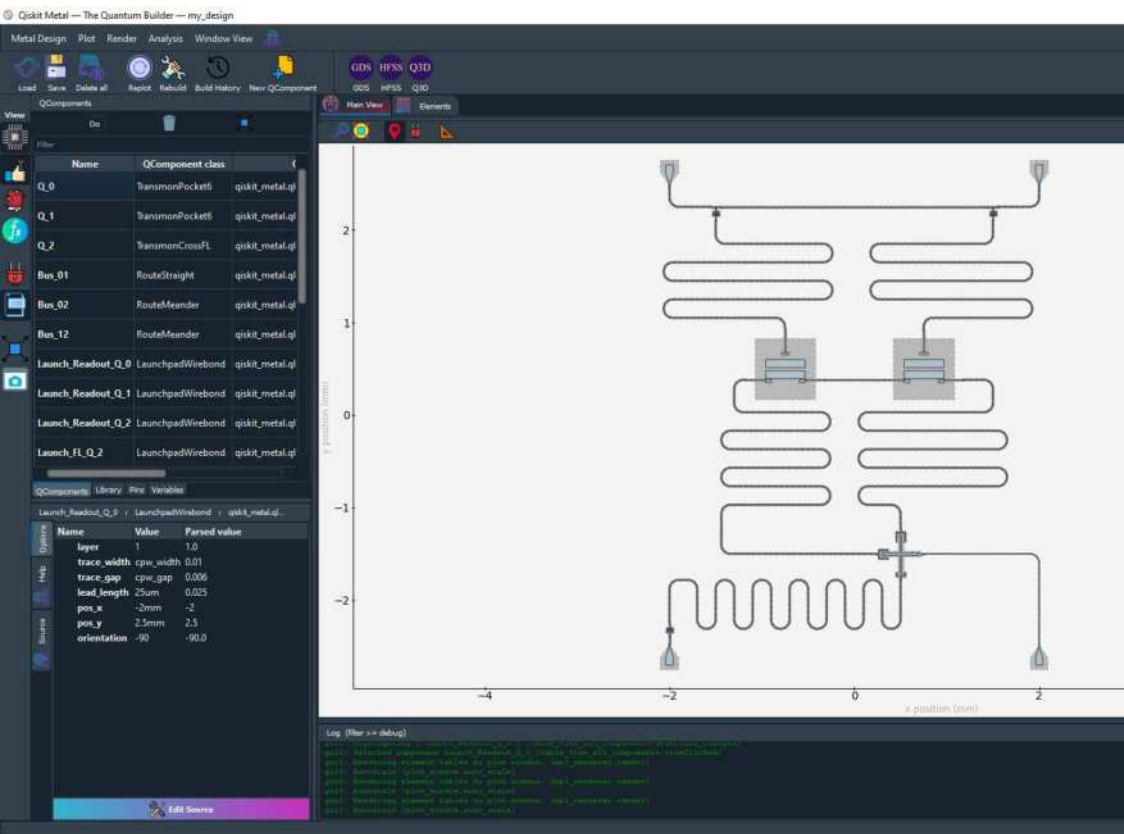
Ansys HFSS

```
metal.analysis.lumped_model.analyze('Q1')
```



Capacitive simulation

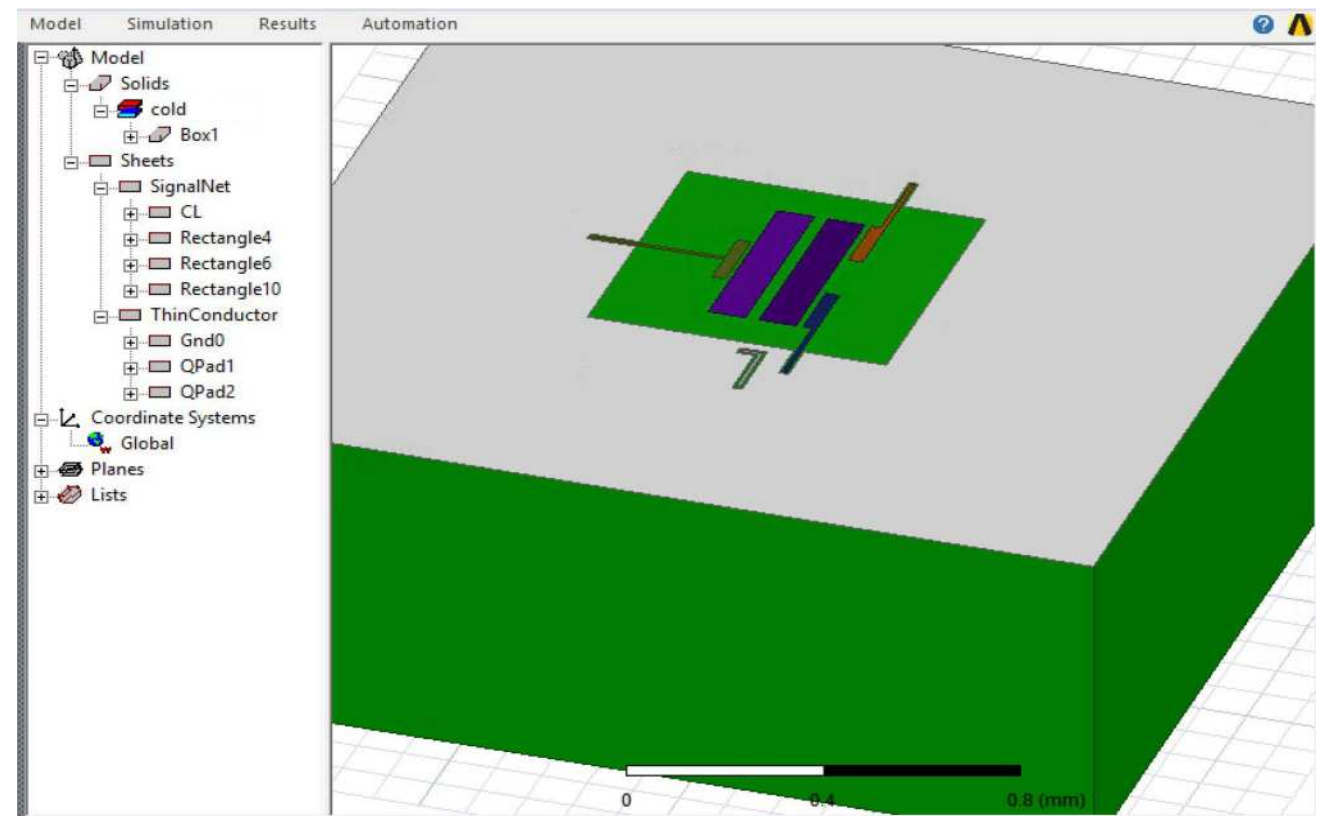
device layout



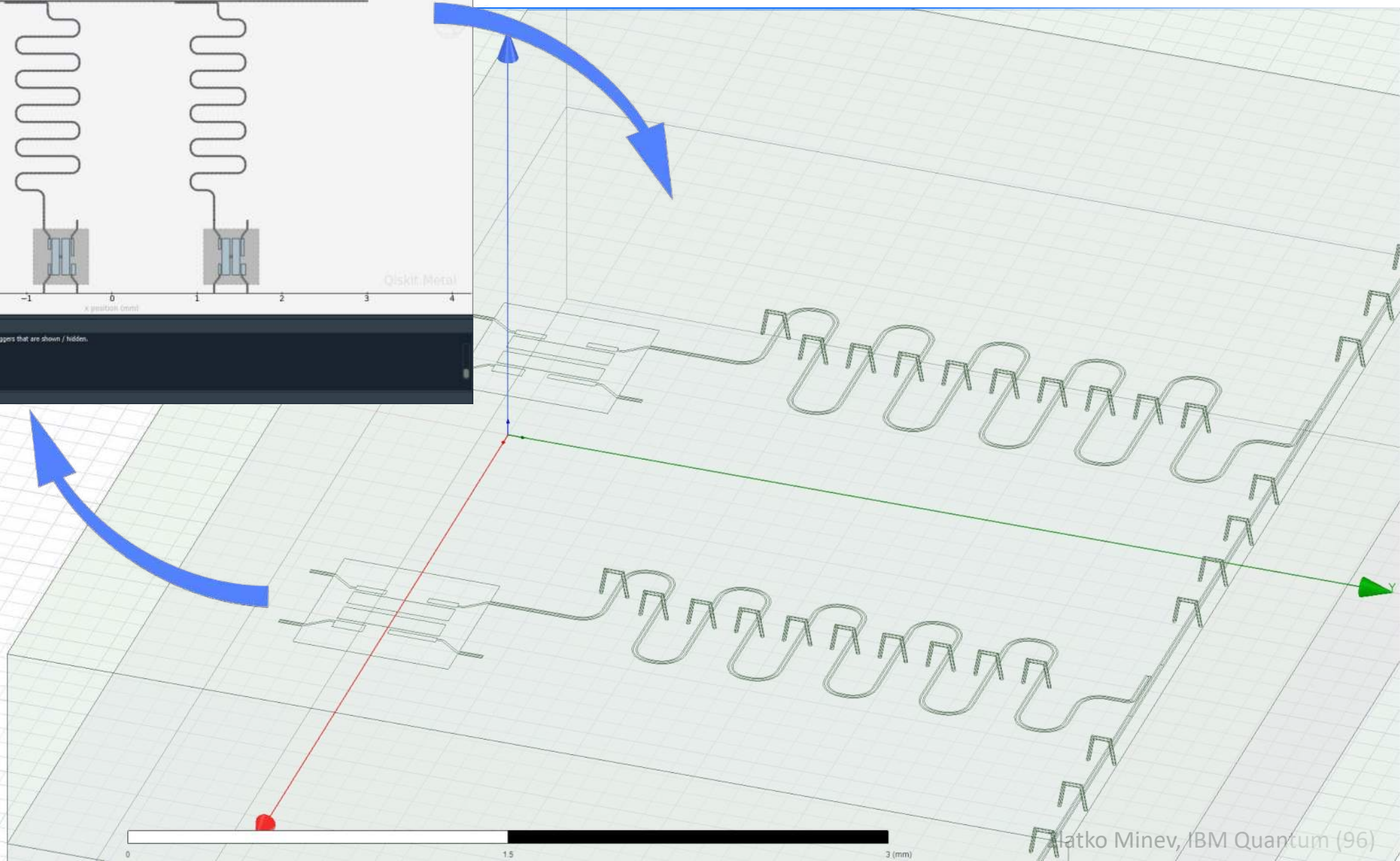
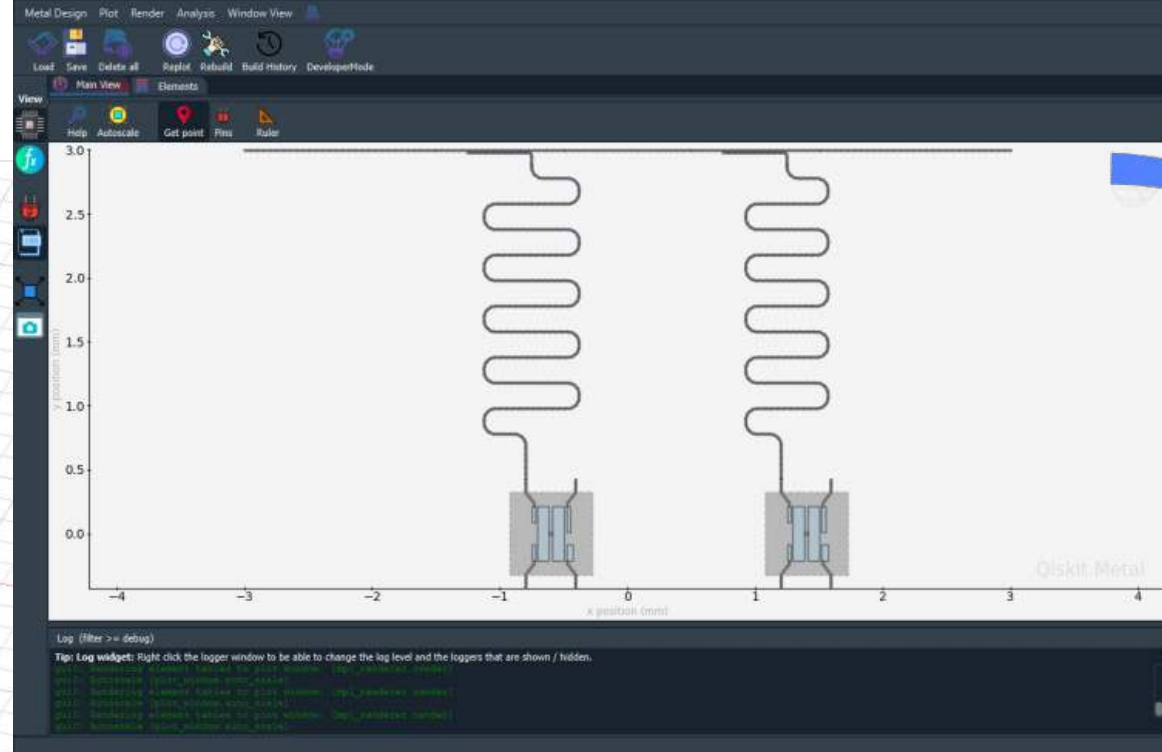
Automated with

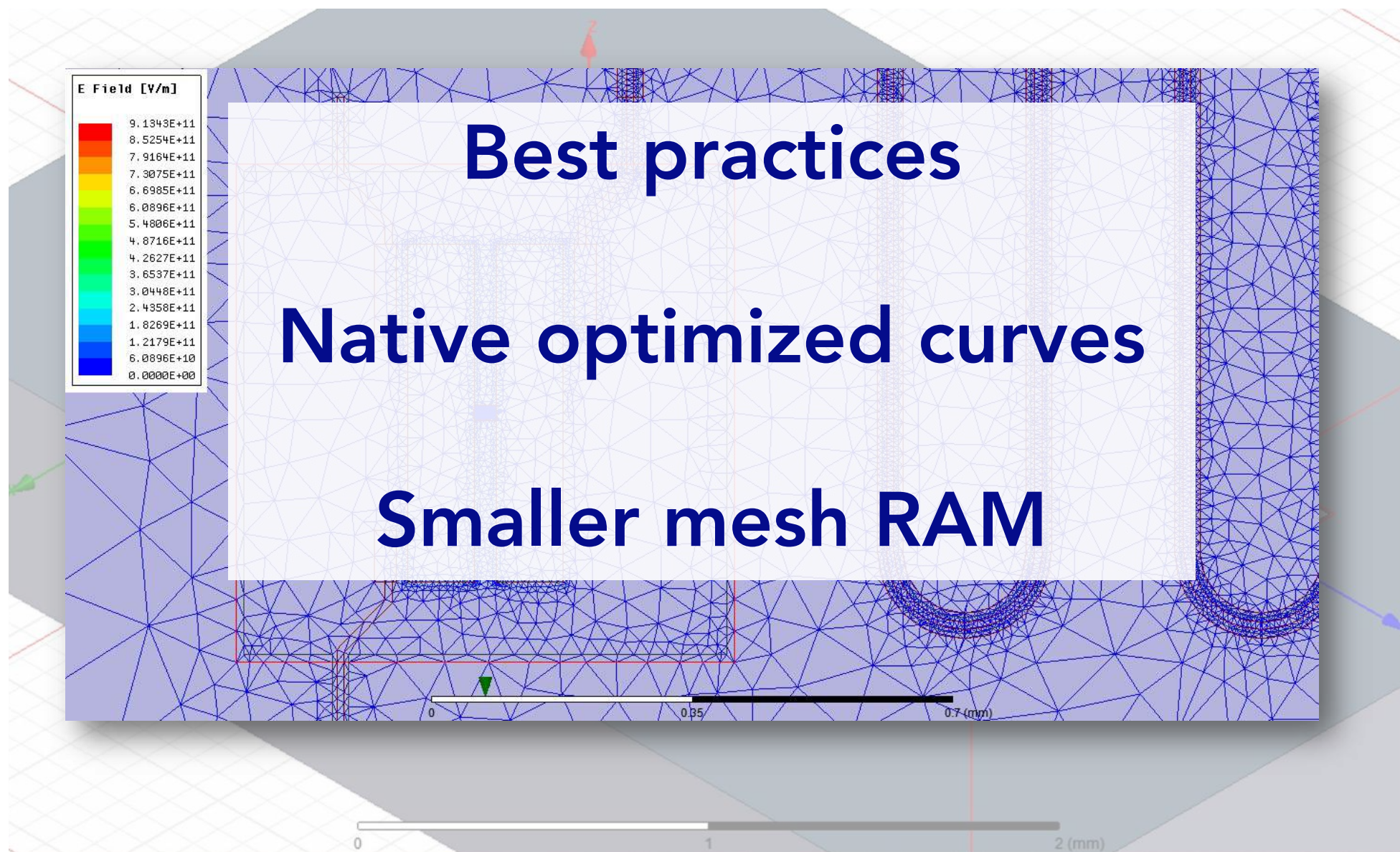
Qiskit | quantum device design

extractor model

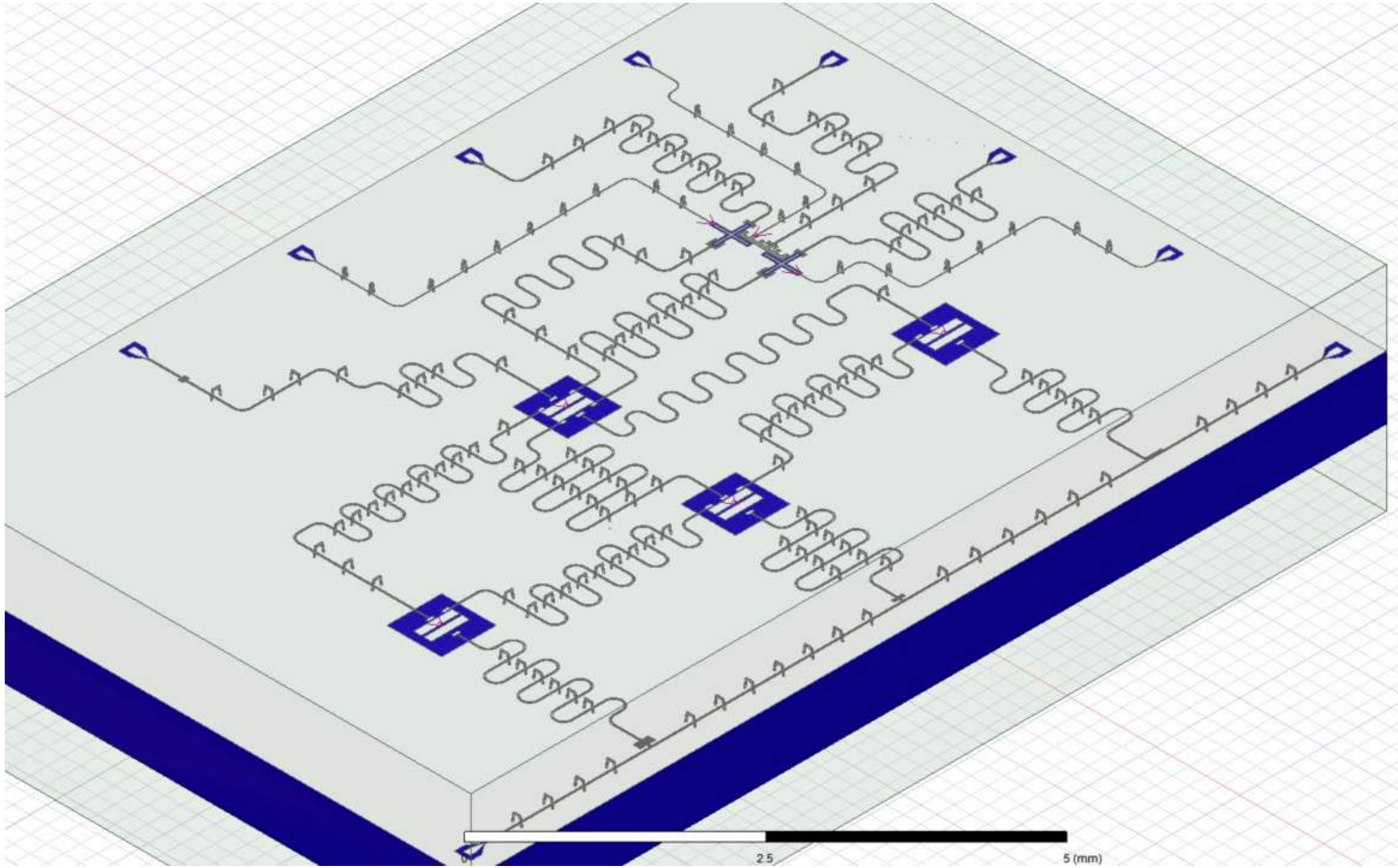


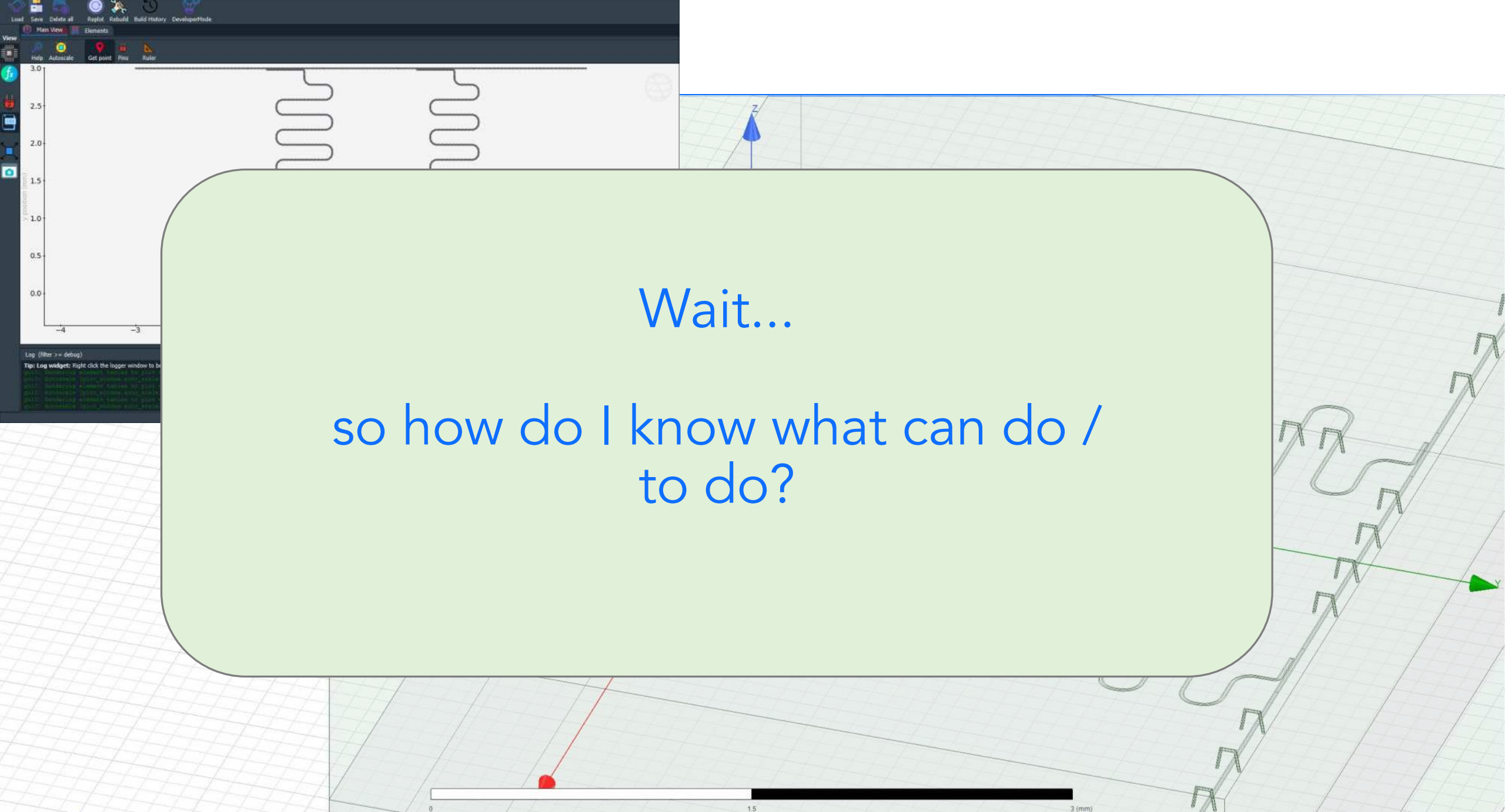
More designs





More complex quantum chips – see next tutorial

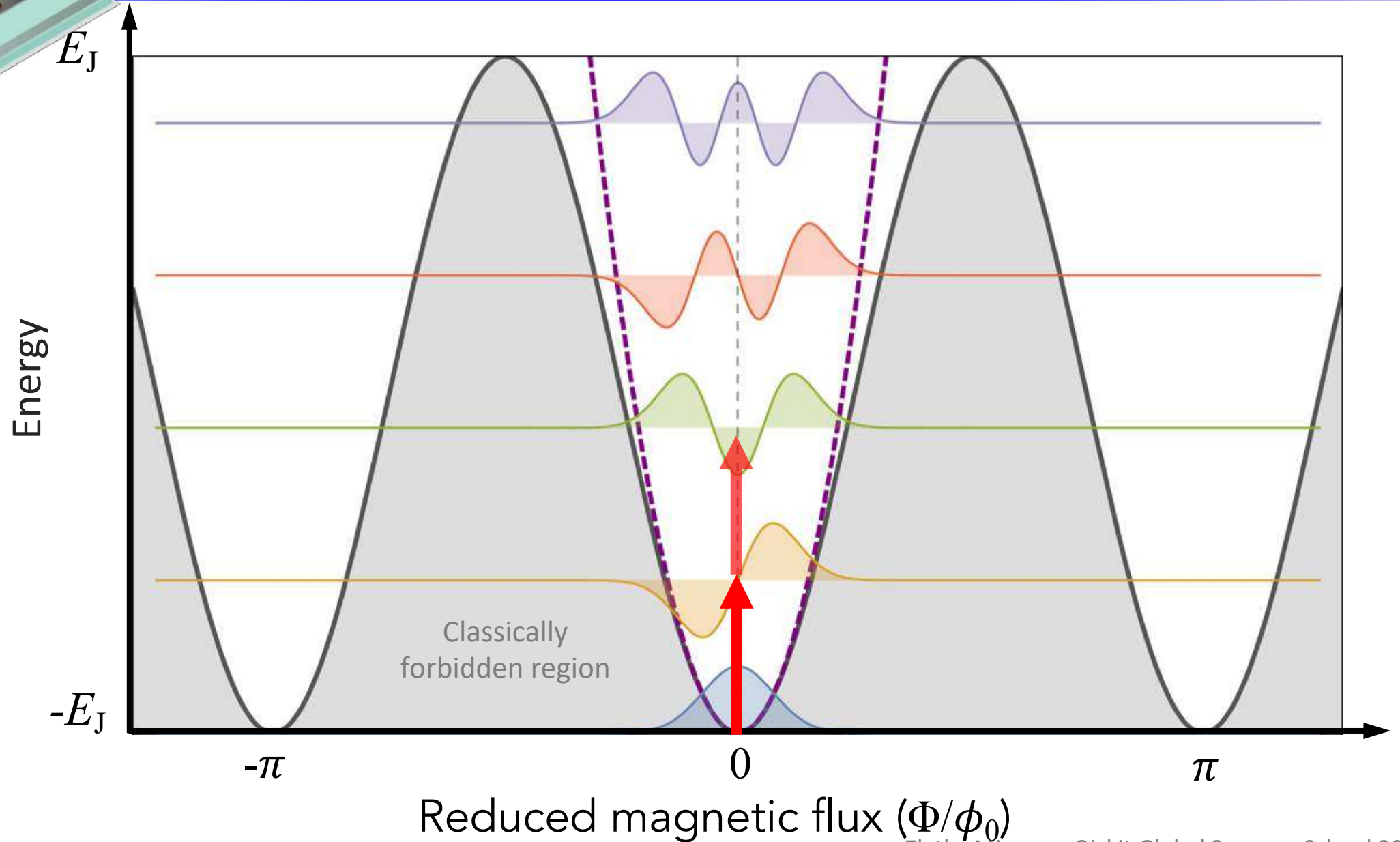
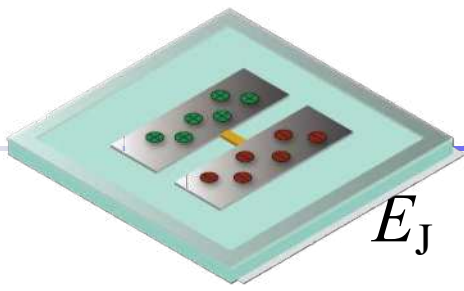




Wait...

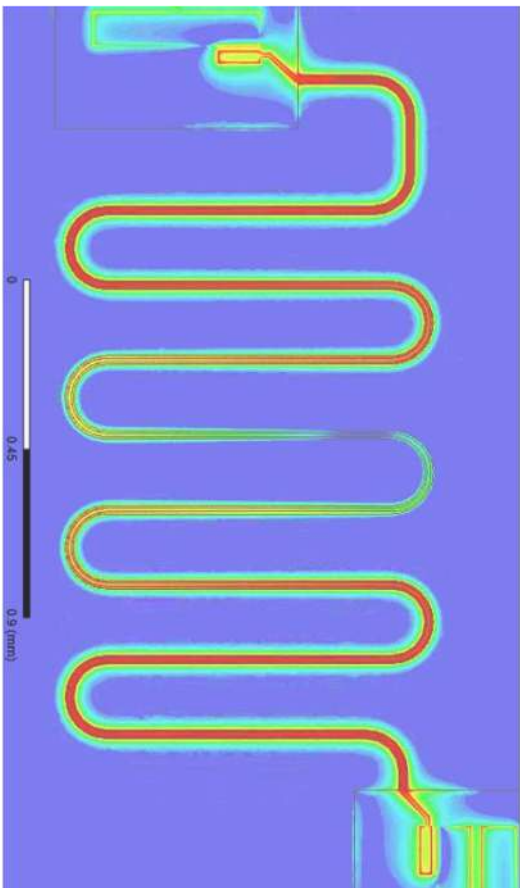
so how do I know what can do /
to do?

Exploring a transmon qubit

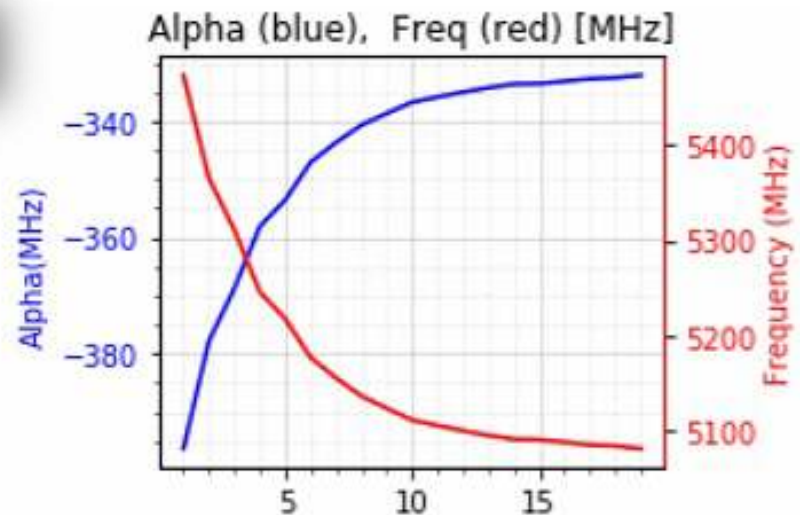


Automated analysis and reports

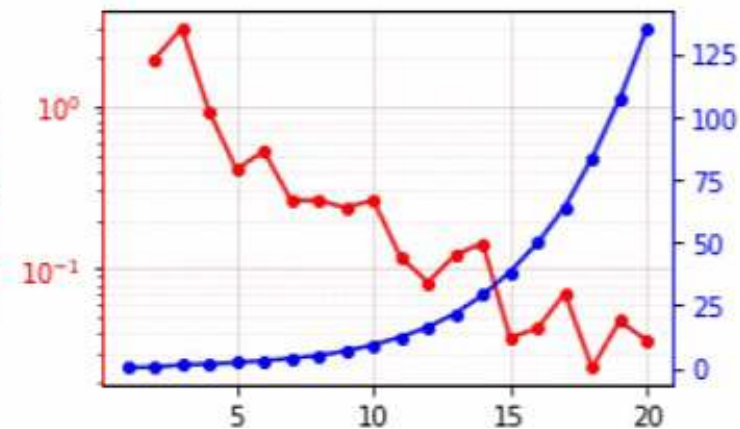
```
metal.analysis.lumped_model.analyze('Q1')
```



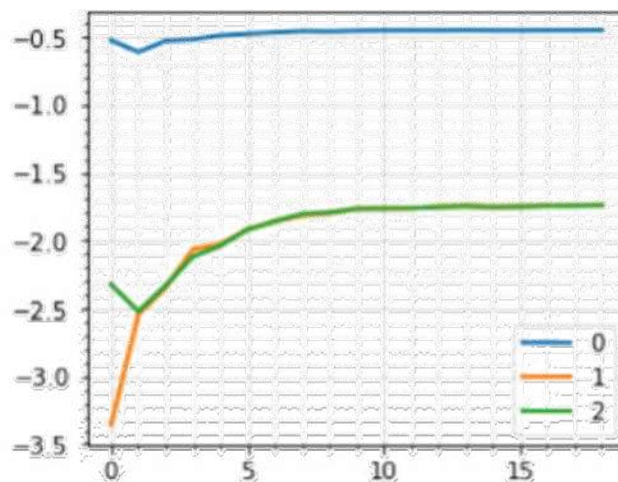
Qubit frequency & anharmonicity



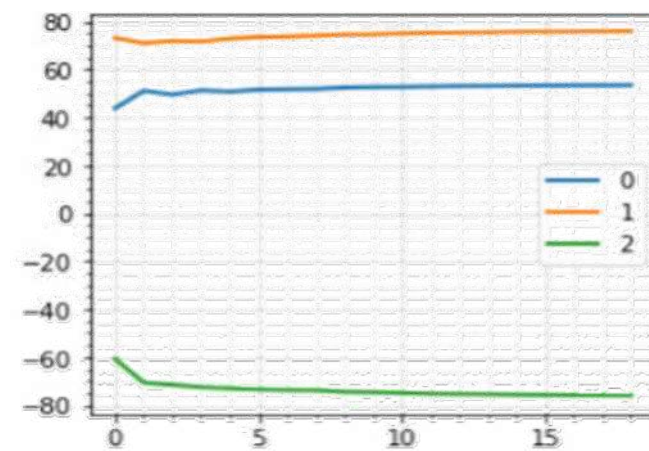
FE simulation convergence



cross-Kerr χ coupling (MHz)



Linear g coupling (MHz)



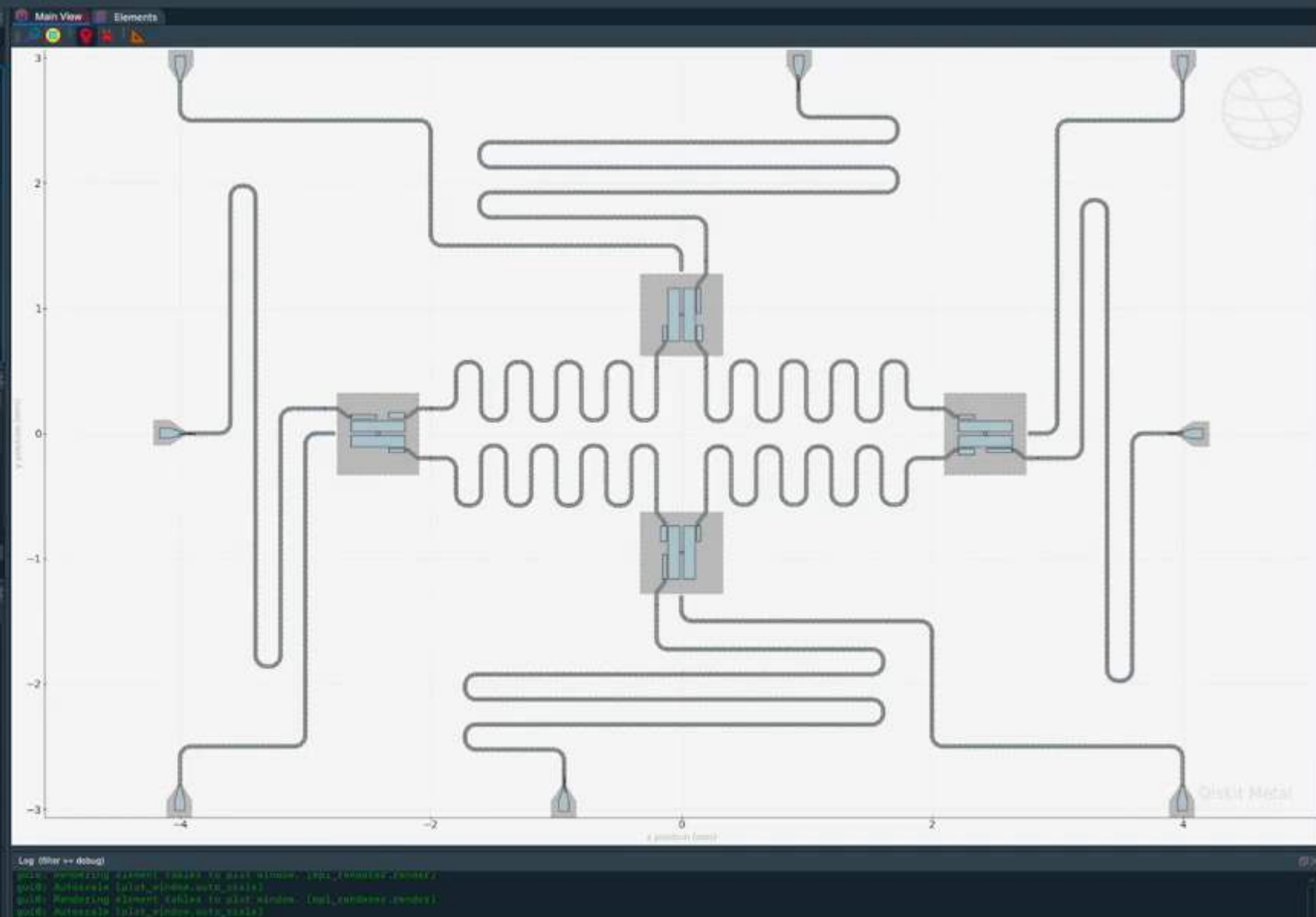
Analysis pass number

The screenshot shows the Qiskit Metal QComponents window. At the top, there is a toolbar with icons for List, Save, Create all, Import, Rebuild, New QComponent, QOS, HFSS, and QTD. Below the toolbar is a table of components. The table has columns for Name, QComponent class, QComponent module, Build status, and ID. The components listed are Q1 through Q11, all of which are TransmonPockets or RouteMeanders, and all have a build status of 'good'. Below the table, there is a section for Variables, which shows a table with columns for Variable name, Value, and Parsed value (in um). The variables listed are cpx_width (10 um) and cpx_gap (6 um). At the bottom, there is a section for Library, Pins, and Variables, with a button to 'Select a QComponent to edit'.

Name	QComponent class	QComponent module	Build status	ID
Q1	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	1
Q2	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	2
Q3	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	3
Q4	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	4
cpx1	RouteMeander	qiskit_metal.components.interconnects.meandered	good	5
cpx2	RouteMeander	qiskit_metal.components.interconnects.meandered	good	6
cpx3	RouteMeander	qiskit_metal.components.interconnects.meandered	good	7
cpx4	RouteMeander	qiskit_metal.components.interconnects.meandered	good	8
OT01	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	9
OT02	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	10
OT03	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	11

Variable name	Value	Parsed value (in um)
cpx_width	10 um	0.01
cpx_gap	6 um	0.006

Name	Value	Parsed value
Select a QComponent to edit* from the QComponents window		



Use cases Devices made with Qiskit Metal

Korea Hackathon – Metal Projects

Project 1A: Metal - Designing a simple qubit chip

Abstract

The designing of a simple two qubit system using Qiskit Metal.

Description

Using Qiskit Metal, design a single plane, two-qubit chip using superconducting qubits, which are coupled together with a coplanar waveguide (CPW) bus resonator (capacitively), and with each qubit having a CPW readout resonator. The readout resonators should be capacitively coupled to transmission line(s) which connect to launcher pads for wire bonds. All components should be selected from those available in the library.

A simple model of the potential layout is shown below,



The project members are to determine the actual physical layout, with the below target parameters in mind.

Parameters	Target Value	Parameters	Target Value
Freq_Q1	5.5 GHz	Freq_readout_Q1	7.5 GHz
Freq_Q2	5.7 GHz	Freq_readout_Q2	7.7 GHz
Anharmonicity (Ec)	300 MHz	χ	400 kHz
		κ	2 MHz
Freq_bus	6.6 GHz		
g			

The project members can also decide on the chip size, though should consider issues such as, potential box/substrate modes, or cross talk between the different components on the chip. It can be assumed wirebonds are available to be added where desired.

Members

Korea Hackathon – Metal Projects

Project 1B: Metal - Designing a simple qubit chip

Abstract

The designing of a simple two qubit system using Qiskit Metal.

Description

Using Qiskit Metal, design a single plane, two-qubit chip using superconducting qubits, which are coupled together with a coplanar waveguide (CPW) bus resonator, and with each qubit having a CPW readout resonator. The readout resonators should be coupled to transmission line(s) which connect to launcher pads for wire bonds. The qubits should be newly written components, as should the capacitive couplers. Only the CPW transmission line components can be from the available library.

The project members are to determine the actual physical layout, with the below target parameters in mind.

Parameters	Target Value	Parameters	Target Value
Freq_Q1	5.5 GHz	Freq_readout_Q1	7.5 GHz
Freq_Q2	5.7 GHz	Freq_readout_Q2	7.7 GHz
Anharmonicity (Ec)	300 MHz	χ	400 kHz
		κ	2 MHz
Freq_bus	6.6 GHz		
g			

The project members can also decide on the chip size, though should consider issues such as, potential box/substrate modes, or cross talk between the different components on the chip. It can be assumed wirebonds are available to be added where desired.

Members

Deliverables

- The design notebook.
- Documentation showing the successful simulation/analysis of the chip design meeting the desired parameters.

First quantum-hardware hackathon

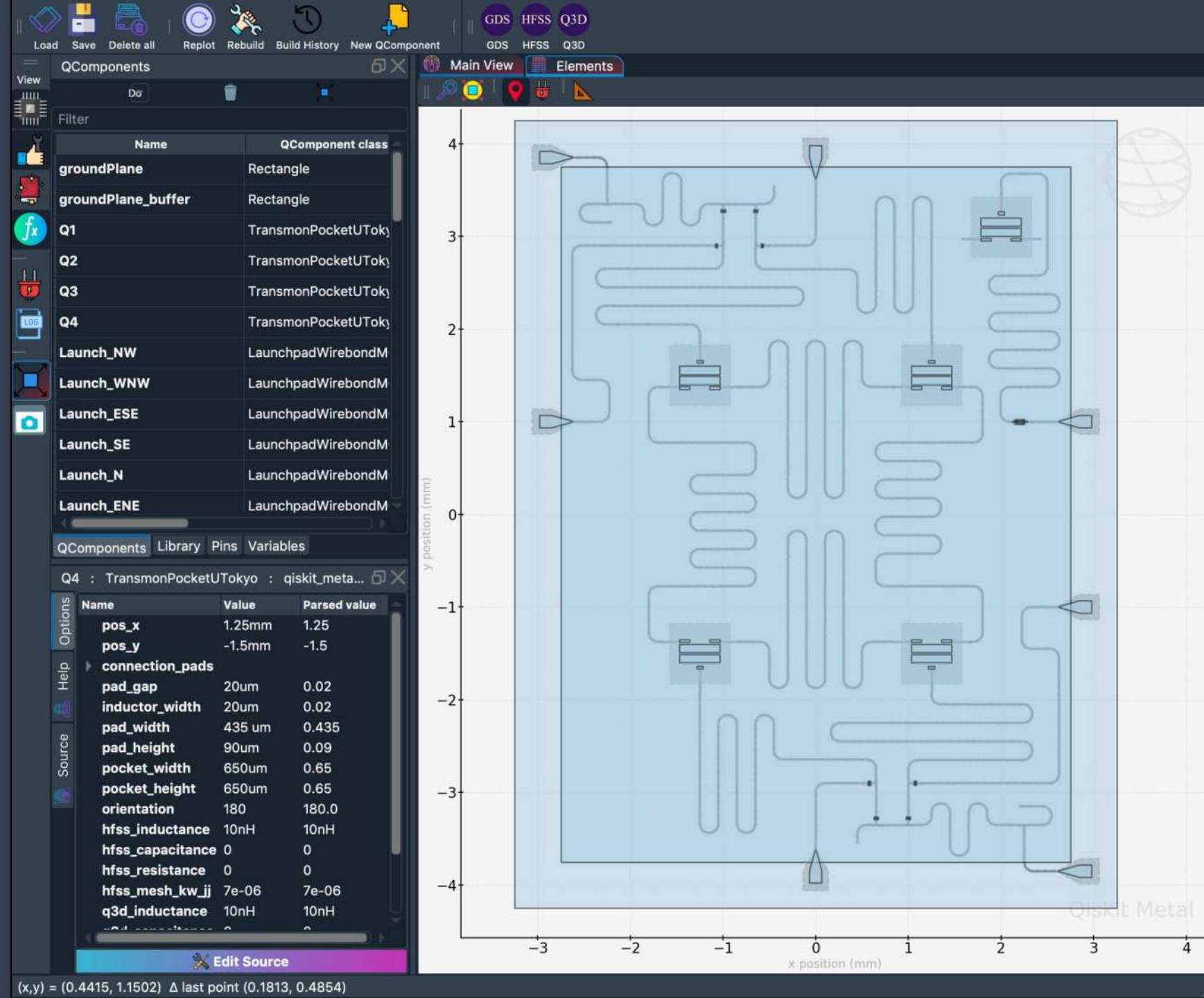
The collage features several key elements:

- Top Left:** A screenshot of a Qiskit circuit diagram with a red box highlighting a section labeled "Readout_Q2".
- Top Center:** A screenshot of a 3D model of a quantum device structure, with the name "Zlatko Minev" displayed below it.
- Top Right:** A list of participants in a video call, including Sam (Co-host), Yuri Kobayashi (Co-host), Atsushi Matsuo, BoSeong Kim, Byung Du Oh, Dongsin Kim, and Eunchan Cho.
- Bottom Left:** A screenshot of a Qiskit simulation window showing a table of results for a quantum circuit. The table has columns for Name, Val, Unit, Evaluation, and Type.
- Bottom Center:** A screenshot of a JupyterLab window showing a code cell with a plot of "Eigenmode f vs. pass [GHz]". The plot shows two curves, one for "hfss1" and one for "hfss2", both increasing and then plateauing as the pass number increases.
- Bottom Right:** A screenshot of a video call window showing a person in a video call, with the name "Thomas M" displayed below it.

BIG BIG THANKS TO Yuri, Samanvay, Hanhee, Junye & team!!
Thomas, Dennis, Metal coaches, ...

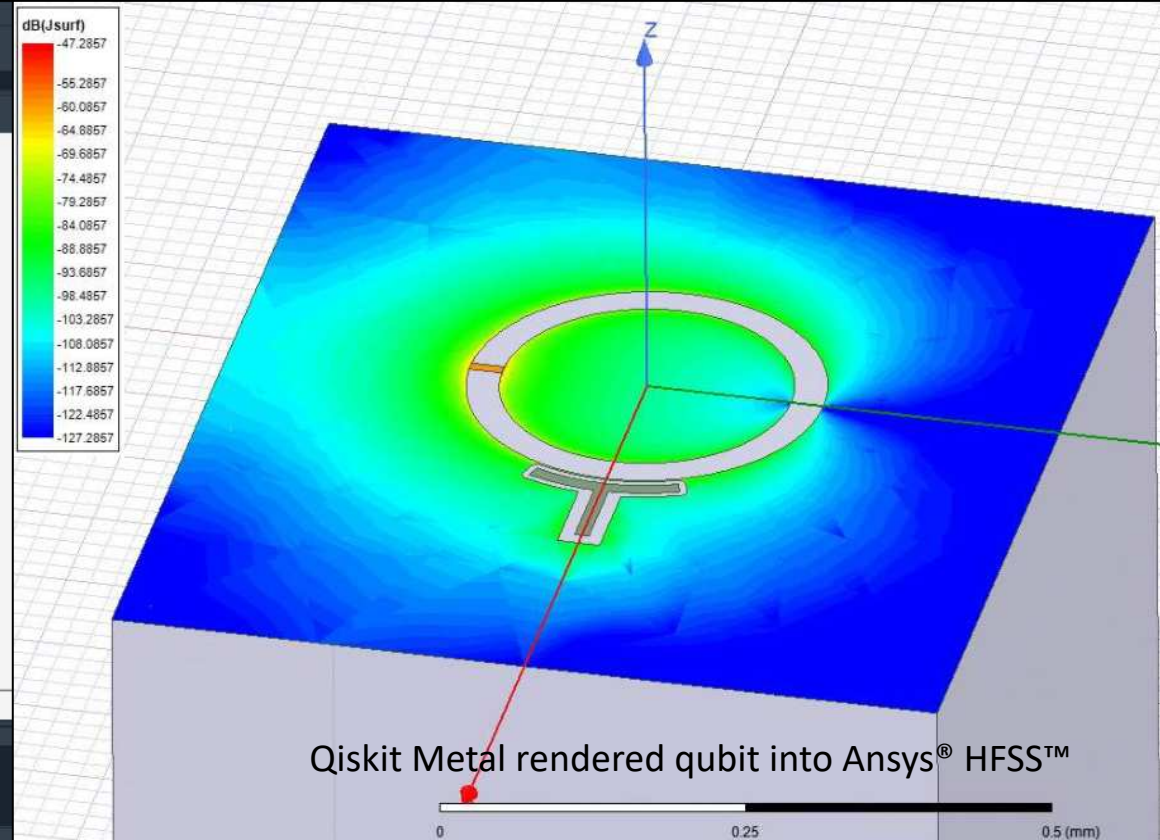
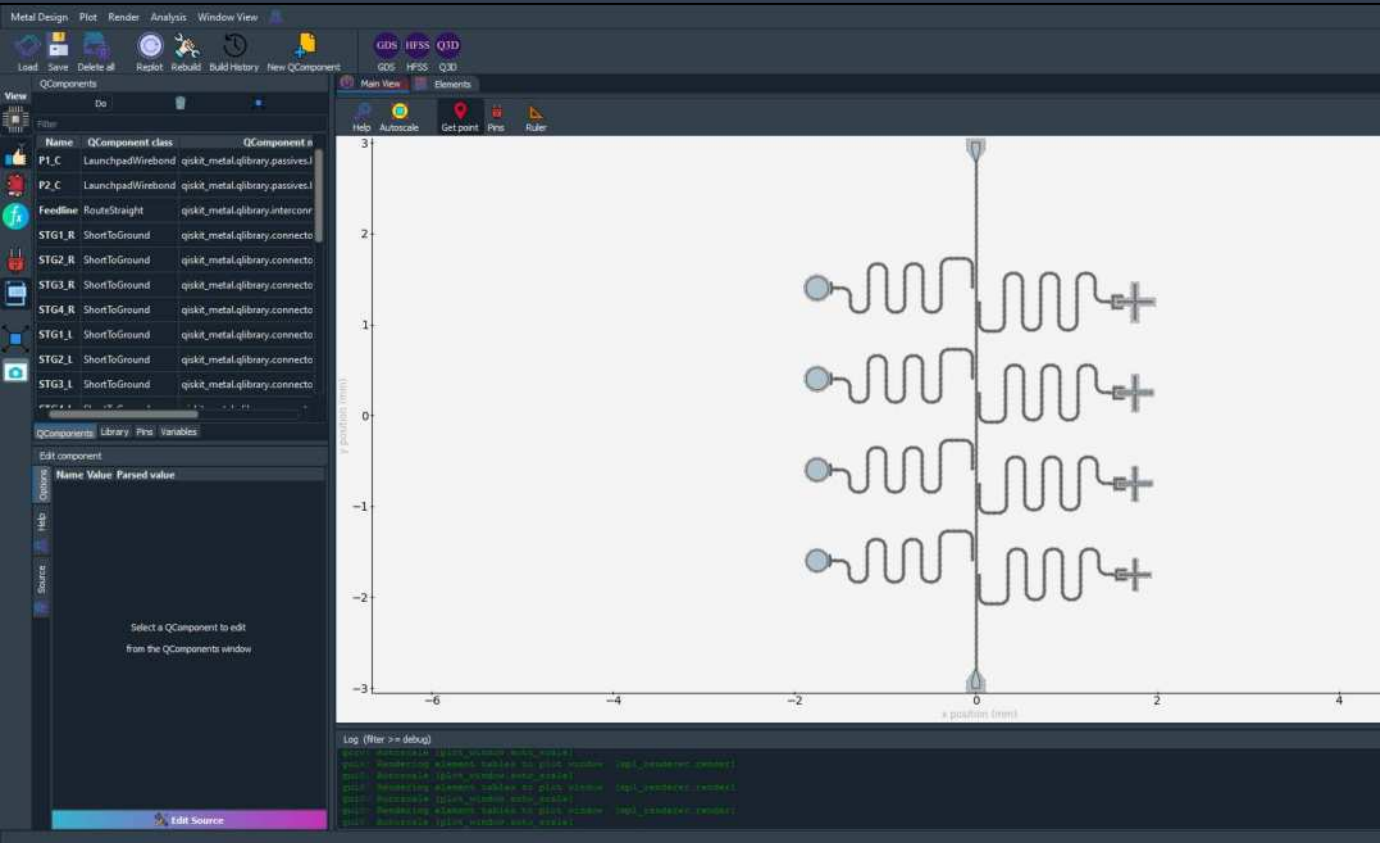
Devices made with Metal

IBM 5Q Tsuru U Tokyo



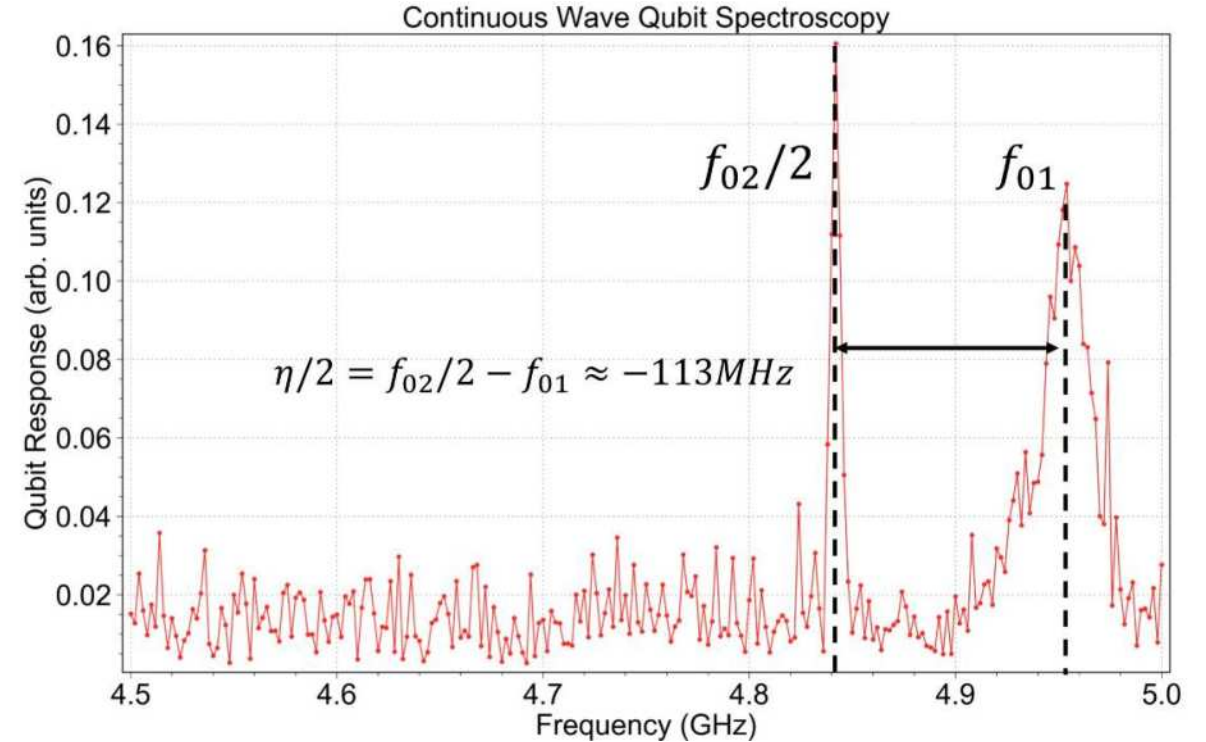
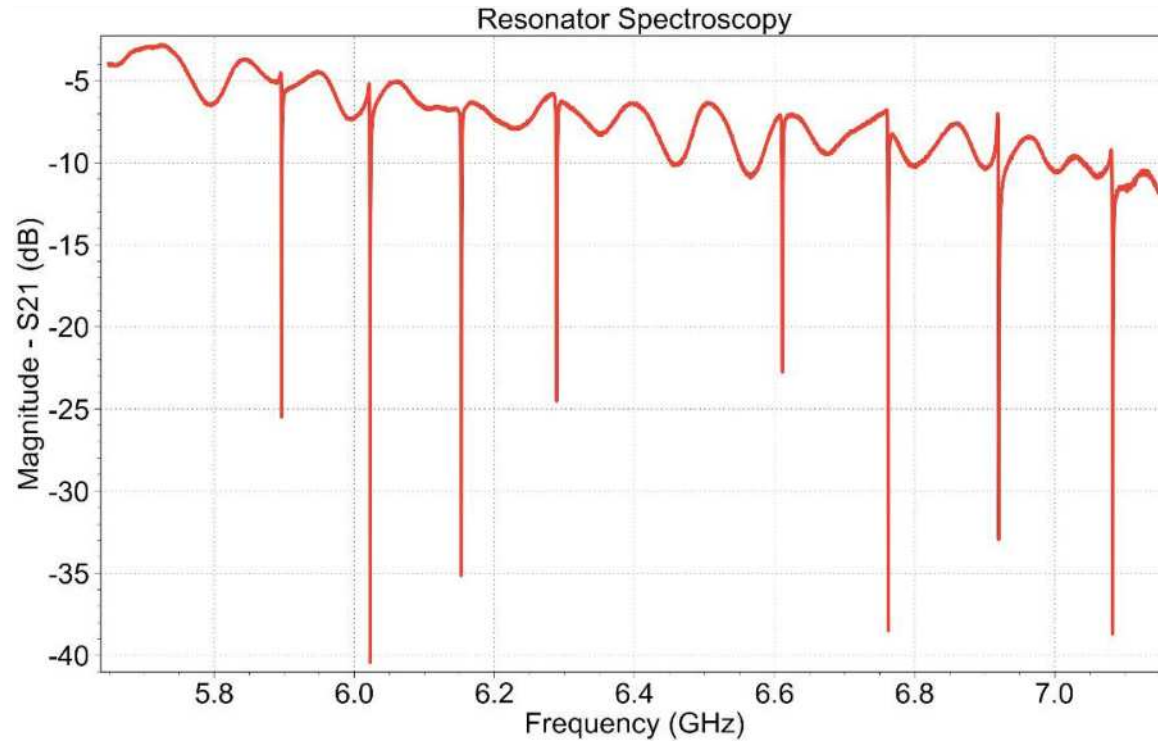
(x,y) = (0.4415, 1.1502) Δ last point (0.1813, 0.4854)

An 8-qubit chip designed by C. Warren, Amr Osman, and Bylander team (Chalmers) with the help of Qiskit Metal



Experimental results of the Chalmers device designed with Qiskit Metal

Data courtesy of *C. Warren* & Amr Osman (Bylander, Chalmers)



Forbes

You Don't Have To Be A Rocket (Or Quantum) Scientist To Design A Quantum Computer Chip Using IBM's New Tool Called Qiskit Metal



KeysightQuantum @KeysightQ

Replying to @zlatko_minev @Fo

🎉 Congrats! 🌟 Is there an eta



Analyst Notes:

- 1.) Considering all its advantages, Qiskit Metal should be a clear long-term winner for IBM and the quantum community.
- 2.) By reducing the complexity of chip design, IBM has eliminated a significant barrier that may make quantum attractive to more people.
- 3.) Qiskit Metal makes it possible for young K-12 students to have an understandable hands-on learning experience with quantum computing. Metal can turn an impossible task into a fun learning experience. A positive early learning experience with Metal could result in thousands of future quantum researchers.



Docs & tutorials

qiskit.org/documentation/metal

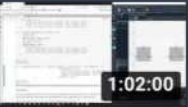
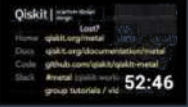
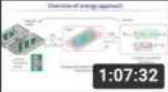
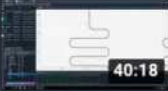
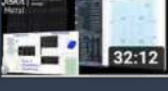
Tutorial videos

YouTube – see docs

Slack

[#metal](#) (qiskit workspace)

Live biweekly tutorials

- 1  **Qiskit Metal E01 - Overview**
Qiskit
- 2  **Welcome to Qiskit metal: download and install instruction**
Qiskit
- 3  **Qiskit Metal E02 - End to end example of Quantum Chip Design - Part 1 of 2**
Qiskit
- 4  **Qiskit Metal E03 End to end example of Quantum Chip Design - Part 2 of 2**
Qiskit
- 5  **Qiskit Metal E04 - QComponents for parametric design**
Qiskit
- 6  **Qiskit Metal E05.1 - Analysis - Capacitance and Frequency Control**
Qiskit
- 7  **Qiskit Metal E05.2 - Analysis - Eigenmode and Energy Participation**
Qiskit
- 8  **Qiskit Metal E05.3 - Analysis - EPR Theory**
Qiskit
- 9  **Qiskit Metal E05.4 - Analysis - Summary EPR Quantization with Code Example**
Qiskit
- 10  **Qiskit Metal E05.5 - Analysis - Finish Eigenmode Start Impedance Analysis**
Qiskit
- 11  **Qiskit Metal E05.6 - Analysis - Extracting S Parameters for a Hanging Resonator**
Qiskit
- 12  **Qiskit Metal E06 - Quantum Analysis 101 Prerequisites**
Qiskit
- 13  **Qiskit Metal E07 - Introduction to the transmon qubit**
Qiskit
- 14  **Qiskit Metal E08 - Physics of the Cooper Pair Box Transmon Qubit**
Qiskit
- 15  **Qiskit Metal E09 - Parametric Sweep Analysis**
Qiskit
- 16  **Qiskit Metal E10 - How to contribute to the code base**
Qiskit
- 17  **Qiskit Metal E11 - Introduction to Energy Participation Ratio (EPR)**
Qiskit

Summer School Lectures

Minev, Z., Lec. 16-22, *Introduction to Quantum Computing and Quantum Hardware*,
url: qiskit.org/learn/intro-qc-qh (2020)



Magnetic flux and inductance

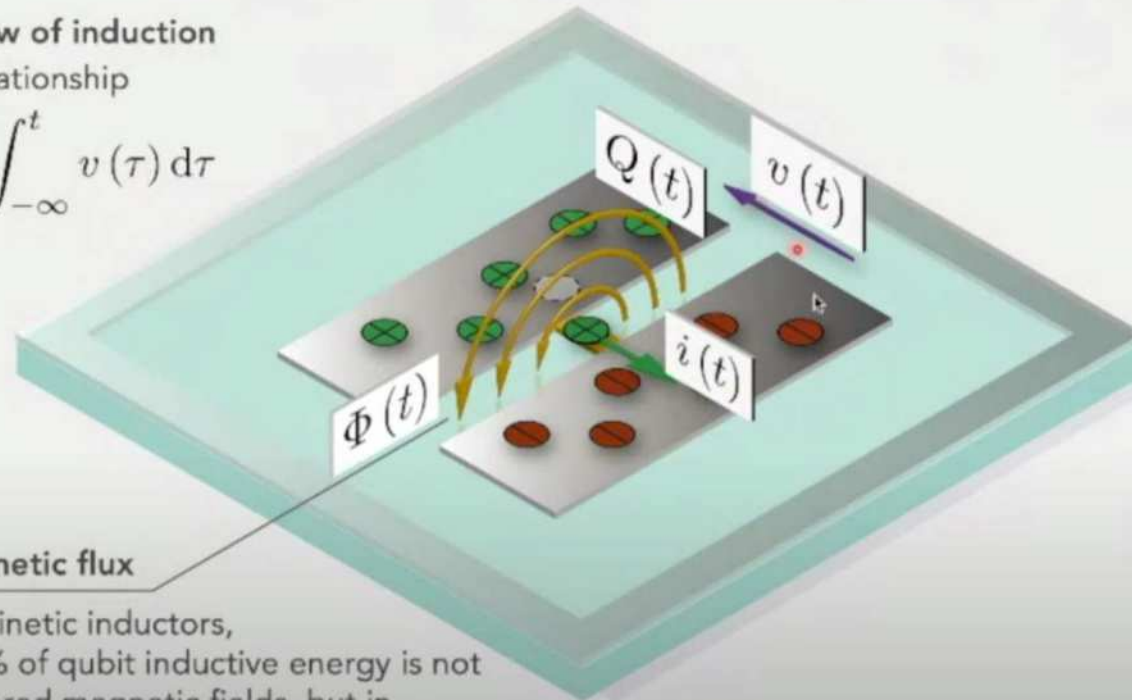
Faraday's law of induction

Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$

Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic induct



It can be related to the voltage across

Image: Zlatko Minev



45:39 / 50:21



For more information, review the **Introduction to Quantum Computing and Quantum Hardware** lectures below

• Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions Part 1	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions Part 2	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions Part 3	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Part 1	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Part 2	Lecture Video	Lecture Notes	Lab
• Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Part 3	Lecture Video	Lecture Notes	Lab

Next steps

Tightly integrated work by Dr. Thomas McConkey!

qiskit.org/metal -> docs -> videos -> E02

Check out references, problems given in the lecture,
dangerous bends

Break away from the rules of today



Thank you!

Zlatko K. Minev



“It is by *logic* that we prove,
but by *intuition* that we discover.”

Henri Poincaré

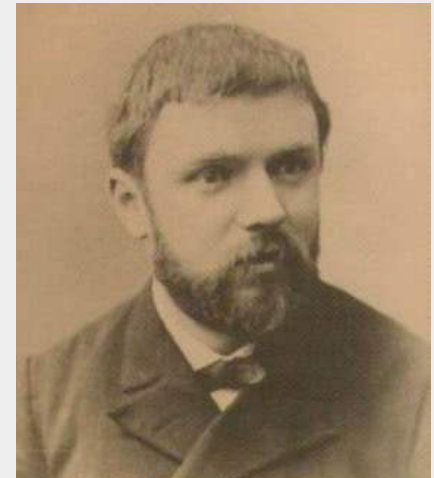


Photo by Eugène Pirou